

An Introduction to Neutron Scattering

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1.1 FUNDAMENTALS

1.1.1 Why Are Neutrons so Unique?

The neutron is a stable subatomic particle (a hadron) when bound in atomic nuclei. Free neutrons are unstable and undergo β decay with a mean (exponential) lifetime of ca. 15 min [1], a limitation which is barely of significance for most research applications. Neutrons have zero charge, mass $m_n = 1.0087$ atomic-mass units [2], one-half spin (i.e., a fermion with $\sigma = 1/2$), and magnetic moment $\mu_n = \gamma \mu_N$, where $\mu_N = e\hbar/2m_p$ is the nuclear magneton and γ is a constant equal to -1.913 . Notwithstanding over 80 years of neutron research following its discovery in the early 1930s [3–5], this particle still remains at the heart of cosmology and particle physics [6,7].

The four fundamental physical properties presented above also make the neutron a highly effective probe of condensed matter. The zero charge and nonzero spin imply that neutrons can interact with materials via nuclear (short-ranged, $\text{fm} = 10^{-15} \text{ m}$) and magnetic interactions, respectively. Both nuclear and magnetic interaction probabilities (or *cross sections*) are small, so the neutron can usually penetrate well into the bulk of the specimen under investigation. In the language of scattering theory, neutron-matter interactions (both nuclear and magnetic) can then be described within the first Born

approximation and thus given in explicit form by relatively simple expressions. We shall return to this particular point in [Section 1.2](#).

Neutrons for condensed-matter research are usually obtained by slowing down or “moderating” high-energy (MeV) neutrons produced by nuclear reactions. The slow neutrons thus produced have kinetic energies on the order of $k_B T$, where T is the temperature of the moderating medium (or “moderator”) and k_B is the Boltzmann’s constant. Given the wave nature of the neutron, there is an associated *de Broglie* wavelength λ_n for a given kinetic energy E_n obeying the well-known relation

$$E_n = \frac{h^2}{2m_n \lambda_n^2} = \frac{\hbar^2 k_n^2}{2m_n} = k_B T, \quad (1.1)$$

where h is the Planck’s constant and k_n is the neutron wave vector. The value of m_n is such that for $T \sim 300$ K, $\lambda_n \sim 2$ Å ($= 2 \times 10^{-10}$ m), a distance comparable to interatomic spacings in solids and dense fluids. Such neutrons are therefore ideally suited to studies of the *atomic structure* of condensed matter via diffraction (elastic scattering) studies, a fact that was recognized soon after the discovery of the neutron by Chadwick in 1932. Furthermore, the associated kinetic energy of such neutrons, $k_B T$, is on the order of 25 meV, a typical energy for collective vibrational modes (or phonons) in condensed-matter systems. Thus, both thermal-neutron wavelengths and energies are ideally suited to studies of the *atomic dynamics* via spectroscopy (inelastic scattering) experiments (cf. [Sections 1.3 and 1.4](#)).

The magnetic moment μ_n of the neutron also makes it a unique probe of *magnetism* on atomic and molecular scales, as neutrons may be scattered from the magnetic moments associated with unpaired electron density. Again, the wavelength and the energy of typical thermal neutrons are such that both magnetic structure and the associated spin dynamics can be studied via neutron-scattering experiments (cf. [Sections 1.5 and 1.6](#)). The spin of the neutron has further important consequences. When a neutron is scattered by a nucleus with nonzero spin, the strength of the interaction depends on the relative orientation of neutron and nuclear spins. This makes the neutron a unique probe of *nuclear-spin correlations* (both static and dynamic) at low temperatures (cf. [Section 1.7](#)).

[Figure 1.1](#) shows neutron-wavelength spectra for the two most common types of high-intensity source, namely, a fission reactor and an accelerator-driven spallation source. It is customary to classify neutrons by the moderator temperature, that is, cold, thermal ($T = 300$ K), and hot. Neutrons that have not slowed down to characteristic velocities for a given moderator temperature are called epithermal and are readily available at spallation sources. Cold(hot) moderators give rise to spectra centered at longer(shorter) neutron wavelengths λ_n , as shown in [Figure 1.1](#). A detailed comparison between the performance of steady and pulsed neutron sources ultimately depends

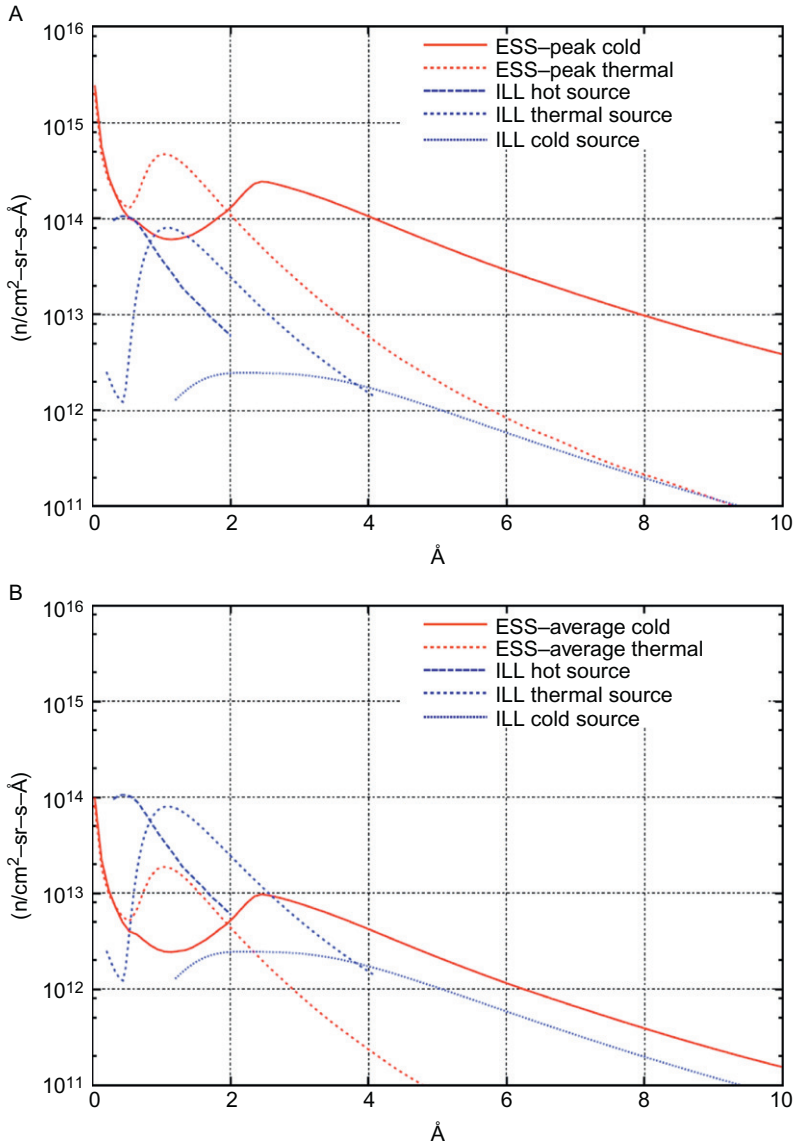


FIGURE 1.1 Neutron flux per unit solid angle per unit wavelength as a function of wavelength for cold, thermal, and hot moderators at the Institut Laue-Langevin (ILL) compared to those expected for the European Spallation Source (ESS), a next-generation pulsed spallation facility. The ILL is a steady reactor source whereas ESS is envisaged to deliver 2.86 ms pulses at a repetition rate of 14 Hz. (a) and (b) A comparison between these two neutron sources in terms of peak and time-averaged fluxes, respectively. *Reprinted with permission from Ref. [8].*

TABLE 1.1 Properties of the Neutron at Selected Kinetic Energies

Quantity	Unit	Definition	Ultracold	Cold	Thermal	Epithermal
Energy E_n	meV ^a		2.5×10^{-4}	1	25	1000
Temperature T	K	E_n/k_B	2.9×10^{-3}	12	290	12,000
Wavelength λ_n ^b	Å	$h/(2m_n E_n)^{1/2}$	570	9.0	1.8	0.29
Wave vector k_n ^c	Å ⁻¹	$(2m_n E_n)^{1/2}/\hbar$	0.011	0.7	3.5	22
Velocity v_n ^d	m s ⁻¹	$(2E_n/m_n)^{1/2}$	6.9	440	2,200	14,000

For definitions of the various symbols and associated numerical values, see the *List of Commonly Used Symbols* and references therein.
^a1 meV = 1.6022×10^{-22} J, the amount of energy necessary to move an electron against a potential difference of one millivolt (1 mV).
^b λ_n (Å) = $9.0446 [E_n \text{ (meV)}]^{-1/2}$.
^c k_n (Å⁻¹) = $0.69469 [E_n \text{ (meV)}]^{1/2}$.
^d v_n (m s⁻¹) = $437.39 [E_n \text{ (meV)}]^{1/2}$.

on application and, therefore, it requires defining the *useful neutron flux* at point of use, as presented in Chapter 3. To a first approximation, the peak neutron flux shown in Figure 1.1a can be used as a rough figure of merit provided that it is possible to exploit the full time structure of the pulsed neutron beam [9]. In this case, next-generation accelerator-based pulsed sources offer the prospect of a significant increase in useful flux relative to current capabilities. Neutron production and moderation are discussed in detail in Chapters 2 and 3.

For reference purposes, Table 1.1 gives the energy, temperature, wavelength, wave vector, and velocity for typical neutrons in each category, along with useful relations to convert between these quantities using commonly used physical units. Likewise, the energies associated with atomic and molecular excitations in condensed matter are expressed in a multitude of ways across the scientific literature, and these can always be related to the amount of energy E transferred to/from the neutron during a scattering experiment. To aid the reader, Table 1.2 provides a conversion table between commonly used units.

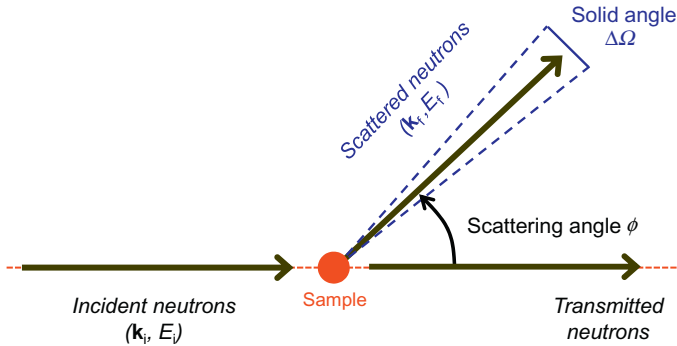
1.1.2 Thermal Neutrons for Condensed Matter Research

We consider a simple scattering experiment as schematically shown in Figure 1.2. Suppose that a beam of neutrons characterized by wave vector $\mathbf{k}_i$ falls on the sample. As pointed out in the previous section, the interaction probability is rather small and in a typical experimental situation most

TABLE 1.2 Common Quantities Used to Denote Energy or Energy Transfer

Quantity	Definition	Value at $E = 1$ meV
Radian frequency ω	$E/\hbar$	$1.5193 \times 10^{12} \text{ rad s}^{-1}$
Frequency ν	E/h	$0.24180 \times 10^{12} \text{ Hz} = 0.24180 \text{ THz}$
Spectroscopic wave number (or kayser) $\tilde{\nu}$	E/hc	8.0655 cm^{-1}
Temperature T	E/k_B	11.605 K

For definitions of the various symbols and associated numerical values, see the *List of Commonly Used Symbols* and references therein.

**FIGURE 1.2** Schematic diagram of a generic arrangement for neutron-scattering experiments.

neutrons are transmitted without any interaction. Some, however, will be scattered and can be measured with a neutron detector placed, let us say, along direction $\hat{\mathbf{k}}_f$.

If the incident beam is characterized by a uniform flux Φ (neutrons crossing per unit area and per unit time), the sample has N identical atoms in the beam, and a detector of efficiency η subtends a solid angle $\Delta\Omega$, the count rate C on the detector is proportional (if $\Delta\Omega$ is small enough) to all these quantities. The proportionality constant is called the *differential scattering cross section* (denoted as $d\sigma/d\Omega$). Mathematically, we write

$$C = \eta \Phi N \left(\frac{d\sigma}{d\Omega} \right) \Delta\Omega. \quad (1.2)$$

The differential cross section is a function of both $\mathbf{k}_i$ and $\mathbf{k}_f$ (technically speaking, it describes the correlation between these two vector quantities).

As such, it is a property of the sample being measured and it may also depend on the spin state of incident and detected neutrons. The most important types of interaction for condensed-matter studies are the nuclear interaction and the magnetic-dipole interaction. Other (much weaker) interactions are discussed in detail in the [Appendix](#).

1.1.2.1 Atomic Nuclei

The interaction between a slow neutron and an atom through the nuclear force can be expressed in a very simple form, provided that the neutron wavelength is much longer than the range of the nuclear force. This condition is easily fulfilled by thermal neutrons (for a more detailed discussion, see the [Appendix](#)). To illustrate this feature, we consider the case where the atoms in the sample are identical and noninteracting. In this case, the differential cross section is a constant:

$$\frac{d\sigma}{d\Omega} = b^2, \quad (1.3)$$

where the scattering length b is a property only of the nucleus (characterized by atomic and atomic-mass numbers Z and weight A , respectively) and, in general, its spin state relative to that of the neutron. Typical values of b are on the order of 1–10 fm and the resulting cross sections are proportional to the square of the scattering length. Scattering cross sections are measured in barn (b), where $1 \text{ b} = 100 \text{ fm}^2 = 10^{-28} \text{ m}^2$. Physically speaking, Equation (1.3) tells us that neutron deflection following a collision is equally probable in all directions, that is, isotropic scattering. It is noteworthy that scattering lengths for thermal neutron scattering are for the most part (e.g., away from resonances) independent of collision energy. Therefore, scattering observables such as total and differential cross sections can be linked quite effortlessly to the properties of the material under investigation via the dependence of scattering probabilities on energy and momentum transfers and not their absolute values (see discussion in [Section 1.1.3](#)). This invariance of thermal neutron-scattering lengths with energy has also important consequences in the design of neutron-scattering instruments, that is, the equivalence between *so-called* direct- and indirect-geometry instrumentation, covered in [Chapter 3](#).

The scattering length is a quantity that depends on the details of the interaction between the neutron and the constituents of the nucleus. For this reason, both the magnitude and sign of b change in a rather irregular fashion as a function of Z or A , with a relatively mild increase in absolute magnitude as one moves across the periodic table of the elements (see [Figure 1.3](#)). In contrast, the atomic scattering length for X-ray scattering is a monotonically increasing function of Z , as X-ray scattering from an isolated electron is characterized by a (bound) scattering length equal to the classical electron radius $r_0 \approx 2.8 \text{ fm}$. These features of neutron scattering relative to X-rays have some

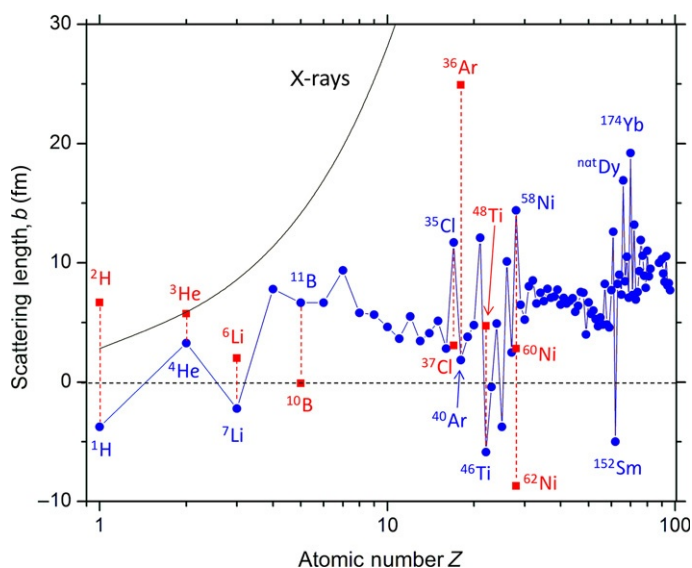


FIGURE 1.3 Neutron (blue circles and red squares) and X-ray (black) scattering lengths across the periodic table of the elements. For neutrons, the blue circles correspond to the most abundant isotope for a given element or its average value across naturally occurring isotopes where these data are not available. Red squares show the scattering lengths for specific isotopes of a given element, so as to illustrate their variation with A for a given Z . The X-ray data assume electrically neutral species (i.e., Z =number of electrons). Neutron-scattering lengths have been taken from the [Appendix](#).

powerful consequences, making neutrons sensitive to the presence of light atoms, most notably hydrogen, and to the difference between atoms with similar atomic number, for example, adjacent transition metals. The variation in scattering length between different isotopes of the same element can be large, as in the case of hydrogen, lithium, chlorine, argon, titanium, or nickel, and can be exploited in so-called *isotopic-substitution* experiments. The most widely used case corresponds to hydrogen versus deuterium, discussed in [Chapter 4](#) and the [Appendix](#). Notwithstanding the merits of this approach for the study of hydrogenous matter (organic materials, polymers, biomolecules, etc.), physicochemical changes associated with deuteration must always be kept in check, as recently reviewed in Ref. [10]. An in-depth discussion of isotopic substitution beyond the case of hydrogen can be found in Ref. [11] (see, in particular, [Table 1.1](#) of this work).

In many cases, the scattering lengths for the two spin states ($I \pm 1/2$) resulting from the interaction of a nucleus of spin I with the neutron are also quite different, which, in general, leads to “incoherent” scattering as discussed in [Section 1.2](#). If the nuclei are polarized, the spin dependence of the scattering

length can be exploited to measure nuclear-spin correlations and structure (Section 1.7). In some cases, the scattering length can also be a complex quantity, where the imaginary part describes the absorption of neutrons by a particular nuclide. This situation arises when there is a low-lying resonance of the neutron–nucleus system, leading to scattering and absorption cross sections that depend on wavelength. More details can be found in the Appendix.

Also, we point out that the outcome of the scattering experiment discussed above and leading to isotropic scattering, cf. Equation (1.3), will depend on whether the nuclei are fixed or free to recoil during the scattering process. The scattering length measured in the laboratory-fixed frame is smaller by a factor $(A/A + 1)$ if the nucleus is free to recoil [12]. For most condensed-matter neutron-scattering experiments, the fixed atom is a more appropriate limiting case, and it is conventional to quote the corresponding “bound-atom” values for scattering lengths and cross sections. We will return to the free-atom case when discussing scattering from the ideal gas in Section 1.4.

A list of scattering lengths and corresponding cross sections is given in the Appendix, which also includes a theoretical discussion and a description of the various methods to measure these quantities.

1.1.2.2 Electron and Nuclear Spins

With a magnetic moment $\mu_n = -1.913\mu_N$, neutrons can interact with unpaired electrons in magnetic materials. For the simple case of a paramagnet with electrons localized on specific ions with random spin orientations and no external magnetic field, the differential cross section is given by

$$\frac{d\sigma}{d\Omega} = (\gamma r_0)^2 [f(\mathbf{k}_i - \mathbf{k}_f)]^2 S(S + 1), \quad (1.4)$$

where r_0 is the classical electron radius defined above and $|\gamma r_0| \approx 5.4$ fm, S is the spin quantum number for the ions, and $f(\mathbf{k}_i - \mathbf{k}_f)$, the so-called *magnetic form factor*, is given by the Fourier transform of the spatial distribution of unpaired-electron density about the ion centre. The form factor appears because this spatial distribution is of comparable dimensions to interatomic distances and, thus, to neutron wavelengths. This situation is in stark contrast to the short-range nuclear force, characterized by a delta-function-like spatial distribution relative to thermal-neutron wavelengths (i.e., a constant). Unlike nuclear scattering lengths, however, magnetic form factors can often be calculated quite accurately from the electronic wave functions of either atoms or molecules. A classic example from the pioneering work of Shull and Yamada is shown in Figure 1.4. Conversely, measurements of the form factor via the use of Equation (1.4) yield direct information about the wave functions associated with magnetic electrons. This particular case is discussed in more detail in Section 1.5.3.1.

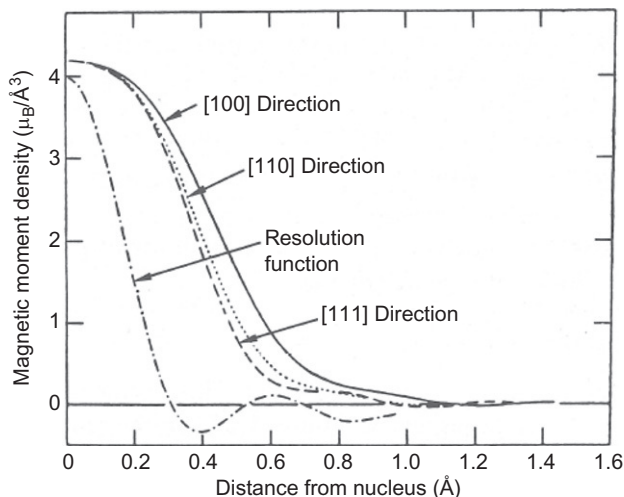


FIGURE 1.4 Distance dependence of the magnetic moment density for bulk iron along different crystallographic directions. These data have been obtained from magnetic-form-factor measurements using neutron scattering. Reprinted with permission from Ref. [13].

In addition to scattering from intrinsic electron spins S , the magnetic moment of the neutron can also interact with the current associated with a moving electron, giving rise to an orbital contribution to the magnetic interaction. In view of the more complicated form of the expressions for magnetic scattering, we develop the formalism of the following sections in terms of the nuclear interaction, returning to magnetism in [Sections 1.5 and 1.6](#).

1.1.2.3 Nuclear Absorption

At the beginning of this section, we discussed two possible outcomes for a neutron incident on a sample: transmission and scattering. A third possibility is neutron absorption by a nucleus, either through a direct nuclear reaction or as a result of compound-nucleus formation. In most scattering experiments, these phenomena can be treated as a simple attenuation of the transmitted and scattered beams via the addition of an extra (multiplicative) factor in Equation (1.2), along with other “nonideal” contributions such as beam removal due to scattering and multiple-scattering events involving the same neutron. Strong absorption due to resonant capture is a special case that leads to a wavelength-dependent (complex) scattering length. Resonance scattering by epithermal neutrons has been proposed as a means to access higher-order correlations than those accessible via thermal neutron scattering [14], yet no experimental implementation of this approach has been demonstrated to date. In general, there is also a dependence on the spin state of the neutron–nucleus system. Absorption cross sections are listed along with scattering cross sections in the [Appendix](#).

1.1.3 Conservation Laws

Returning to our schematic experimental arrangement shown in Figure 1.2, we see that the incident neutron wave vector $\mathbf{k}_i$ undergoes a change in direction during scattering. This deflection implies a change in the neutron linear momentum, which, as we know from classical mechanics, must be exchanged with the sample. The momentum transferred to the sample is conventionally described in terms of the corresponding scattering vector $\mathbf{Q}$, and the law of conservation of linear momentum is written as

$$\mathbf{k}_i - \mathbf{k}_f = \mathbf{Q}. \quad (1.5)$$

The momentum $\hbar\mathbf{Q}$ may be taken up by the scattering atom and subsequently shared with the rest of the sample, or in certain cases as discussed below, the sample as a whole can recoil; either outcome depends on the specific circumstances of the scattering event. In general, both the magnitude and direction of the linear momentum $\mathbf{k}$ will change, and so energy is also exchanged with the sample. Energy conservation implies

$$\frac{\hbar^2 k_i^2}{2m_n} - \frac{\hbar^2 k_f^2}{2m_n} = E_i - E_f = E, \quad (1.6)$$

where the neutron energy transfer E is defined to be positive when the sample takes energy from the neutron (neutron energy loss). The two terms on the left-hand side represent the initial (E_i) and final (E_f) neutron energies. A process whereby the neutron is scattered from $\mathbf{k}_i$ to $\mathbf{k}_f$ is therefore associated with a set of values of $\mathbf{Q}$ and E . The probability of scattering as a function of the variables ($\mathbf{Q}$, E) is a property of the particular sample and its environment (temperature, pressure, magnetic field, etc.) in the scattering experiment. Most of this book is devoted to a description of the form and properties of this function for different materials and environments.

1.1.4 The Structure of Materials

Let us consider a very simple case where the sample consists of individual noninteracting scattering centers of mass M . Then, the relation between E and $\mathbf{Q}$ is of the form

$$E = \frac{\hbar^2 Q^2}{2M}. \quad (1.7)$$

We note that Equations (1.5) and (1.6) represent the familiar two-body collision problem [15], with the solution

$$E = \frac{2E_i m_n}{M} (1 - \cos \phi) + O\left(\frac{m_n}{M}\right)^2, \quad (1.8)$$

where ϕ is the angle between $\mathbf{k}_i$ and $\mathbf{k}_f$ and E_i is the incident neutron energy. Thus, E tends to zero as the mass M of the scattering unit becomes large. If the sample is a solid, a significant amount of the scattering is “elastic,” that is, $E=0$ exactly, and the sample recoils as a rigid unit. This phenomenon is analogous to recoil-less emission in Mossbauer spectroscopy where the scattering unit becomes the entire sample (or at least a macroscopic fraction of it). Hereafter, we shall also assume that $u=m_n$, where u is the unified atomic-mass unit defined in the *List of Commonly Used Symbols*.

From Equation (1.6), we must have $|\mathbf{k}_i|=|\mathbf{k}_f|$ in the case of purely elastic scattering, and thus from Equation (1.5)

$$2k_i \sin(\phi/2) = Q. \quad (1.9)$$

For crystalline materials, elastic scattering is strong when $\mathbf{Q}$ is equal to a reciprocal lattice vector of the crystal:

$$\mathbf{Q} = 2\pi \left(\frac{h}{a}, \frac{k}{b}, \frac{l}{c} \right) \quad \text{or} \quad Q = \frac{2\pi}{d}, \quad (1.10)$$

where d is the spacing of the (h, k, l) set of crystal planes. Putting $k_i = 2\pi/\lambda_i$, we derive the familiar *Bragg condition* for diffraction:

$$\lambda_i = \lambda_f = 2d \sin(\phi/2) = 2d \sin \theta. \quad (1.11)$$

where θ is the Bragg angle. This expression describes the situation when waves scattered from successive crystallographic planes interfere constructively to give an intensity maximum in the diffraction pattern. In the case of a single crystal, there is a further condition that the crystal orientation be such that $\mathbf{Q}$ is parallel to the reciprocal lattice vector. The study of crystalline materials with thermal neutrons, leading to the identification of plane spacings from the peaks in the diffraction pattern, and the inference of crystal structures from these data represent a major research activity at neutron sources around the world. This particular application of neutron scattering is discussed in more detail in [Chapter 4](#).

Equation (1.11) shows that there are two possible methods of scanning through the different plane spacings in the material: varying ϕ at constant λ_i or varying λ_i at constant ϕ . Both methods are used extensively. In both cases, the value of λ_i must be either defined by the experimental configuration or measured during the experiment for each scattering event. The most commonly used methods are based on a determination of either neutron wavelength using the Bragg condition for a single-crystal monochromator or analyzer (cf. Equation 1.11), or of velocity by measuring the time a neutron takes to travel a known distance. These methods are discussed in detail in [Chapter 3](#).

Strictly speaking, to measure truly elastic scattering one should also determine the wavelength of both incident and scattered beams to ensure that

$|\mathbf{k}_i| = |\mathbf{k}_f|$. In practice, however, structural measurements in crystalline materials are usually made under conditions such that elastic scattering is the dominant contribution and the *total scattering* corresponding to a certain λ_i and ϕ is measured regardless of energy transfer. The inelastic ($E \neq 0$) contribution in this case is either neglected or somehow accounted for during data analysis. For noncrystalline samples such as glasses or liquids, the total scattering is usually the quantity of interest, although we shall see that, in practice, inelasticity corrections still have to be made to account for $E \neq 0$ scattering. Figure 1.5 shows typical time-dependent diffraction data for a polycrystalline (powder) sample of a hydrogen-storage material during *in situ* gas desorption. These data have been measured with the variable- λ_i , constant- ϕ method at a pulsed spallation source. The development of large-area detector arrays over the past couple of decades has made these detailed kinetic studies on complex multicomponent systems quite routine at high-intensity neutron sources.

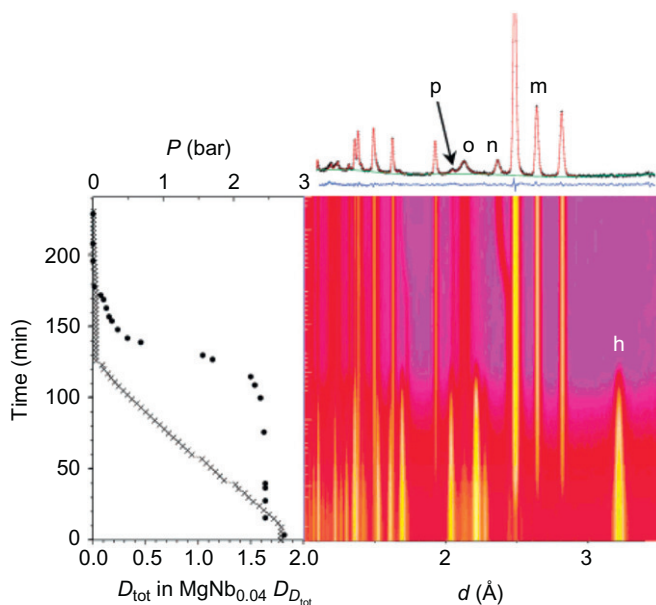


FIGURE 1.5 Time-resolved neutron diffraction data during the *in situ* desorption of D_2 from niobium-doped nanostructured magnesium at 593 K. The graph on the left shows the total deuterium content in the sample as inferred from these neutron measurements (crosses), as well as the corresponding gas pressures (black circles). The representative diffraction pattern displayed at the top shows characteristic reflections from the five different phases present during gas desorption: Mg (m), MgD_2 (h), MgO (o), Nb (n), and a perovskite phase (p). The red line in this diffraction pattern corresponds to a fit to the experimental data using Rietveld refinement methods. *Reprinted with permission from Ref. [16].*

1.1.4.1 Comparison with Other Probes

It is instructive at this juncture to provide rough estimates of the total count rate in a typical neutron diffraction experiment. Combining Equations (1.2) and (1.3), we can write

$$C = \eta \Phi N \left(\frac{d\sigma}{d\Omega} \right) \Delta\Omega \approx \eta \Phi(\lambda_i) \Delta\lambda_i N b^2 \Delta\Omega, \quad (1.12)$$

where $\Phi(\lambda_i)$ is the incident neutron flux per unit wavelength and $\Delta\lambda_i$ is the wavelength bandwidth used in the measurement. If we require a 1% resolution, for a typical diffractometer we have $\Phi(\lambda_i) \Delta\lambda_i \approx 10^9 \text{ n/cm}^2 \text{ s}$, $\Delta\lambda_i / \lambda_i \approx 10^{-2}$, $b^2 \approx 2 \times 10^{-25} \text{ cm}^2$, $\Delta\Omega \approx 10^{-4}$, and so $C \approx 10^{-22} N$. For an average count rate of 1 count per second we therefore require $\sim 10^{22}$ atoms or $\sim 1 \text{ g}$ of sample. Of course these numbers can vary considerably, and the efficiency of the measurement can be improved with devices such as multiplexed and position-sensitive detection systems to lower these requirements by orders of magnitude, as detailed in Chapter 3. It remains true, however, that the amount of sample required for neutron diffraction is generally in the milligram-to-gram range. The corresponding sample dimensions lie in the range of millimeter to centimeter, which are not a limitation as long as the material is available, since neutrons easily penetrate such distances in most materials.

The above situation is in marked contrast to the use of X-ray or electron scattering techniques where much smaller samples can be used and where the penetration depth is generally considerably less [17,18]. Experimentally, the much-higher scattering and absorption probabilities of X-rays and electrons must be taken into account when designing sample containers and other sample-environment equipment, a requirement that can turn into a real limitation when using laboratory sources or typical synchrotrons. For X-rays with energies on the order of 100 keV (available at third-generation synchrotron sources) absorption can be less of an issue.

When assessing the merits and strengths of (widely available) X-ray diffraction techniques relative to neutron scattering, the following considerations might be useful:

- Unlike neutron scattering (governed by the scattering length b and delta-like nuclear interactions), the analogous X-ray form factors $f(Q)$ display an intrinsic dependence on momentum transfer Q , falling off rapidly with increasing Q . Therefore, measurements at high- Q values ($> 5 \text{ \AA}^{-1}$) become difficult, and can severely limit the information content of the data, particularly when calculating Fourier transforms (see also our discussion of total scattering techniques from disordered materials in Section 1.4.3).
- For a given element, X-ray form factors are practically insensitive to A , and the scattering is therefore coherent. Neutron scattering can probe both coherent and incoherent (single-particle) correlations, as covered in Section 1.2.5.

- X-ray form factors display a strong energy dependence around absorption edges, described by both real (dispersive) and imaginary (dissipative) components. This behavior is termed *anomalous scattering* and can be exploited to distinguish and isolate specific elements in the sample, akin to isotope substitution and resonant scattering with neutrons—see also below.
- X-rays do not have a magnetic moment and so the interaction with magnetic materials is quite weak. X-ray polarization techniques such as circular dichroism can still be exploited to this end.

An important limitation is that neutrons are generally sensitive to all elements in a sample, compared with element-selective techniques using X-ray absorption edges [19–21]. In X-ray absorption, beam attenuation by the sample is measured as a function of incident energy, or sometimes it is more convenient to probe the resulting (background-free) fluorescence. In materials science, the important energy regions are near the edge (X-ray absorption near-edge spectroscopy, XANES) as well as the energy range above the edge (extended X-ray absorption fine-structure spectroscopy, EXAFS). Since the absorption edge is associated with an electronic transition from a core level to a free state, a detailed energy dependence of the XANES spectrum gives information on the electronic structure of valence and conduction electrons. EXAFS, on the other hand, is essentially a diffraction phenomenon whereby the photoelectron is scattered back from neighboring atoms such that the backscattered wave interferes with that of the primary photoelectron leading to a change in absorption probability. The higher the X-ray energy above the absorption edge, the larger the wave vector k of the photoelectron and, hence, the scattering wave vector $2k$ characterizing the diffraction process. The absorption spectrum as a function of the magnitude of the scattering vector then shows oscillations similar to that of the structure factor $S(Q)$ in X-ray or neutron diffraction experiments (see Section 1.4). EXAFS data therefore provide element-specific structural information by tuning to different absorption edges, yet it can be difficult to gain insight beyond nearest neighbors. Another approach possible with the advent of high-intensity synchrotron sources involves the use of anomalous X-ray scattering techniques. In this case, however, there is a correlation between the energy of the absorption edge and the maximum momentum transfer ($Q_{\max}$), ultimately dictating the spatial resolution of the measurement. For the case of light elements (e.g., those lighter than germanium with a K-edge energy of $E_k = 11.1$ keV, $Q_{\max} = 11.25 \text{ \AA}^{-1}$), other methods must be used including EXAFS, neutron diffraction (including isotopic substitution) or a combination of these and possibly other techniques. Other advantages of EXAFS include access to three-body particle correlations [22,23]. Also, measurements are relatively fast and, therefore, structural changes can be followed during rapid variations of temperature, pressure, or other external stimuli.

Element specificity with neutron-scattering techniques is primarily achieved using isotope-substitution techniques, hydrogen being the most notorious case.

Other methods include the use of nuclear resonances leading to anomalous dispersion (see, e.g., Ref. [24]), recoil scattering by epithermal neutrons (introduced in Section 1.4.3.2), and nuclear-spin alignment (see Section 1.7).

1.1.5 Adding Motion: Dynamics and Spectroscopy

In the general case where both momentum $\hbar\mathbf{Q}$ and energy E are exchanged with the sample, the conservation laws for these quantities are still given by Equations (1.5) and (1.6). The energy transferred to (or from) the sample may be taken up (or given off) by a single elementary excitation, for example, a quantum of vibration, in which case the variable E is often replaced by $\hbar\omega$, where ω is the mode frequency (or by $\hbar\nu$, $\hbar c\tilde{\nu}$, etc., as illustrated in Table 1.2). In other cases, the scattering event may involve multiple excitations and it is more convenient to refer to the total energy transfer E —note that E and $\hbar\omega$ will be used interchangeably throughout this work. It is worth emphasizing at this stage that the primary objective of inelastic neutron-scattering experiments is to measure scattered intensities as a function of both $\mathbf{Q}$ (a vector quantity) and E . As such, the physics underlying any inelastic-neutron-scattering event can be viewed as a Doppler shift in neutron velocities following their interaction with a moving target atom in the laboratory frame—we recall that the neutron speed in the center-of-mass scattering frame must remain the same before and after the collision so as to conserve linear momentum during scattering. Thus, these processes are not inelastic *per se* as far as the neutron is concerned, as such a possibility would necessarily imply the excitation of its internal degrees of freedom (quarks!).

Denoting as before the scattering angle by ϕ , Equation (1.5) can be rewritten as

$$k_i^2 - 2k_i k_f \cos \phi + k_f^2 = Q^2, \quad (1.13)$$

Combining this expression with Equation (1.6) and eliminating k_f , we obtain

$$2 \left[1 - (1 - E/E_i)^{1/2} \cos \phi \right] - E/E_i = Q^2/k_i^2 \quad (1.14)$$

which generates a family of curves of E/E_i versus Q/k_i , with ϕ as a parameter. To illustrate these relationships, Figure 1.6 shows curves for fixed incident energies $E_i = 5, 25$, and 100 meV. It can be seen that, for any value of E_i , it is possible to make measurements over a range of values of Q and E by varying the scattering angle ϕ . The accessible ranges for both of these variables increase if E_i is increased. However, the resolution in Q and E is generally compromised as k_i , and thus E_i , are increased. The values of E_i and ϕ should therefore be optimized for the particular experiment at hand.

An inelastic-scattering experiment involves the determination of the magnitudes of $\mathbf{k}_i$ and $\mathbf{k}_f$ and the scattering angle ϕ for each detected neutron. For

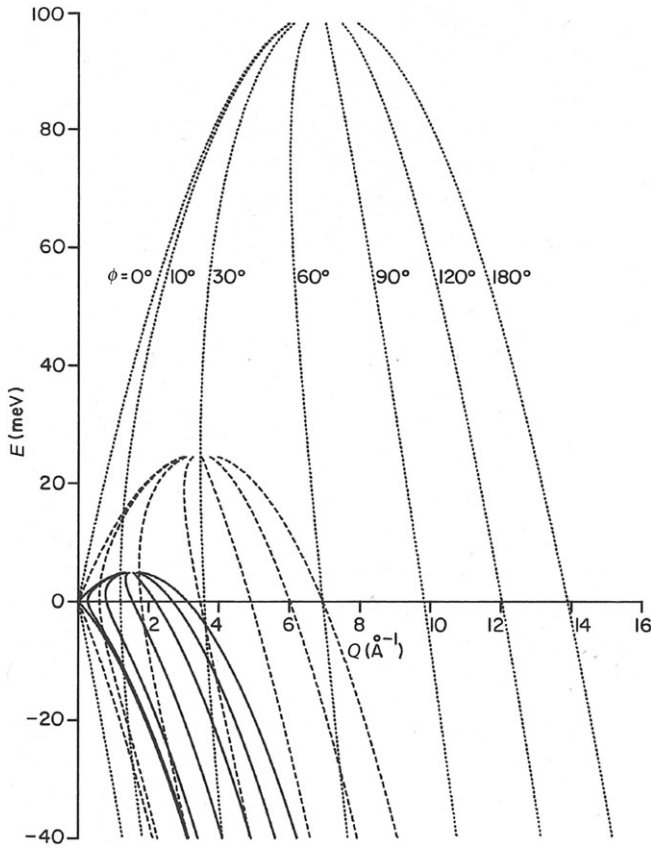


FIGURE 1.6 Relationship between energy transfer E and wave vector Q for three different incident energies $E_i = 5$ meV (solid curves), 25 meV (dashed curves), and 100 meV (dotted curves). For each E_i , loci are shown for different values of the scattering angle ϕ .

single-crystal measurements, the direction relative to the crystal lattice, that is, the direction of $\mathbf{Q}$ in the crystal coordinate system, must also be known. As discussed in [Chapter 3](#), the magnitudes of $\mathbf{k}_i$ and $\mathbf{k}_f$ may be determined by measuring either wavelength or velocity, while the directions are determined from the geometry of the experimental arrangement.

Mathematically, inelastic experiments are described by a *double differential cross section* given by

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{C}{\Phi N(\Delta\Omega)\eta\Delta E}, \quad (1.15)$$

where C represents the count rate for a given (typically small) interval of energy transfers ΔE and solid angles $\Delta\Omega$. The double differential cross section (derived more rigorously in [Section 1.2.1](#)) is a function of the magnitudes and

directions of both $\mathbf{k}_i$ and $\mathbf{k}_f$. It is customarily defined in terms of final-state (measurable) variables for the projectile, for example, Ω and E_f , but not E as sometimes written [15]. Integration of this observable over all energy transfers E at a fixed k_i and fixed direction $\hat{\mathbf{k}}_f$, gives the (single) differential cross section introduced in Equation (1.2):

$$\frac{d\sigma}{d\Omega} = \int_{-\infty}^{E_0} \frac{d^2\sigma}{d\Omega dE_f} dE. \quad (1.16)$$

Thus, $d\sigma/d\Omega$ depends in an integral fashion on the dynamics of the system but does not reveal detailed information about the underlying excitations. Similarly, we may define a total scattering cross section for the system

$$\sigma_t = \int_{4\pi} \frac{d\sigma}{d\Omega} d\Omega, \quad (1.17)$$

which can be measured in a simple transmission experiment. This observable depends only on $\mathbf{k}_i$, and obviously contains even less information about the structure and dynamics of the sample than $d\sigma/d\Omega$. Nevertheless, the ease with which it can be measured is sometimes a decisive advantage, particularly in technological applications.

Figure 1.7 illustrates the exquisite level of detail currently possible with inelastic-neutron-scattering techniques in the study of low-energy excitations in quantum matter. The data are represented in the form of (Q, E) color maps, making it self-evident that the observed spectral features are dispersive in nature. The ability to explore such low energy transfers with high resolution over wide regions of (Q, E) space remains a distinct advantage of neutron scattering relative to other techniques. The analysis and interpretation of this type of data is the subject of later sections.

1.1.5.1 Comparison with Other Spectroscopic Techniques

The count rate in a typical inelastic neutron-scattering experiment can be estimated along the same lines as in Section 1.1.4.1, except that we must account for a factor of order $\sim \Delta k_f/k_f$ because of the need to determine k_f as well as k_i for each scattering event. This additional requirement will lower the count rate by a factor of $\sim 10^{-2}$ if, for example, an energy-transfer resolution $\Delta E \approx 0.02E_i$ is required and other experimental parameters remain the same. Also, scans must be made over a range of energy transfers E by either fixing incident or final energies. As a result, sample sizes are typically in the range of grams to tens of grams, and experiments can take from hours to several days to complete. Often, the limiting factor is the time available to an experimenter at a particular spectrometer.

As pointed out above, the energy transfers that can be measured with this technique are often well matched to the characteristic time and length scales of excitations in condensed matter. This very important point is brought to

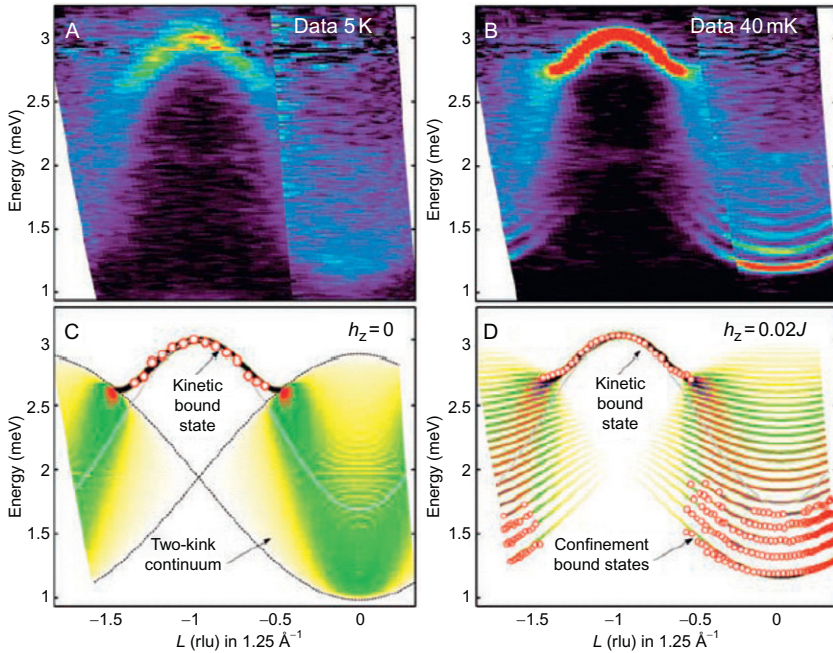


FIGURE 1.7 (a) and (b) Inelastic-neutron-scattering (Q , E) contour maps of spin excitations in CoNb_2O_6 . (b) A progression of closely spaced Zeeman levels associated with an ordered spin phase in the material. (c) and (d) Model calculations to be compared with (a) and (b), respectively. Q is given in terms of reciprocal lattice units L of $2\pi/c$. Reprinted with permission from Ref. [25].

the fore in Figure 1.8, showing the wave-vector and energy ranges accessible with different experimental techniques. It is apparent that the range for neutron scattering covers many dynamic phenomena of interest to condensed-matter research. In particular, the important range of wave vectors centered at $1\text{--}2 \text{ \AA}^{-1}$, corresponding to the Brillouin zone boundary in crystals and the first diffraction peak in glasses and liquids, is spanned by energy transfers extending several orders of magnitude. The extensive coverage of (Q , E) space possible with neutron-scattering techniques constitutes the main driver for the construction and operation of large centralized neutron facilities for condensed-matter research across the globe.

An exhaustive comparison between neutron scattering and the landscape of experimental techniques shown in Figure 1.8 transcends the scope of this introductory chapter, yet a few general remarks are in order. Inelastic X-ray scattering (IXS) can cover wide regions of (Q , E) space, yet it requires energies typically above 10 keV to access an adequate Q range for structural and dynamical studies of both ordered and disordered matter [27,28]. Given these high incident energies, it is also quite challenging to attain an energy

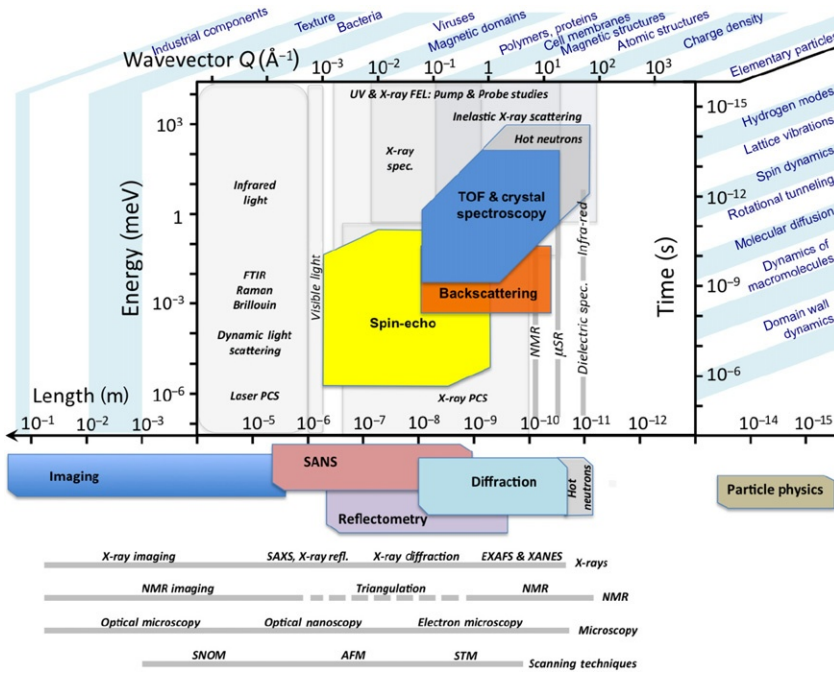


FIGURE 1.8 Regions of (Q, E) space accessible to neutrons and comparison with other experimental probes. For ease of comparison, associated length and time scales are given by the bottom and right axes, respectively. Neutron-scattering techniques include the use of epithermal (hot neutrons), thermal (diffraction as well as time-of-flight and crystal spectroscopy), and cold/ultracold wavelengths (imaging, SANS, reflectometry, backscattering, and spin-echo). Text entries around the main figure illustrate areas of scientific and technological application. *Reprinted with permission from Ref. [26].*

resolution comparable to low-energy inelastic neutron scattering. The present limit for IXS is ca. 1 meV [29,30], whereas neutron spectrometers can routinely access energy transfers which are orders of magnitude below this value. A compensating advantage of IXS resides in that the velocity of X-rays is orders of magnitude higher than sound velocities in materials. This feature avoids the kinematic restrictions that make it hard to perform spectroscopic measurements at low Q and high E with inelastic neutron scattering [30]. The resonant variant of IXS (or RIXS) exploits anomalous X-ray scattering and has gained much popularity in the past decade, particularly in the study of excitations in hard-condensed-matter systems such as semiconductors, insulators, transition-metal and rare-earth metals and oxides, or superconductors [31,32]. To gain access to slower (sub-meV) processes with photon-based probes over momentum transfers similar to those accessible to neutrons, X-ray photon correlation spectroscopy (XPCS) stands as the method of choice with

the advent of third-generation synchrotrons [33,34]. XPCS measures the speckle pattern produced when coherent light is scattered off the sample and therefore allows a direct measurement of temporal correlations—in a sense, it is the X-ray analog of neutron-spin-echo techniques, discussed in more detail in Chapter 3. More recently, the unprecedented brightness afforded by X-ray free-electron lasers and the implementation of novel techniques such as, for example, ultrafast pump-probe diffraction schemes will certainly provide a plethora of yet-to-be-explored scientific opportunities using temporally and spatially coherent X-ray pulses [35].

Light sources in the visible and infrared regions span a broad energy-transfer range yet at the expense of very small momentum transfers ($Q \sim 10^{-5} \text{ \AA}^{-1}$). Techniques include photocorrelation laser spectroscopy, dynamic light scattering, or Brillouin spectroscopy using primarily coherent light sources in the visible range [36], as well as the ubiquitous and well-established infrared and Raman spectroscopies to probe nuclear vibrations and magnetic phenomena at higher energy transfers [37,38]. In the far-infrared, the terahertz (THz $\sim 4 \text{ meV}$) window is now routinely accessible with optical techniques [39], providing good complementarity with well-established inelastic neutron scattering in the study of low-energy excitations and stochastic phenomena, including those underpinning biological activity in proteins [40] or oligopeptides [41]. Above the visible range of the electromagnetic spectrum, photoelectron spectroscopy and, most notably, its angle-resolved variant ARPES can probe the electronic structure of solids, giving detailed information on band dispersion, the Fermi surface, and the nature of many-body correlations [42]. ARPES also finds a natural overlap with high-energy (eV) studies of strongly correlated electron systems using inelastic neutron scattering [43].

To explore short- and intermediate-range structure and relaxation, nuclear magnetic resonance (NMR) stands out as a mature and versatile tool applicable to a large number of nuclides across the periodic table [44]. It also offers a significant degree of overlap with neutron scattering, particularly after the development of pulsed-field-gradient (PFG-NMR) schemes to probe stochastic relaxation processes with spatial resolution [45,46]. Slow dynamics on surfaces can also be probed with ^3He spin-echo spectroscopy [47,48], a technique developed over the past decade to access timescales of picoseconds and length scales of nanometers—a recent review highlighting its overlap with neutron spectroscopy is given in Ref. [49]. Muon-based spectroscopies [50–53] also span a complementary dynamic range in time and length scales to NMR and, in some cases such as open-shell systems like metalloproteins, to electron paramagnetic resonance. The technique is based on the implantation of spin-polarized muons in matter followed by the subsequent detection of the influence of the surrounding medium (be it an atom, molecule, or unpaired electron) on their spin dynamics. Dedicated muon facilities for condensed matter research require the use of a proton accelerator for muon production, and these are often collocated with a spallation neutron source [54–57].

Last but not least, the simplicity and unrivalled dynamic range of dielectric (or impedance) spectroscopy techniques can be used to survey dynamical processes over many orders of magnitude in the frequency domain, from 10^{-4} all the way up to $>10^{+10}$ Hz [58]. Motions that can be followed via the measurement of the frequency-dependent dielectric constant (a complex quantity with both real and imaginary components) include fast rotations in dynamically disordered solids and liquids, all the way to sluggish molecular and supramolecular reorientations in glasses and biological systems. Although the technique has been used extensively for well over a century, a detailed theoretical description of the frequency-dependent dielectric susceptibility over such a wide frequency range remains a theoretical challenge, and closed-form expressions are mostly restricted to idealized cases.

The use of “computational experiments” to predict the structural and dynamical properties of materials overarches the space–time diagram shown in Figure 1.8. The field has experienced an explosive growth [59], from its early days of molecular dynamics and Monte Carlo simulations deeply rooted in the exploration of simple and paradigmatic condensed-matter systems in the 1960s, to the development of first-principles methods (most notoriously those based on density functional theory) [60] and multiscale techniques to probe disparate length and time scales [61]. In this respect, neutron scattering stands out as the experimental tool of choice for a direct comparison with computational and theoretical predictions, a recurring theme throughout this work.

In closing this section, and for reference purposes, Table 1.3 summarizes some of the *pros* and *cons* of neutron scattering in structural and dynamical investigations.

1.2 SCATTERING FOUNDATIONS

We proceed to establish a quantitative basis for the concepts introduced in Section 1.1. For a rigorous development of the subject, interested readers are encouraged to consult the excellent monographs by Squires [62] or Lovesey [63].

1.2.1 The Master Formula and Fermi’s Golden Rule

Recalling the scattering experiment in Figure 1.2 and the concept of the differential cross section introduced in Section 1.1.2, we now allow explicitly for a change in state of the target (sample) from a state τ_i to a state τ_f , and also a possible change in neutron spin from σ_i to σ_f as a result of the scattering process. For an ideal detector of unit efficiency, Equation (1.2) can be written as

$$\left(\frac{d\sigma}{d\Omega}\right)_{\mathbf{k}_i\sigma_i\tau_i\rightarrow\mathbf{k}_f\sigma_f\tau_f} = \frac{1}{N\Phi\Delta\Omega} W_{\mathbf{k}_i\sigma_i\tau_i\rightarrow\mathbf{k}_f\sigma_f\tau_f}, \quad (1.18)$$

where $W_{\mathbf{k}_i\sigma_i\tau_i\rightarrow\mathbf{k}_f\sigma_f\tau_f}$ ($=C/\eta$ in Equation 1.2) is the number of transitions per second from an initial state $\mathbf{k}_i\sigma_i\tau_i$ of the combined neutron-target system to

TABLE 1.3 *Pro et Contra* of Neutron-Scattering Techniques for Condensed Matter Research*Pro*

1. Neutrons carry no charge and therefore have high penetrating power into bulk samples
2. Thermal neutron wavelengths and energies match interatomic distances and excitation energies in condensed matter
3. Neutron-scattering lengths vary irregularly with Z —generally good for light atoms (especially hydrogen) and for discriminating between nearby elements in the periodic table
4. Scattering lengths vary irregularly with A —isotopic substitution can be used to provide information about a particular element, most notably hydrogen. Resonant and recoil scattering have also been used to achieve element specificity. See [Section 1.4.3.2](#) and [Chapters 4–6](#)
5. Scattering intensities can be *quantitatively* related to the structural and dynamical properties of the sample—the main subject of later sections in this chapter
6. Scattering from nuclei can be both incoherent and coherent, thus sensitive to single- and two-particle spatiotemporal correlations in condensed matter. See [Section 1.2.5](#)
7. Scattering from unpaired electrons constitutes a unique probe of magnetism. See [Sections 1.5 and 1.6](#)
8. Scattering from nuclei with nonzero spin can be used to probe nuclear magnetism. See [Section 1.7](#)
9. Neutron-scattering experiments offer a direct link to the predictions of computational experiments—a recurring theme throughout the present and future volumes in this series

Contra

1. High-intensity neutron beams can only be produced at central facilities, i.e., you cannot (yet!) perform neutron-scattering experiments on your lab bench. See [Chapter 2](#) for the most recent developments
2. Beam intensities may be too low to measure very small specimens, although current neutron instrumentation has improved significantly over the past two decades. See [Chapter 3](#)
3. Being neutral, neutrons are difficult to manipulate, e.g., focus into small spot sizes. See [Chapter 3](#)

a final state $\mathbf{k}_f \sigma_f \tau_f$. The transition rate $W_{\mathbf{k}_i \sigma_i \tau_i \rightarrow \mathbf{k}_f \sigma_f \tau_f}$ on the right-hand side of Equation (1.18) may be evaluated within first-order perturbation theory using Fermi's Golden Rule:

$$W_{\mathbf{k}_i \sigma_i \tau_i \rightarrow \mathbf{k}_f \sigma_f \tau_f} = \frac{2\pi}{\hbar} |\langle \mathbf{k}_f \sigma_f \tau_f | V | \mathbf{k}_i \sigma_i \tau_i \rangle|^2 \rho_{\mathbf{k}_f \sigma_f}(E_f), \quad (1.19)$$

where V is the interaction potential between the neutron and the target and $\rho_{\mathbf{k}_f\sigma_f}(E_f)$ is the density of final neutron-scattering states per unit energy interval. To arrive at this expression, we note that first-order perturbation theory is valid for nuclear scattering because the neutron–nucleus interaction potential is short range and thermal energies are sufficiently low such that only s-wave (zero orbital angular momentum) scattering applies provided we are far away from resonances. Consequently, the Fermi pseudopotential (cf. Equation 1.27) used in conjunction with the Golden Rule gives the correct result. The magnetic scattering potential is not short range but it is weak and the usual conditions for the validity of first-order perturbation theory still apply. A more detailed discussion is given in Ref. [63].

If we consider the neutron and target to be in a large box of volume V_0 , the incident and scattered neutron wave functions are the product of the spatial (plane-wave) and spin functions, that is, $V_0^{-1/2}e^{i\mathbf{k}_i\cdot\mathbf{r}}|\sigma_i\rangle$ and $V_0^{-1/2}e^{i\mathbf{k}_f\cdot\mathbf{r}}|\sigma_f\rangle$, respectively. Then the number of states over a final-energy interval dE_f is

$$\rho_{\mathbf{k}_f\sigma_f}(E_f)dE_f = \frac{V_0}{8\pi^3}d\mathbf{k}_f = \frac{V_0}{8\pi^3}k_f^2 dk_f \Delta\Omega. \quad (1.20)$$

Considering the relationship between kinetic energy and its associated linear momentum $dE_f = \hbar^2 k_f dk_f / m_n$ we can write

$$\rho_{\mathbf{k}_f\sigma_f}(E_f) = \frac{V_0}{8\pi^3} \frac{m_n k_f}{\hbar^2} \Delta\Omega. \quad (1.21)$$

The incident beam flux is the normalized number density times velocity v_i :

$$\Phi = \frac{1}{V_0} v_i = \frac{\hbar k_i}{V_0 m_n} \quad (1.22)$$

Using Equations (1.19)–(1.22) to rewrite Equation (1.18), we find

$$\left(\frac{d\sigma}{d\Omega} \right)_{\mathbf{k}_i\sigma_i\tau_i \rightarrow \mathbf{k}_f\sigma_f\tau_f} = \left(\frac{1}{N} \right) \frac{k_f}{k_i} \left(\frac{m_n V_0}{2\pi\hbar^2} \right)^2 |\langle \mathbf{k}_f\sigma_f\tau_f | V | \mathbf{k}_i\sigma_i\tau_i \rangle|^2, \quad (1.23)$$

where the ratio of final to incident neutron wave vectors k ensures flux conservation before and after scattering. Furthermore, energy conservation applies to both neutron and target

$$E = E_i - E_f = E_{\tau_f} - E_{\tau_i}, \quad (1.24)$$

resulting in a delta-function condition that must be fulfilled by the double differential scattering cross section

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{\mathbf{k}_i\sigma_i\tau_i \rightarrow \mathbf{k}_f\sigma_f\tau_f} = \left(\frac{1}{N} \right) \frac{k_f}{k_i} \left(\frac{m_n V_0}{2\pi\hbar^2} \right)^2 |\langle \mathbf{k}_f\sigma_f\tau_f | V | \mathbf{k}_i\sigma_i\tau_i \rangle|^2 \delta(E + E_{\tau_i} - E_{\tau_f}). \quad (1.25)$$

Finally, we sum over all final states of the sample τ_f and final neutron-spin states σ_f as well as average over all initial states of the target occurring with probability p_{τ_i} , and over the initial spin states of the neutron (probability p_{σ_i}) to derive the *double differential cross section*:

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{\mathbf{k}_i \rightarrow \mathbf{k}_f} = \left(\frac{1}{N} \right) \frac{k_f}{k_i} \left(\frac{m_n V_0}{2\pi\hbar^2} \right)^2 \sum_{\tau_i \sigma_i} p_{\tau_i} p_{\sigma_i} \sum_{\tau_f \sigma_f} |\langle \mathbf{k}_f \sigma_f \tau_f | V | \mathbf{k}_i \sigma_i \tau_i \rangle|^2 \delta(E + E_{\tau_i} - E_{\tau_f}). \quad (1.26)$$

Equation (1.26) has been termed the *master formula* [63] and it forms the basis for the interpretation of all neutron-scattering experiments.

1.2.2 Nuclear Scattering

The spatial range of the nuclear potential (fm) is several orders of magnitude smaller than typical thermal neutron wavelengths ($\text{\AA} = 10^{-5}$ fm). It is, therefore, appropriate to describe it in terms of the *Fermi pseudopotential* involving a delta function:

$$\frac{2\pi\hbar^2}{m_n} b_{\sigma\tau} \delta(\mathbf{r} - \mathbf{R}), \quad (1.27)$$

where b is the scattering length introduced in Section 1.1.2.1 (also dependent on the spin state of the neutron-nucleus system), and $\mathbf{r}$ and $\mathbf{R}$ represent the instantaneous positions of the neutron and nucleus, respectively. The total interaction potential between the neutron and the sample is then obtained by summing over all atoms in the sample:

$$V(\mathbf{r}) = \frac{2\pi\hbar^2}{m_n} \sum_j b_j \delta(\mathbf{r} - \mathbf{R}_j). \quad (1.28)$$

Averaging Equation (1.28) over incident and final (translational) neutron wave functions we get

$$\langle k_f | V | k_i \rangle = \frac{2\pi\hbar^2}{m_n V_0} \sum_j b_j \int e^{-i\mathbf{k}_f \cdot \mathbf{r}} \delta(\mathbf{r} - \mathbf{R}_j) e^{i\mathbf{k}_i \cdot \mathbf{r}} d\mathbf{r} = \frac{2\pi\hbar^2}{m_n} \sum_j b_j e^{i\mathbf{Q} \cdot \mathbf{R}_j}, \quad (1.29)$$

where $\mathbf{Q} = \mathbf{k}_i - \mathbf{k}_f$ as in Section 1.1.3. The corresponding master formula (cf. Equation 1.26) then reads

$$\frac{d^2\sigma}{d\Omega dE_f} = \left(\frac{1}{N} \right) \frac{k_f}{k_i} \sum_{\tau_i \sigma_i} p_{\tau_i} p_{\sigma_i} \left| \sum_{\tau_f \sigma_f} \sum_j b_j \langle \sigma_f \tau_f | e^{i\mathbf{Q} \cdot \mathbf{R}_j} | \sigma_i \tau_i \rangle \right|^2 \delta(E + E_{\tau_i} - E_{\tau_f}), \quad (1.30)$$

where, for convenience, the subscript $\mathbf{k}_i \rightarrow \mathbf{k}_f$ has been omitted on the left-hand side.

The majority of applications of neutron–nuclear scattering use unpolarized beams where both neutron-spin states are equally probable for both incident and scattered beams. This situation requires summation over σ_i and σ_f to give

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \frac{k_f}{k_i} \sum_{\tau_i} p_{\tau_i} \left| \sum_{\tau_f} \sum_j b_j \langle \tau_f | e^{i\mathbf{Q}\cdot\mathbf{R}_j} | \tau_i \rangle \right|^2 \delta(E + E_{\tau_i} - E_{\tau_f}). \quad (1.31)$$

We shall return to Equation (1.30) when dealing with polarized neutrons for studies of electronic (Section 1.5.2) and nuclear magnetism (Section 1.7).

1.2.3 The Double Differential Cross Section in the Time Domain

Equation (1.31) gives the *double differential cross section* in terms of matrix elements of the operators $e^{i\mathbf{Q}\cdot\mathbf{R}_i}$ between initial and final target states τ_i and τ_f . Although this observable is not a time-dependent quantity as it describes an asymptotic scattering experiment, its evaluation becomes more transparent if we consider the explicit time evolution of particle positions $\mathbf{R}_j$. To achieve this, we replace the delta function in Equation (1.31) by an integral over time using its well-known integral representation:

$$\delta(E + E_{\tau_i} - E_{\tau_f}) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \exp\left[-i\frac{t}{\hbar}(E + E_{\tau_i} - E_{\tau_f})\right] dt, \quad (1.32)$$

so that Equation (1.31) becomes

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE_f} = & \frac{1}{N} \left(\frac{k_f}{k_i}\right) \frac{1}{2\pi\hbar} \sum_{\tau_i} p_{\tau_i} \sum_{\tau_f} \sum_{jj'} b_j^* b_{j'} \int_{-\infty}^{\infty} \langle \tau_i | e^{i\mathbf{Q}\cdot\mathbf{R}_j} | \tau_f \rangle \\ & \langle \tau_f | e^{iHt/\hbar} e^{i\mathbf{Q}\cdot\mathbf{R}_{j'}} e^{-iHt/\hbar} | \tau_i \rangle e^{-iEt/\hbar} dt, \end{aligned} \quad (1.33)$$

where H denotes a Hamiltonian (time evolution) operator such that

$$e^{iHt/\hbar} | \tau_i \rangle = e^{iE_i t/\hbar} | \tau_i \rangle, \quad (1.34)$$

in addition to an analogous expression for the final target states τ_f . From the closure relation $\sum_{\tau_f} | \tau_f \rangle \langle \tau_f | = 1$, the expression above simplifies to:

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \left(\frac{k_f}{k_i}\right) \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \sum_{jj'} e^{-iEt/\hbar} \left\langle b_j^* b_{j'} e^{-i\mathbf{Q}\cdot\mathbf{R}_i} e^{iHt/\hbar} e^{i\mathbf{Q}\cdot\mathbf{R}_{j'}} e^{-iHt/\hbar} \right\rangle_{\tau_i} dt, \quad (1.35)$$

where the angular brackets represent a thermal average over the initial states τ_i . Noting that the term in brackets contains time-propagation terms of the form,

$$e^{i\mathbf{Q}\cdot\mathbf{R}_j(t)} \equiv e^{iHt/\hbar} e^{i\mathbf{Q}\cdot\mathbf{R}_j} e^{-iHt/\hbar} \quad (1.36)$$

we finally get

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \left(\frac{k_f}{k_i} \right) \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \sum_{jj'} \left\langle b_j^* b_{j'} e^{-i\mathbf{Q}\cdot\mathbf{R}_j(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt, \quad (1.37)$$

which is a real-time representation of the double differential cross section in terms of thermal averages of a product of two operators, one involving a particle j at time 0, and the other a particle j' at time t . We are justified in putting the origin at time $t=0$ in Equation (1.37) since the time (Heisenberg) propagator depends only on time (or energy) differences between initial and final states. In a classical picture, $\mathbf{R}_j(t)$ can be equated to the actual (instantaneous) position of a given particle at a given time, averaged over configuration space. In other words, Equation (1.37) tells us that the double differential cross section (what we measure) contains information on the spatial (R) and temporal (t) evolution of the target system, weighted by pair products of nuclear-scattering lengths b . In the language of classical mechanics, such a result implies that Equation (1.37) may be used to calculate neutron-scattering cross sections from a history of the temporal and spatial development of the system, as obtained, for example, from a computer simulation. Much of the progress of neutron science over the past few decades has relied on such a synergy between experimental and computational techniques.

1.2.4 Farewell to Nuclear Physics

Despite the insights brought forward by Equation (1.37), it still represents a sum of averages for each pair of atoms where the nuclear states enter explicitly through their respective scattering lengths b . These quantities depend on the isotope and spin state of the j th nucleus, as presented earlier in Section 1.1.2.1 and, for all practical purposes, they are deeply rooted in nuclear physics, not condensed matter.

In cases where the atomic state is independent of nuclear variables (i.e., when nuclear spins and isotopes for a given element are randomly distributed), it is possible to represent the scattering process in terms of a function only dependent on atomic coordinates $\mathbf{R}_j(t)$. In most cases (bar a few exceptions like molecular hydrogen), the energy differences associated with different nuclear-spin orientations are $\sim 10^{-4}$ K per atom and, therefore, these spins are effectively random for temperatures above 1 mK—nuclear-spin correlations at ultralow temperatures is a special subject introduced in Section 1.7. Moreover, fractional energy differences associated with different isotopic masses are $\sim \Delta A/A$ and so the isotope distribution is generally random except for the lightest atoms at low temperatures, for example $^3\text{He}/^4\text{He}$ mixtures, which again must be treated as a special case. In general, therefore,

we may average over initial nuclear-spin and isotope distributions in Equation (1.37) and write

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \left(\frac{k_f}{k_i} \right) \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \sum_{dd'} \sum_{j \in d, j' \in d'} \overline{b_j^* b_{j'}} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt, \quad (1.38)$$

where d and d' refer to different elements and the bar represents an average over spin and isotope distributions for the corresponding pair of elements. In the case of isotope substitution experiments, we note that the index d may also refer to a particular isotope of an element, for example, if some hydrogen atoms in the sample are systematically exchanged for deuterium and the others left as hydrogen. For this reason, we will refer to “atom type” rather than “element” in the following sections.

1.2.5 Coherent and Incoherent Scattering

For conciseness, we define

$$S_{jj'}(\mathbf{Q}, E) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt, \quad (1.39)$$

and rewrite the double differential cross section in Equation (1.38) as

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \left(\frac{k_f}{k_i} \right) \sum_{dd'} \sum_{j \in d, j' \in d'} \overline{b_j^* b_{j'}} S_{jj'}(\mathbf{Q}, E). \quad (1.40)$$

If the nuclear spins and isotopes are uncorrelated, then

$$\overline{b_j^* b_{j'}} = \begin{cases} \overline{b_d^* b_{d'}}, & j \neq j', \\ \left| \overline{b_d^2} \right|, & j = j', \end{cases} \quad (1.41)$$

or more succinctly using tensor notation

$$\overline{b_j^* b_{j'}} = \overline{b_d^* b_{d'}} + \left(\left| \overline{b_d^2} \right| - \left| \overline{b_d} \right|^2 \right) \delta_{jj'} \delta_{dd'}. \quad (1.42)$$

Substituting into Equation (1.40) we obtain

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE_f} &= \frac{1}{N} \left(\frac{k_f}{k_i} \right) \sum_{dd'} \sum_{j \in d, j' \in d'} \left[\overline{b_d^* b_{d'}} + \left(\left| \overline{b_d^2} \right| - \left| \overline{b_d} \right|^2 \right) \delta_{jj'} \delta_{dd'} \right] S_{jj'}(\mathbf{Q}, E) \\ &= \frac{1}{N} \left(\frac{k_f}{k_i} \right) \sum_{dd'} \overline{b_d^* b_{d'}} \sum_{j \in d, j' \in d'} S_{jj'}(\mathbf{Q}, E) + \frac{1}{N} \left(\frac{k_f}{k_i} \right) \sum_d \left(\left| \overline{b_d^2} \right| - \left| \overline{b_d} \right|^2 \right) \sum_{j \in d} S_{jj}(\mathbf{Q}, E). \end{aligned} \quad (1.43)$$

The first term represents a sum over all possible pairs of atoms (j, j'), each term representing a two-body correlation between the position of atom

j at time 0 and that of atom j' at time t ; it therefore contains the interference terms in the total scattering and gives rise to *coherent scattering*. Coherent scattering is therefore sensitive to the relative arrangement of atoms (structure).

The second term represents a sum over all atoms j one at a time, each term representing a correlation between the position of atom j at time 0 and that of the same atom at time t , that is, *incoherent scattering*, sensitive to single-particle spatiotemporal correlations. If for a particular atom type d there is only one possible scattering length b_d for all possible nuclear-spin states, then $|\bar{b}_d^2| = |\bar{b}_d|^2$ and there is no incoherent scattering from that atom type. Incoherent scattering arises from the random distribution of scattering lengths about the mean. Important examples that conform to this rule include all nuclides with zero nuclear spin such as ^{12}C and ^{16}O . We also note that these specific cases (must!) also coincide with NMR-inactive nuclides, all of which must also be pure coherent neutron scatterers.

Since isotopic compositions are generally uniform, coherent, incoherent, and total scattering cross sections may be regarded as universal properties of an atom type. These are defined as (see also the [Appendix](#))

$$\begin{aligned}\sigma_c^d &= 4\pi |\bar{b}_d|^2 \text{ (coherent cross section),} \\ \sigma_i^d &= 4\pi \left[|\bar{b}_d^2| - |\bar{b}_d|^2 \right] \text{ (incoherent cross section),} \\ \sigma_t^d &= 4\pi \left[|\bar{b}_d^2| \right] \text{ (total scattering cross section).}\end{aligned}\tag{1.44}$$

To avoid confusion, we note that σ_i above no longer relates to the initial spin-state of the neutron. With these definitions, Equation (1.43) is then written

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \left(\frac{k_f}{k_i} \right) \sum_{dd'} \bar{b}_d^* \bar{b}_{d'} \sum_{j \in d, j' \in d'} S_{jj'}(\mathbf{Q}, E) + \frac{1}{N} \left(\frac{k_f}{k_i} \right) \sum_d \frac{\sigma_i^d}{4\pi} \sum_{j \in d} S_{jj}(\mathbf{Q}, E),\tag{1.45}$$

where for a single atom type $\bar{b}_d^* \bar{b}_{d'}$ may be replaced by $\sigma_c^d/4\pi$.

For a particular atom type that consists of different isotopes a with abundance c_a and nuclear spin I_a , the cross sections can be calculated from

$$\begin{aligned}\sigma_c &= 4\pi \left\{ \sum_a \frac{c_a}{2I_a + 1} [(I_a + 1)b_a^+ + I_a b_a^-] \right\}^2, \\ \sigma_s &= 4\pi \sum_a \frac{c_a}{2I_a + 1} [(I_a + 1)(b_a^+)^2 + I_a (b_a^-)^2], \\ \sigma_i &= \sigma_s - \sigma_c.\end{aligned}\tag{1.46}$$

Values of σ_c and σ_i for most nuclides are listed in the [Appendix](#). For non-standard compositions as those used for isotopic-substitution experiments, the

appropriate cross sections can be calculated if the values for the individual isotopes are known.

1.2.6 Scattering Functions

To represent the sums over the $S_{jj'}(\mathbf{Q}, E)$ in Equation (1.45), we define the *coherent dynamic structure factor* or *scattering function*

$$\begin{aligned} S_c^{dd'}(\mathbf{Q}, E) &= \frac{1}{(N_d N_{d'})^{1/2}} \sum_{j \in d, j' \in d'} S_{jj'}(\mathbf{Q}, E) \\ &= \frac{1}{(N_d N_{d'})^{1/2}} \left(\frac{1}{2\pi\hbar} \right) \sum_{j \in d, j' \in d'} \int_{-\infty}^{\infty} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt, \end{aligned} \quad (1.47)$$

and the *incoherent dynamic structure factor* or *scattering function*

$$S_i^d(\mathbf{Q}, E) = \frac{1}{N_d} \sum_{i \in d} S_{ii}(\mathbf{Q}, E) = \frac{1}{N_d} \left(\frac{1}{2\pi\hbar} \right) \sum_{j \in d} \int_{-\infty}^{\infty} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_j(t)} \right\rangle e^{-iEt/\hbar} dt. \quad (1.48)$$

In terms of these functions, the double differential cross section has the form

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{k_f}{k_i} \left(\sum_{dd'} \left[c_d^{1/2} c_{d'}^{1/2} \bar{b}_d^* \bar{b}_{d'} \right] S_c^{dd'}(\mathbf{Q}, E) + \sum_d \left[c_d \frac{\sigma_d}{4\pi} \right] S_{dd}(\mathbf{Q}, E) \right), \quad (1.49)$$

where $c_d = N_d/N$ is the concentration of atom type d .

The time-dependent functions under the integral signs in Equations (1.47) and (1.48) are known as the *intermediate scattering function*

$$I^{dd'}(\mathbf{Q}, t) = \frac{1}{(N_d N_{d'})^{1/2}} \sum_{j \in d, j' \in d'} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_{j'}(t)} \right\rangle, \quad (1.50)$$

and the *self-intermediate scattering function*

$$I_s^d(\mathbf{Q}, t) = \frac{1}{N_d} \sum_{j \in d} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_j(t)} \right\rangle. \quad (1.51)$$

The intermediate scattering functions are dimensionless, while the scattering functions themselves are of dimension E^{-1} , as defined via Equations (1.38) and (1.40). As shown explicitly by Equation (1.49), scattering functions are obtained via the measurement of double differential cross sections. Intermediate scattering functions are directly accessible with neutron-spin-echo techniques, first introduced by Mezei in the early 1970s [64] and described in more detail in Chapter 3.

1.3 CANONICAL SOLIDS

The formalism developed in [Section 1.2](#) finds its most natural and immediate application in the study of the solid state. At this point, it would be customary to consider the neutron-scattering response of a high-symmetry single crystal with just one atom per unit cell (Bravais lattice), as treated in depth in well-established monographs [\[62,63\]](#). Notwithstanding the merits of this approach, much of the current interest focuses on disordered and amorphous systems, thus we choose to start out with a consideration of a general solid without the assumption of long-range order or periodicity. To this end, a solid is defined, operationally speaking, as a system in which each atom has a well-defined (and fixed) equilibrium position over the duration of the measurement. This somewhat restricted definition excludes quantum solids such as the condensed phases of helium, where atoms undergo quantum-mechanical exchange and are inherently delocalized in space. In this case, special methods are required to deal with a truly many-body problem, as described by Glyde [\[65\]](#). Nor does it include solids in which atoms or ions undergo translational diffusion. The treatment of these cases requires considering the conceptual framework underpinning the scattering response of liquids, as introduced later on in [Section 1.4](#).

1.3.1 Normal Modes of Vibration

According to our operational definition of a solid, each atom can be uniquely associated with a specific equilibrium site at position $\mathbf{j}$. The position of each atom around this site can be expressed in terms of the instantaneous displacement from equilibrium:

$$\mathbf{R}_j(t) = \mathbf{j} + \mathbf{u}_j(t), \quad j = 1, \dots, n. \quad (1.52)$$

Using Equations [\(1.47\)](#) and [\(1.48\)](#), the corresponding coherent and incoherent scattering functions considering only one type of atom in the lattice become

$$S_c(\mathbf{Q}, E) = \frac{1}{N} \left(\frac{1}{2\pi\hbar} \right) \sum_{jj'} \int_{-\infty}^{\infty} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{u}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{u}_{j'}(t)} \right\rangle e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} e^{-iEt/\hbar} dt \quad (1.53)$$

and

$$S_i(\mathbf{Q}, E) = \frac{1}{N} \left(\frac{1}{2\pi\hbar} \right) \sum_j \int_{-\infty}^{\infty} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{u}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{u}_j(t)} \right\rangle e^{-iEt/\hbar} dt. \quad (1.54)$$

To proceed, we assume that the forces in the solid are *harmonic*, that is, the force that acts to return each atom to its equilibrium position is a linear function of the displacement. In this case, the displacements $\mathbf{u}_j$ can be expressed as a superposition of $3N$ normal modes [\[66\]](#),

$$\mathbf{u}_j(t) = \sum_k \left(\frac{\hbar}{2M_k\omega_k} \right)^{1/2} \left[\mathbf{e}_j^k e^{-\omega_k t} a_k + \mathbf{e}_j^{k*} e^{\omega_k t} a_k^+ \right], \quad (1.55)$$

where ω_k is the frequency of the k th mode and $\mathbf{e}_j^k$ is the mode polarization vector for the j th atom in this mode; the $\mathbf{e}_j^k$ s are conventionally defined to be an orthonormal vector set. The frequencies and polarization vectors are eigenvalues and eigenvectors of the dynamical matrix $\mathbf{D}$, respectively, which is given by

$$\mathbf{D}_{jj'} = \frac{1}{(M_j M_{j'})^{1/2}} f_{jj'}, \quad (1.56)$$

where $f_{jj'}$ are force constants representing the coefficients of the displacements in the expansion of the potential energy of the solid, that is,

$$U - U_0 = \frac{1}{2} \mathbf{u}_j f_{jj'} \mathbf{u}_{j'}. \quad (1.57)$$

Thus, the dynamical matrix and hence the frequencies and polarization vectors can be calculated from the interatomic potential of the solid; a_j and a_j^+ are quantum-mechanical operators expressing, respectively, the annihilation and creation of one quantum of vibrational energy.

1.3.2 Scattering Under the Harmonic Approximation

To evaluate the thermal averages in Equations (1.53) and (1.54), we make recourse to the main results of the quantum-mechanical theory of a simple harmonic oscillator [62,67,68]. In particular, we recall that

$$\left\langle e^{-i\mathbf{Q}\cdot\mathbf{u}_j(0)} e^{i\mathbf{Q}\cdot\mathbf{u}_{j'}(t)} \right\rangle = e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} e^{\langle \mathbf{Q}\cdot\mathbf{u}_j(0) \mathbf{Q}\cdot\mathbf{u}_{j'}(t) \rangle}, \quad (1.58)$$

where the first (time-independent) factor is a function of $\mathbf{Q}$, usually referred to as the *Debye–Waller factor*:

$$e^{-W_j(\mathbf{Q})} = e^{-1/2 \langle [\mathbf{Q}\cdot\mathbf{u}_j(0)]^2 \rangle}. \quad (1.59)$$

For small (harmonic) displacements, the second factor in Equation (1.58) can be expanded to read [69,70]

$$e^{\langle \mathbf{Q}\cdot\mathbf{u}_j(0) \mathbf{Q}\cdot\mathbf{u}_{j'}(t) \rangle} = 1 + \langle \mathbf{Q}\cdot\mathbf{u}_j(0) \mathbf{Q}\cdot\mathbf{u}_{j'}(t) \rangle + \frac{1}{2!} \langle \mathbf{Q}\cdot\mathbf{u}_j(0) \mathbf{Q}\cdot\mathbf{u}_{j'}(t) \rangle^2 + \dots \quad (1.60)$$

The Debye–Waller factor acts as a multiplying factor for each term in the expansion of the scattering cross section. Using the normal-mode expansion of Equation (1.55), it can be shown that

$$W_j(\mathbf{Q}) = \frac{\hbar}{3M_j} \sum_k \frac{|\mathbf{Q}\cdot\mathbf{e}_j^k|^2}{\omega_k} \langle 2n_k + 1 \rangle, \quad (1.61)$$

where $\langle n_k \rangle$ is the Bose–Einstein population number for the k th mode, given by

$$\langle n_k \rangle = \frac{1}{e^{\hbar\omega_k/k_B T} - 1}. \quad (1.62)$$

The reader is referred to the work of Sears and Shelley [71] describing a simple and insightful model for the calculation of Debye–Waller factors. For isotropic systems (e.g., polycrystalline samples), integration over all orientations leads to a Q -dependent Debye–Waller factor of the form

$$2W(Q) = \frac{1}{3}Q^2 \langle u^2 \rangle, \quad (1.63)$$

where $\langle u^2 \rangle$ is the mean-square displacement. Equation (1.63) is used extensively in the study of disordered materials including glasses, polymers, and biological systems. It is also a quantity readily available from computer simulations using molecular dynamics methodologies. For example, in the particular case of incoherent (proton-dominated) neutron scattering from proteins, it has been used to quantify protein flexibility and how it may relate to biological function and activity [72].

1.3.3 Purely Elastic Events

Neglecting all time-dependent terms in the expansion shown by Equation (1.60), the coherent scattering function (Equation 1.53) becomes

$$\begin{aligned} S_{c,el}(\mathbf{Q}, E) &= \frac{1}{N} \left(\frac{1}{2\pi\hbar} \right) \sum_{jj'} \int_{-\infty}^{+\infty} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} e^{-iEt/\hbar} dt \\ &= \left[\frac{1}{N} \sum_{jj'} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} \right] \delta(E), \end{aligned} \quad (1.64)$$

where we have exploited the integral representation of the Dirac delta-function introduced in Equation (1.32). The presence of $\delta(E)$ implies purely elastic scattering and justifies the intuitive introduction of this concept given in Section 1.1.3. We note that this delta function in energy is the result of the fact that the argument in the correlation function of Equation (1.58) remains finite at infinite times, as implied by the concept of equilibrium sites (i.e., the existence of a well-defined time-averaged structure).

Integration of Equation (1.64) over energy gives the *elastic structure factor*

$$S_{el}(\mathbf{Q}) = \frac{1}{N} \sum_{jj'} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})}. \quad (1.65)$$

An analogous procedure using Equation (1.54) leads to similar expressions for the incoherent scattering function:

$$S_{i,\text{el}}(\mathbf{Q}, E) = \left[\frac{1}{N} \sum_j e^{-2W_j(\mathbf{Q})} \right] \delta(E), \quad (1.66)$$

and integration over energy gives

$$S_{i,\text{el}}(\mathbf{Q}) = \int S_{i,\text{el}}(\mathbf{Q}, E) dE = \frac{1}{N} \sum_j e^{-2W_j(\mathbf{Q})}, \quad (1.67)$$

which contains the average Debye–Waller factor for all atoms in the target system.

For a crystalline solid with long-range order, equilibrium lattice sites $\mathbf{j}$ may be described by the following vector relation

$$\mathbf{j} = \mathbf{l} + \mathbf{d}, \quad (1.68)$$

where $\mathbf{l}$ refers to the center of the unit cell and $\mathbf{d}$ to the relative position of a lattice site within the cell. Allowing for the presence of atom type d at a given position $\mathbf{d}$ in the unit cell, the set of vectors $\mathbf{l}$ makes up a crystal lattice with perfect long-range order. In this case the elastic structure factor becomes

$$S(\mathbf{Q})_{\text{el}}^{\text{dd}'} = \frac{(2\pi)^3}{v_0} e^{-[W_d(\mathbf{Q}) + W_{d'}(\mathbf{Q})]} e^{i\mathbf{Q} \cdot (\mathbf{d}' - \mathbf{d})} \sum_{\boldsymbol{\tau}} \delta(\mathbf{Q} - \boldsymbol{\tau}), \quad (1.69)$$

where the vectors $\boldsymbol{\tau}$ are the reciprocal lattice vectors of the crystal and v_0 is the unit cell volume. Equation (1.69) renders itself to a simple physical interpretation whereby a series of (infinitely sharp) peaks appear every time $\mathbf{Q}$ is equal to a reciprocal lattice vector $\boldsymbol{\tau}$ of the crystal. These are the *Bragg peaks*, and Equation (1.69) is the exact formulation of the Bragg condition introduced in Equation (1.11).

From Equations (1.69) and (1.49) we may derive the *elastic coherent scattering cross section*

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{c,el}} = \frac{(2\pi)^3}{v_0} \sum_{\boldsymbol{\tau}} |F(\boldsymbol{\tau})|^2 \delta(\mathbf{Q} - \boldsymbol{\tau}), \quad (1.70)$$

where

$$F(\boldsymbol{\tau}) = \sum_{\mathbf{d}} \bar{b}_{\mathbf{d}} e^{-W_{\mathbf{d}}(\boldsymbol{\tau})} e^{i\boldsymbol{\tau} \cdot \mathbf{d}} \quad (1.71)$$

is the *unit-cell structure factor* for the crystal. Equations (1.70) and (1.71) form the basis of neutron crystallography, covered in [Chapter 4](#).

Similarly, the incoherent elastic differential cross section is given by

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{i,el}} = \sum_{\mathbf{d}} c_{\mathbf{d}} \sigma_{\mathbf{i}}^{\mathbf{d}} e^{-2W_{\mathbf{d}}(\mathbf{Q})}. \quad (1.72)$$

In the interpretation of neutron-diffraction experiments from crystalline materials, it is crucial to note that Equation (1.68) assumes a single value of $\mathbf{d}$ relative to the center of the unit cell. If (as in most cases!), there is a variation in the values of $\mathbf{d}$ through the crystal, translational invariance is no longer guaranteed and one needs to reconsider the general expression given by Equation (1.65). An immediate consequence of this loss of translational invariance is the emergence of scattering intensity (so-called *elastic diffuse scattering*) at all values of $\mathbf{Q}$, not just at the reciprocal lattice points as implied by Equation (1.69). Section 1.4.4 discusses diffuse scattering in more depth. For systems with no long-range order, like glasses and other amorphous solids, all the elastic scattering is diffuse and one must (in principle) know all the time-averaged atomic positions in order to calculate $S_{\text{el}}(\mathbf{Q})$. This task can be efficiently achieved via the calculation of the differential cross section from computer models of the material under investigation [73,74].

1.3.4 Inelastic (One-Phonon) Scattering

Returning to Equation (1.53), straight substitution of the second term in the expansion given by Equation (1.60) gives

$$S_{\text{c},1}(\mathbf{Q}, E) = \frac{1}{N} \left(\frac{1}{2\pi\hbar} \right) \sum_{jj'} \int_{-\infty}^{+\infty} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} \langle \mathbf{Q} \cdot \mathbf{u}_j(0) \mathbf{Q} \cdot \mathbf{u}_{j'}(t) \rangle e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} e^{-iEt/\hbar} dt \quad (1.73)$$

for the coherent scattering function, hereafter $S_{\text{c},1}(\mathbf{Q}, E)$. Dimensionally speaking, we note that this second-order term (describing harmonic vibrations) is quadratic in Q . Using the definition of normal modes given by Equation (1.55), we can rewrite it as

$$S_{\text{c},1}(\mathbf{Q}, E) = \frac{1}{N} \left(\frac{1}{4\pi} \right) \sum_{jj'} \int_{-\infty}^{+\infty} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} \sum_k \frac{(\mathbf{Q} \cdot \mathbf{e}_j^k)^* (\mathbf{Q} \cdot \mathbf{e}_{j'}^k)}{(M_j M_{j'})^{1/2} \omega_k} \times \\ \times [e^{-i\omega_k t} \langle n_k + 1 \rangle + e^{i\omega_k t} \langle n_k \rangle] e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} e^{-iEt/\hbar} dt. \quad (1.74)$$

We express once more the time integral as a delta function in the energy domain to write this expression as the sum of two time-independent terms $S_{\text{c},\pm 1}$ given by

$$S_{\text{c},+1}(\mathbf{Q}, E) = \frac{1}{2N} \sum_{jj'} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} \times \\ \times \sum_k \frac{\hbar^2 (\mathbf{Q} \cdot \mathbf{e}_j^k)^* (\mathbf{Q} \cdot \mathbf{e}_{j'}^k)}{(M_j M_{j'})^{1/2} E_k} \langle n_k + 1 \rangle \delta(E - E_k) \quad (1.75)$$

and

$$S_{c,-1}(\mathbf{Q}, E) = \frac{1}{2N} \sum_{jj'} e^{-[W_j(\mathbf{Q}) + W_{j'}(\mathbf{Q})]} e^{i\mathbf{Q} \cdot (\mathbf{r}' - \mathbf{j})} \times \\ \times \sum_k \frac{\hbar^2 (\mathbf{Q} \cdot \mathbf{e}_j^k)^* (\mathbf{Q} \cdot \mathbf{e}_{j'}^k)}{(M_j M_{j'})^{1/2} E_k} \langle n_k \rangle \delta(E + E_k) \quad (1.76)$$

where $E_k = \hbar \omega_k$ corresponds to normal-mode energy eigenvalues. These two scattering functions predict peaks in the scattering function whenever E corresponds to $\pm E_k$, that is, to the creation (+1) or annihilation (−1) of a phonon from the scattering event, leading to the loss or gain of energy by the scattered neutron, respectively.

The corresponding *incoherent scattering function* also contains one-phonon emission and absorption peaks and it has a sensibly simpler functional form:

$$S_{i,+1}(\mathbf{Q}, E) = \frac{1}{2N} \sum_j e^{-W_j(\mathbf{Q})} \sum_k \frac{\hbar^2 |\mathbf{Q} \cdot \mathbf{e}_j^k|^2}{M_j E_k} \langle n_k + 1 \rangle \delta(E - E_k) \quad (1.77)$$

and

$$S_{i,-1}(\mathbf{Q}, E) = \frac{1}{2N} \sum_j e^{-W_j(\mathbf{Q})} \sum_k \frac{\hbar^2 |\mathbf{Q} \cdot \mathbf{e}_j^k|^2}{M_j E_k} \langle n_k \rangle \delta(E + E_k). \quad (1.78)$$

If the solid has long-range order, $\mathbf{j} = \mathbf{l} + \mathbf{d}$ as before, the normal modes can be described as plane waves characterized by a reduced wave vector $\mathbf{q}$ and polarization vectors given by

$$\mathbf{e}_d^k = \frac{1}{N_c^{1/2}} \sum_{\mathbf{q}} \mathbf{e}_d^k(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{l}}, \quad (1.79)$$

where N_c is the number of unit cells in the crystal. Then, for example, the one-phonon contribution to the coherent scattering function (cf. Equation 1.47) is given by

$$S_{c, \mathbf{d}\mathbf{d}'}^{\mathbf{d}\mathbf{d}'}(\mathbf{Q}, E) = \frac{1}{2N_c} (2\pi)^3 e^{-[W_d(\mathbf{Q}) + W_{d'}(\mathbf{Q})]} \sum_{\mathbf{q}\mathbf{k}} \frac{\hbar^2 (\mathbf{Q} \cdot \mathbf{e}_d^k(\mathbf{q}))^* (\mathbf{Q} \cdot \mathbf{e}_{d'}^k(\mathbf{q}))}{(M_d M_{d'})^{1/2} E_k(\mathbf{q})} \times \\ \times e^{i\mathbf{Q} \cdot (\mathbf{d}' - \mathbf{d})} \langle n_{\mathbf{q}\mathbf{k}} + 1 \rangle \sum_{\boldsymbol{\tau}} \delta(\mathbf{Q} - \mathbf{q} - \boldsymbol{\tau}) \delta(E - E_k(\mathbf{q})) \quad (1.80)$$

and the corresponding double differential cross section

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{c,+1} = \frac{k_f}{k_i} \left[\frac{(2\pi)^3}{2N_c v_0} \right] \sum_{\boldsymbol{\tau}} \sum_{\mathbf{q}\mathbf{k}} |F_1(\mathbf{Q}, \mathbf{q}\mathbf{k})|^2 \frac{\hbar^2 \langle n_{\mathbf{q}\mathbf{k}} + 1 \rangle}{E_k(\mathbf{q})} \\ \delta(\mathbf{Q} - \mathbf{q} - \boldsymbol{\tau}) \delta(E - E_k(\mathbf{q})), \quad (1.81)$$

where it is implied that the double differential cross section is both a function of $\mathbf{Q}$ and E . In analogy with Equation (1.71), we have also defined a one-phonon structure factor

$$F_1(\mathbf{Q}, \mathbf{q}, k) = \sum_{\mathbf{d}} \frac{\bar{b}_{\mathbf{d}}}{M_{\mathbf{d}}^{1/2}} e^{-W_{\mathbf{d}}(\mathbf{Q})} e^{i\mathbf{Q} \cdot \mathbf{d}} [\mathbf{Q} \cdot \mathbf{e}_{\mathbf{d}}^k(\mathbf{q})]. \quad (1.82)$$

The expression for one-phonon annihilation (neutron-energy gain) is analogous to Equation (1.81) with $\langle n_k \rangle$ instead of $\langle n_k + 1 \rangle$ and the signs of $\mathbf{q}$ and E_k reversed. The delta-functions in Equation (1.81) imply that, for a single crystal, one-phonon scattering is only possible for values of $\mathbf{Q}$ and E satisfying

$$\mathbf{Q} = \boldsymbol{\tau} \pm \mathbf{q}, \quad E = E_k(\mathbf{q}), \quad (1.83)$$

for some set of values of $\mathbf{q}$ and k . These restrictions lead to well-defined spectral features in the inelastic scattering observed from such crystals. Measurements of this kind are illustrated in Figure 1.9 and can be used to derive the *phonon dispersion relations* for a crystalline lattice, $E_k(\mathbf{q})$. In comparison with optical probes like infrared and Raman spectroscopy, neutron scattering can explore wide regions of the Brillouin zone, thus providing stringent tests to theoretical predictions.

Deviations in the structure from a given equilibrium position will give rise to diffuse scattering as the selection rule $\mathbf{Q} = \boldsymbol{\tau} \pm \mathbf{q}$ no longer applies. For a solid without long-range order, all the inelastic scattering is diffuse and we need to revert to the general expressions in Equations (1.75) and (1.76). It

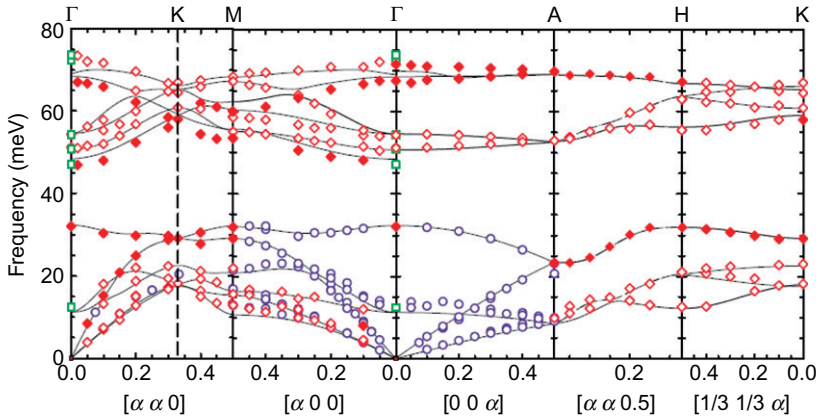


FIGURE 1.9 Phonon-dispersion relations for zinc oxide along the main symmetry directions. Diamonds and circles (red and blue) correspond to inelastic neutron-scattering data. Raman data at the Γ point ($Q=0$) are depicted by squares (green). *Ab initio* calculations are shown by the solid curves. Reprinted with permission from Ref. [75].

should be noted, however, that as long as the harmonic approximation applies, the phonon expansion is still valid, but the normal modes are no longer described by plane waves with a well-defined value of $\mathbf{q}$.

Inelastic incoherent scattering for a crystal is given by

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{i,+1} = \frac{k_f}{k_i} \left(\frac{1}{2N} \right) \sum_{\mathbf{d}} \frac{1}{M_{\mathbf{d}}} \frac{\sigma_i^{\mathbf{d}}}{4\pi} e^{-2W_{\mathbf{d}}(\mathbf{Q})} \sum_{qk} \frac{\hbar^2 |\mathbf{Q} \cdot \mathbf{e}_{\mathbf{d}}^k(\mathbf{q})|^2}{E_k(\mathbf{q})} \langle n_{qk} + 1 \rangle \delta(E - E_k(\mathbf{q})), \quad (1.84)$$

with an analogous expression for phonon annihilation. Equation (1.84) represents a continuous function of $\mathbf{Q}$ and E since only one selection rule applies, namely

$$E = \pm E_k(\mathbf{q}). \quad (1.85)$$

We can therefore express the cross section in terms of a *one-phonon vibrational density of states* $Z(E)$, where $Z(E)dE$ is the fraction of normal modes with energies in the range $(E, E + dE)$. For neutron-energy loss, this becomes

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{i,+1} = \frac{k_f}{k_i} \sum_{\mathbf{d}} \frac{3}{24\pi} \frac{\sigma_i^{\mathbf{d}}}{M_{\mathbf{d}}} e^{-2W_{\mathbf{d}}(\mathbf{Q})} \frac{\overline{\hbar^2 |\mathbf{Q} \cdot \mathbf{e}_{\mathbf{d}}^k(\mathbf{q})|^2}}{E} \langle n + 1 \rangle Z(E), \quad (1.86)$$

where the bar represents an average over all the modes at frequency E and

$$\langle n + 1 \rangle = \frac{1}{2} \left[1 + \coth \left(\frac{E}{2k_B T} \right) \right]. \quad (1.87)$$

For a one-component system of mass M in which the anisotropy in the Debye–Waller factor can be neglected, Equation (1.86) takes a simpler form

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{i,+1} = \frac{k_f}{k_i} \left(\frac{\hbar^2 Q^2}{2M} \right) e^{-2W(Q)} \frac{\langle n + 1 \rangle}{E} Z(E), \quad (1.88)$$

so that the vibrational density of states $Z(E)$ can be deduced directly from the incoherent one-phonon scattering. This result holds for a one-component disordered system as well. For multicomponent systems, it is no longer the case since the scattering also depends on the polarization vectors of the different atom types through the factor $|\mathbf{Q} \cdot \mathbf{e}_{\mathbf{d}}^k(\mathbf{q})|^2$, which in general will be frequency-dependent. A more detailed discussion of vibrational densities of states and how these can be obtained from neutron-scattering experiments is given by Taraskin and Elliott [76]. Equation (1.88) also constitutes the starting point for the analysis of inelastic neutron-scattering experiments on proton-containing materials where incoherent scattering from hydrogen

dominates the spectral response, as detailed in Ref. [77]. Readers interested in the use of inelastic neutron-scattering techniques for the study of hydrogen-containing materials are also referred to Ref. [78] for an extensive compilation of publicly available data.

1.3.5 Multiphonon Scattering

Higher-order terms in Equation (1.60) formally represent scattering processes in which two or more phonons are involved. The n -phonon contribution scales as $|\mathbf{Q} \cdot \mathbf{u}|^{2n}$, and for values of $\mathbf{Q}$ commonly encountered in thermal neutron scattering, it generally appears as a relatively small and smooth background under the one-phonon peaks; it tends to a continuous function for coherent as well as for incoherent scattering, particularly at high momentum transfers.

Using Equations (1.53), (1.54), and (1.58), calculations of the inelastic cross section can (in principle) be carried out to all orders in terms of multiphonon contributions provided the phonon-dispersion relations are known. For the coherent cross section, this is a tedious calculation and it is usual to assume that multiphonon terms are equal to the incoherent cross section (*incoherent approximation*). For crystals with only one type of atom, an expression similar to Equation (1.88) can be used to give the exponent in Equations (1.53) and (1.54) as a weighted integral of the one-phonon density of states; alternatively, approximations such as that of Sjölander [63,79] can also be used.

In the language of the spectroscopist, multiphonon scattering corresponds to the appearance of overtone and combination bands associated with the simultaneous excitation of more than one quantum of vibration. These effects are best illustrated by considering the inelastic neutron scattering response from molecular systems where intermolecular (lattice) and intramolecular vibrational modes are typically well separated from each other in energy transfer. The color contour plot in Figure 1.10 shows inelastic neutron-scattering data for Rb_2PtH_6 , a metal-hydride complex with an unusually high hydrogen content. The scattering is dominated by the strong incoherent response from hydrogen and the observed (nondispersive) bands at well-defined energy transfers correspond to either fundamental or overtone/combination (multiphonon) transitions from the ground vibrational state. The intensity of a given band as a function of momentum transfer (i.e., the *inelastic form factor*) scales as $Q^{2n} e^{-2W(Q)}$ with $n=1$ for a fundamental transition (cf. Equation 1.88), $n=2$ for a first overtone or binary combination, $n=3$ for a third overtone or ternary combination, etc. The specific value of n for a given transition can then be obtained directly from the experimental data by locating the position of Q_{max} corresponding to the maximum in the inelastic form factor, as within the harmonic approximation we can always write $n = Q_{\text{max}}^2 \langle u^2 \rangle / 3$ for a polycrystalline system (cf. Equation 1.63). These simple considerations provide a unique method of discriminating between fundamentals and higher-order vibrational transitions using inelastic neutron

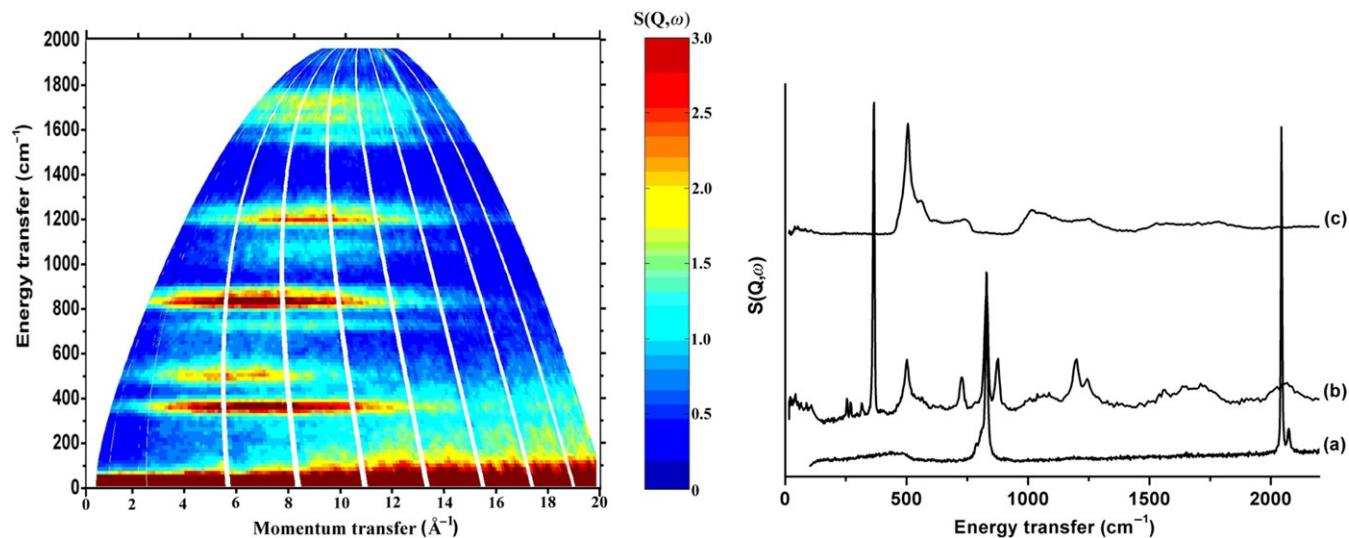


FIGURE 1.10 The contour color plot on the left shows inelastic neutron-scattering data for the metal-hydride complex Rb_2PtH_6 using an incident neutron energy of 2000 cm^{-1} ($\sim 250 \text{ meV}$). The figure on the right compares vibrational spectra using (a) Raman scattering and (b) inelastic neutron scattering along the proton-recoil line for the same compound. The top trace (c) corresponds to the inelastic neutron-scattering response from the reference material RbH , also present as a reaction by-product in (b). The broad energy features in the Raman data below $\text{ca. } 500 \text{ cm}^{-1}$ arise from fluorescence backgrounds. *Reprinted with permission from Ref. [80].*

scattering. The relative simplicity of these data on the (Q, E) plane arises from the predominance of incoherent scattering from hydrogen. In this situation, it is possible to focus data collection preferentially along the proton-recoil line so as to obtain a hydrogen-projected vibrational density of states (see spectra on the right-hand side of [Figure 1.10](#)). This approach provides a powerful means of boosting count rates by orders of magnitude with a much-improved spectral resolution, comparable to infrared spectroscopy and Raman scattering. It is also noteworthy that the neutron data contains far more spectral features than the corresponding optical spectrum, as the former is not subjected to hard selection rules and it can be directly related to the vibrational density of states. This feature of inelastic neutron scattering makes comparison with theoretical predictions relatively straightforward.

When Q becomes large, the expansion shown by Equation (1.60) is no longer meaningful, and a more useful limiting case is the *impulse approximation* whereby the target atom is assumed to recoil as an effectively free particle. Andreani *et al.* have discussed the transition between these two limits for the case of molecular hydrogen [81]. We will also return to this case in [Section 1.4.3](#) in connection with the spectral moments of the scattering function.

1.3.6 Beyond Harmonic Vibrations

At higher temperatures (or even at the lowest temperatures for quantum solids), atomic displacements can be such a large fraction of the interatomic distance that the harmonic approximation is no longer valid (for reviews of anharmonic effects in crystals see, e.g., Refs. [82–84]). This departure from harmonic behavior has two primary consequences. First, higher-order terms appear in the expansion of $\langle e^{-iQ \cdot u_i(0)} e^{iQ \cdot u_i(t)} \rangle$ in Equation (1.58), as recently revisited in the context of the analysis of diffraction data [85]. Numerical estimates [86] have shown that these effects are generally small in normal crystals at high temperatures. In quantum crystals, they appear to be more pronounced and lead to interference terms between one- and multiphonon scattering [65,87]. The second consequence of anharmonicity involves corrections to the thermal averages given in [Section 1.3.4](#). One immediate effect is that the delta functions in Equations (1.75) and (1.76) are replaced by spectral distributions with temperature-dependent energies and widths that arise from phonon–phonon scattering. A formal separation into elastic, one-phonon, and multiphonon processes is however still valid. Direct experimental access to these (typically tiny) spectral shifts and line widths (excitation lifetimes) of phonons [88], magnetic excitations [89], and molecular rotations [90] has been made possible via the use of inelastic-spin-echo neutron scattering [91]. These data can now be compared with first-principles predictions, particularly those within the framework of density functional theory [92,93].

1.4 BEYOND CANONICAL SOLIDS

1.4.1 Space–Time (Van Hove) Correlation Functions

For systems exhibiting particle diffusion within experimental timescales (e.g., gases, liquids, and some solids like plastic crystals and battery materials), the operational definition of a solid introduced in [Section 1.3](#) based on the concept of well-defined equilibrium sites is no longer applicable. We must then return to the general expressions for the scattering functions given in [Section 1.2.6](#) and, in particular, to the definition of the intermediate scattering functions $I(\mathbf{Q}, t)$ and $I_s(\mathbf{Q}, t)$ introduced in Equations (1.50) and (1.51) at the end of that section. These two functions are intimately related to space–time correlation functions $G(\mathbf{r}, t)$, first introduced by Van Hove [94], and often named after him:

$$G^{dd'}(\mathbf{r}, t) = \frac{1}{(2\pi)^3} \int I^{dd'}(\mathbf{Q}, t) e^{-i\mathbf{Q}\cdot\mathbf{r}} d\mathbf{Q}, \quad (1.89)$$

$$G_s^d(\mathbf{r}, t) = \frac{1}{(2\pi)^3} \int I_s^d(\mathbf{Q}, t) e^{-i\mathbf{Q}\cdot\mathbf{r}} d\mathbf{Q}. \quad (1.90)$$

Using again the integral representation of the delta function for spatial variables, and the definitions of $I(\mathbf{Q}, t)$ and $I_s(\mathbf{Q}, t)$ (Equations 1.50 and 1.51), these spatial Fourier transforms of the intermediate scattering functions are equal to

$$G^{dd'}(\mathbf{r}, t) = \frac{1}{(N_d N_{d'})^{1/2}} \sum_{\substack{j \in d \\ j' \in d'}} \int \langle \delta[\mathbf{r}' - \mathbf{R}_j(0)] \delta[\mathbf{r}' + \mathbf{r} - \mathbf{R}_{j'}(t)] \rangle d\mathbf{r}', \quad (1.91)$$

$$G_s^d(\mathbf{r}, t) = \frac{1}{N_d} \sum_{j \in d} \int \langle \delta[\mathbf{r}' - \mathbf{R}_j(0)] \delta[\mathbf{r}' + \mathbf{r} - \mathbf{R}_j(t)] \rangle d\mathbf{r}'. \quad (1.92)$$

The physical meaning of Equation (1.91) becomes more transparent when written in terms of particle-density operators $\rho(\mathbf{r}, t)$ of the form

$$\rho_d(\mathbf{r}, t) = \sum_{j \in d} \delta[\mathbf{r} - \mathbf{R}_j(t)]. \quad (1.93)$$

so that the total scattering function reads

$$G^{dd'}(\mathbf{r}, t) = \frac{1}{(N_d N_{d'})^{1/2}} \int \langle \rho_d(\mathbf{r}', t) \rho_{d'}(\mathbf{r}' + \mathbf{r}, t) \rangle d\mathbf{r}'. \quad (1.94)$$

Similarly, we introduce particle-density operators in momentum space

$$\rho_d(\mathbf{Q}, t) = \int \rho_d(\mathbf{r}, t) e^{i\mathbf{Q}\cdot\mathbf{r}} d\mathbf{r} = \sum_{j \in d} e^{i\mathbf{Q}\cdot\mathbf{R}_j(t)}, \quad (1.95)$$

so that the intermediate scattering function becomes

$$I^{dd'}(\mathbf{Q}, t) = \frac{1}{(N_d N_{d'})^{1/2}} \langle \rho_d(-\mathbf{Q}, 0) \rho_{d'}(\mathbf{Q}, t) \rangle. \quad (1.96)$$

From the above, it becomes clear that $G_{dd'}(\mathbf{r}, t)$ and $I_{dd'}(\mathbf{Q}, t)$ represent spatiotemporal particle-density correlations in the system. As such, they embody all the information accessible via inelastic neutron-scattering experiments from atomic nuclei.

1.4.2 Pair Distribution Functions

Using the property that at $t=0$ the operators $\mathbf{R}_j(t)$ and $\mathbf{R}_{j'}(t)$ commute for a pair of distinguishable particles, Equation (1.91) simplifies to

$$G^{dd'}(\mathbf{r}, 0) = \left(\frac{N_d}{N_{d'}} \right) \sum_{\substack{j' \in d' \\ (0 \in d)}} \langle \delta[\mathbf{r}' - \mathbf{R}_0(0) + \mathbf{R}_{j'}(0)] \rangle, \quad (1.97)$$

where we neglect surface effects and assume that all d atoms are equivalent to the one at the origin. *Pair-density functions* are defined such that

$$\rho^{dd'}(\mathbf{r}) = \left(\frac{N_d}{N_{d'}} \right) \sum_{\substack{j' \in d' \\ (0 \in d)}} \langle \delta[\mathbf{r}' - \mathbf{R}_0(0) + \mathbf{R}_{j'}(0)] \rangle, \quad (1.98)$$

which give the average instantaneous density of particles of type d' with respect to one atom of type d sitting at the origin. $G(\mathbf{r}, 0)$ then becomes

$$G^{dd'}(\mathbf{r}, 0) = \delta_{dd'} \delta(\mathbf{r}) + \left(\frac{N_d}{N_{d'}} \right) \rho^{dd'}(\mathbf{r}) \quad (1.99)$$

and

$$G_s^d(\mathbf{r}, 0) = \delta(\mathbf{r}). \quad (1.100)$$

If we now define structure factors representing the energy integrals of the dynamic structure factor $S(\mathbf{Q}, E)$:

$$S^{dd'}(\mathbf{Q}) = \int_{-\infty}^{\infty} S_c^{dd'}(\mathbf{Q}, E) dE = I^{dd'}(\mathbf{Q}, 0), \quad (1.101)$$

then Equations (1.89) and (1.90) lead to

$$\begin{aligned} \int S^{dd'}(\mathbf{Q}) e^{-i\mathbf{Q}\cdot\mathbf{r}} d\mathbf{Q} &= \int I^{dd'}(\mathbf{Q}, 0) e^{-i\mathbf{Q}\cdot\mathbf{r}} d\mathbf{Q} \\ &= (2\pi)^3 \left[\delta_{dd'} \delta(\mathbf{r}) + \left(\frac{N_d}{N_{d'}} \right)^{1/2} \rho^{dd'}(\mathbf{r}) \right] \end{aligned} \quad (1.102)$$

or equivalently

$$\frac{1}{(2\pi)^3} \int [S^{dd'}(\mathbf{Q}) - \delta_{dd'}] e^{-i\mathbf{Q}\cdot\mathbf{r}} d\mathbf{Q} = \left(\frac{N_d}{N_{d'}} \right)^{1/2} \rho^{dd'}(\mathbf{r}). \quad (1.103)$$

As shown by Equation (1.103), the pair-density function and its associated structure factor constitute the most fundamental quantities in a detailed analysis of both ordered and disordered solid-state structure using pair-distribution-function (PDF) methods [95]. Systems amenable to PDF analysis fall in the category of “canonical solids” as discussed in Section 1.3.

For incoherent scattering, we note from Equation (1.51) that

$$\int_{-\infty}^{\infty} S_i^d(\mathbf{Q}, E) dE = I_s^d(\mathbf{Q}, 0) = 1. \quad (1.104)$$

The structure factors $S^{dd'}(\mathbf{Q})$ are measured in a *total scattering* experiment in which the energy-integrated scattering at constant $\mathbf{Q}$ is determined according to

$$\begin{aligned} \left. \frac{d\sigma}{d\Omega} \right|_{\mathbf{Q}} &= \int_{-\infty}^{E_i} \frac{d^2\sigma}{d\Omega dE_f} dE_f \approx \sum_{dd'} c_d^{1/2} c_{d'}^{1/2} \overline{b_d^* b_{d'}} S^{dd'}(\mathbf{Q}) + \sum_d c_d \frac{\sigma_i^d}{4\pi} \\ &= \sum_{dd'} c_d^{1/2} c_{d'}^{1/2} \overline{b_d^* b_{d'}} [S^{dd'}(\mathbf{Q}) - \delta_{dd'}] + \sum_d c_d \frac{\sigma^d}{4\pi}, \end{aligned} \quad (1.105)$$

where the second term incorporates the total scattering cross section σ^d and the first term represents the “distinct” scattering excluding the self term. Equation (1.105), relating the differential cross section $d\sigma/d\Omega$ to the partial structure factors $[S^{dd'}(\mathbf{Q}) - \delta_{dd'}]$, and Equation (1.103), relating the latter to the pair-density functions, provide the basis for the measurement of the instantaneous structure of dense fluids with neutron diffraction. Of course, scattering measurements observe an ensemble and time average of such instantaneous structures.

A series of challenges arise in such experiments:

- (1) The condition of integration over energy at constant $\mathbf{Q}$ is not easy to achieve. In a typical experimental arrangement as shown in Figure 1.2, measurements at constant scattering angle ϕ and energy transfer E involve a curved locus in $(\mathbf{Q}, E)$ (cf. Figure 1.6). Conversion of these data

to “constant- $\mathbf{Q}$ ” cuts might not be a trivial task. Also, Equation (1.105) is not exact unless E_i is replaced by ∞ in the upper limit of the integral and a (k_f/k_i) factor is included in the integrand. The so-called static approximation neglects these limitations (see Section 1.4.4).

- (2) The incoherent scattering cross sections σ_i^d must be known to sufficient accuracy, as well as their associated scattering lengths $\bar{b}_d$.
- (3) A single measurement for a multicomponent system gives a linear combination of the various partial structure factors $S^{dd'}$, and multiple measurements with different contrasts (i.e., values of $\bar{b}_d$), for example, by varying isotopic compositions, are required to separate these out.

In spite of these hurdles, neutron diffraction on liquids and dense gases has become a mature tool for the study of an increasing number of complex disordered materials, including liquids and glasses. A classic result, for liquid argon, is shown in Figure 1.11. We may also measure the structure factors for glasses, subject to the same limitations. Note that these measurements give information about the instantaneous positions of the atoms in the glass as defined by Equation (1.98), compared with measurements of the purely elastic scattering (Equation 1.65), which only yield information about the equilibrium sites. The differences between the two depend on the details of the atomic vibrations. For fluids, there is no purely elastic scattering and the functions $S_c^{dd'}(\mathbf{Q}, E)$ and $S_i^d(\mathbf{Q}, E)$ continuously approach well-defined limits as $E \rightarrow 0$. To illustrate their typical behavior, these two scattering functions are shown in Figure 1.12 for liquid argon. For $E=0$, $S_i^d(\mathbf{Q}, E)$ peaks at $Q=0$

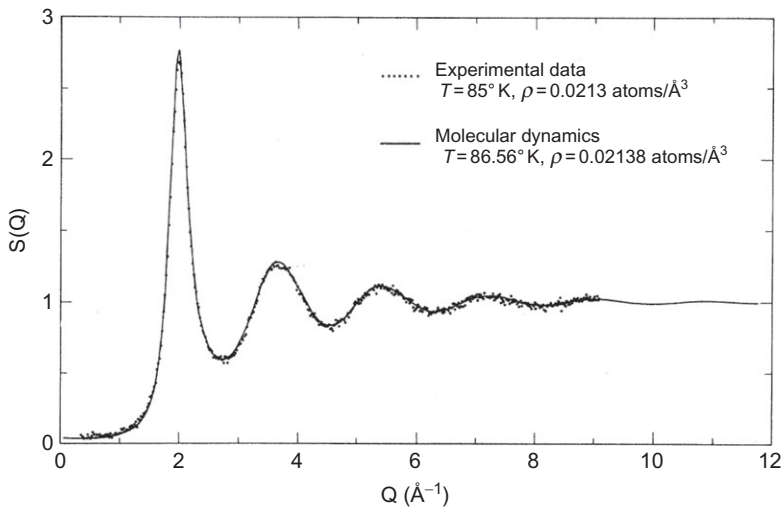


FIGURE 1.11 Experimental (black circles) structure factor for liquid ^{36}Ar at 85 K, and comparison with the predictions of molecular dynamics simulations (solid line) for a Lennard-Jones fluid. Reprinted with permission from Ref. [96].

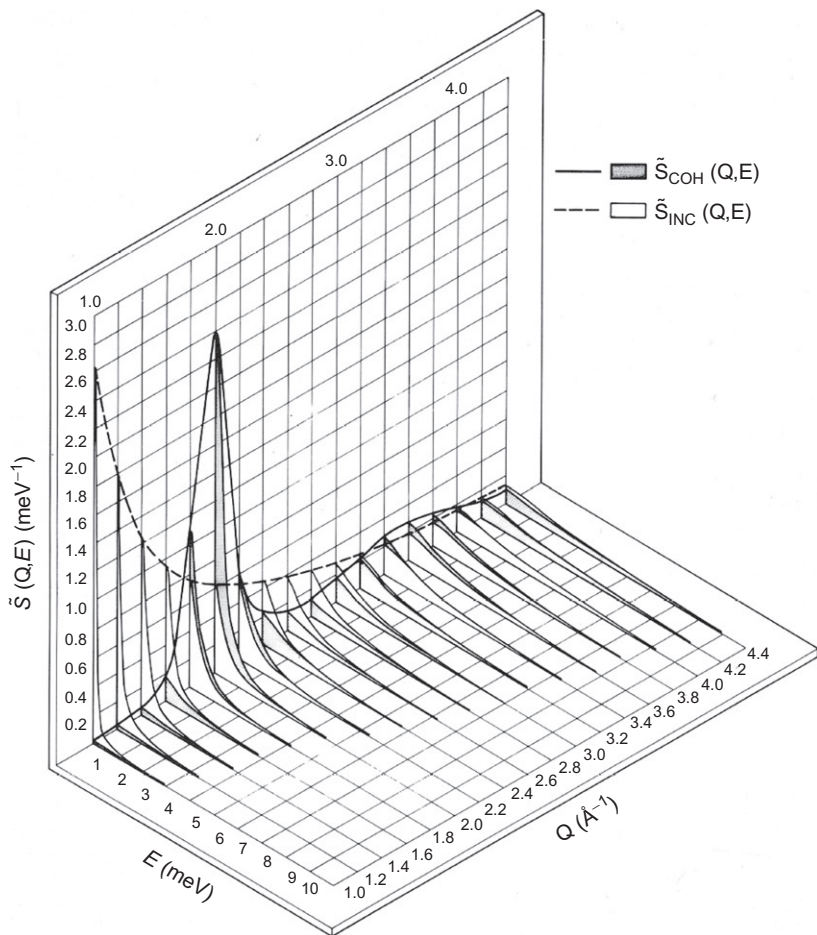


FIGURE 1.12 Coherent (solid line) and incoherent (dashed line) scattering functions for liquid argon. Reprinted with permission from Ref. [97].

and it decreases monotonically with Q . This behavior is offset by a concomitant increase in spectral widths, such that the total intensity remains constant when integrated over all energies (cf. Equation 1.112). We shall see in Section 1.4.5.1 that the Q dependence of the energy widths provides a direct measure of the self-diffusion coefficient at low momentum transfers. The coherent scattering function $S_c^{dd'}(\mathbf{Q}, E)$, on the other hand, displays a strong maximum at $Q \sim 2 \text{ \AA}^{-1}$, followed by lower-amplitude oscillations at higher momentum transfers. This behavior mimics the one shown earlier in Figure 1.11 for the same system. These intensity oscillations are related to characteristic interatomic distances in the liquid, as implied by the Fourier relationship given by Equation (1.103). The attentive reader will also note that

the spectral width around the first maximum in $S_c^{dd'}(\mathbf{Q}, E)$ is narrower than adjacent energy spectra shown at lower and higher momentum transfers. This behavior is termed *de Gennes narrowing*, and necessarily arises from the physical properties of the coherent dynamic structure factor, as we shall see below. Intuitively, it might be rationalized as the arrest of atomic motions in the vicinity of characteristic distances associated with the solvation cage around a given atom. For further details, we strongly encourage a careful reading of the seminal (and very elegant) work of de Gennes, applicable to both nuclear [98] and magnetic [99] neutron scattering in liquids and paramagnets, respectively. A quantitative interpretation of $S_c^{dd'}(\mathbf{Q}, E)$ is not as straightforward as that given above for $S_i^d(\mathbf{Q}, E)$ and is deferred to our discussion of stochastic processes in Section 1.4.5.

In closing this discussion, we refer the reader to Ref. [100] for an online collection of examples of neutron-derived structure factors of disordered materials (both glasses and liquids) using total-scattering methods.

1.4.3 Properties of the Dynamic Structure Factor

The dynamic structure factors $S_c(\mathbf{Q}, E)$ and $S_i(\mathbf{Q}, E)$, and by implication the intermediate and space–time correlation functions, must satisfy certain general relations regardless of the details of the system under consideration. These relations are important for several reasons as they provide:

- (1) A consistency check on experiment data.
- (2) A necessary constraint on theoretical models.
- (3) An indication as to how a classically based calculation, for example, using computer simulation, can be related to the experimentally observed neutron scattering where quantum effects might be at play.

Although these relations are valid for all types of systems, it is in the field of neutron scattering from liquids that they are most often invoked and it is therefore appropriate to introduce them here.

- (a) $S(\mathbf{Q}, E)$ is real (i.e., an experimental observable)

$$S_c^{dd'}(\mathbf{Q}, E) = S_c^{dd'*}(\mathbf{Q}, E), \quad (1.106)$$

$$S_i^d(\mathbf{Q}, E) = S_i^{d'*}(\mathbf{Q}, E). \quad (1.107)$$

These relations follow from the requirement that arbitrary linear combinations of the partial scattering functions with real coefficients lead to cross sections that must be real and positive; from the same reasoning, S_c^{dd} and S_i^d must be everywhere positive, but not necessarily $S_c^{dd'}$ when $d \neq d'$.

- (b) $S(\mathbf{Q}, E)$ must satisfy detailed balance

$$S_c^{dd'}(\mathbf{Q}, E) = e^{E/k_B T} S_c^{dd'}(-\mathbf{Q}, -E), \quad (1.108)$$

$$S_i^d(\mathbf{Q}, E) = e^{E/k_B T} S_i^d(-\mathbf{Q}, -E). \quad (1.109)$$

Here we allow for the fact that the scattering functions may not be symmetric with respect to time reversal, for example in the presence of external (e.g., magnetic) fields. The condition of detailed balance governs the ratio of the scattering probabilities for two transitions proceeding in opposite directions. The matrix element is the same in both directions, but the probability of the sample being initially in the lower state is a factor $e^{E/k_B T}$ higher than that of it being initially in the higher state. Thus, the scattering is more intense when the neutron loses energy E than when it gains it.

The detailed-balance condition can also be expressed in terms of a relation between the real and imaginary parts of the correlation functions:

$$\text{Im}G(r, t) = -\tanh\left(\frac{\hbar}{2k_B T} \frac{\partial}{\partial t}\right) \text{Re}G(\mathbf{r}, t), \quad (1.110)$$

which is a special case of the *fluctuation–dissipation theorem* expressing the relation between the dissipative response of the system to a perturbation and the equilibrium fluctuations of the system in the absence of the perturbation [101].

(c) *Zeroth moment*

The relations between total and dynamic structure factor introduced in Section 1.4.2 can be rewritten as

$$\int_{-\infty}^{\infty} S_c^{dd'}(\mathbf{Q}, E) dE = S^{dd'}(\mathbf{Q}) \quad (1.111)$$

and

$$\int_{-\infty}^{\infty} S_i^d(\mathbf{Q}, E) dE = 1. \quad (1.112)$$

The limit of $S^{dd'}(\mathbf{Q})$ as $|\mathbf{Q}| \rightarrow 0$ is related to the statistical fluctuation in the number of particles N_d in a fixed volume of dimension $V^{1/3} \gg 2\pi/|\mathbf{Q}|$:

$$S^{dd'}(0) = \frac{\overline{\Delta N_d \Delta N_{d'}}}{\overline{N_d N_{d'}}}. \quad (1.113)$$

This limit is also related to the macroscopic isothermal compressibility for fluids (including their mixtures, see p. 77 in Ref. [103]) such that

$$S(0) = n \left(\frac{\partial V}{\partial P} \right)_T k_B T. \quad (1.114)$$

(d) *First moment*

In the case of a system with velocity-independent interatomic forces, the first energy moments of the scattering functions are given by

$$\int_{-\infty}^{\infty} S_c^{dd'}(\mathbf{Q}, E) E \, dE = \frac{\hbar^2 Q^2}{2M_d} \delta_{dd'}, \quad (1.115)$$

$$\int_{-\infty}^{\infty} S_i^d(\mathbf{Q}, E) E \, dE = \frac{\hbar^2 Q^2}{2M_d}, \quad (1.116)$$

where the right-hand side of these expressions correspond to the recoil energy of an atom of mass M_d .

(e) *Second moment*

The second moment of the incoherent scattering function is

$$\int_{-\infty}^{\infty} S_i^d(\mathbf{Q}, E) E^2 \, dE = \left(\frac{\hbar^2 Q^2}{2M_d} \right)^2 + \hbar^2 Q^2 \left\langle (\mathbf{v} \cdot \hat{\mathbf{Q}})^2 \right\rangle, \quad (1.117)$$

where $\mathbf{v} \cdot \hat{\mathbf{Q}}$ is the velocity in the direction of $\mathbf{Q}$. The second moment of $S_i^d(\mathbf{Q}, E)$ and the higher moments of both scattering functions depend on the interatomic potential. Explicit expressions valid for quantum systems have been given by Rahman *et al.* [104].

The significance of these moment relations is best illustrated by discussing two limiting cases, either fulfilling or violating these fundamental properties of the dynamic structure factor.

1.4.3.1 *Total Scattering and the Static Approximation*

If energy changes in the system are negligible compared with the incident neutron energy E_i , $(E + E_{\tau_i} - E_{\tau_f}) \sim E$, the delta function $\delta(E + E_{\tau_i} - E_{\tau_f})$ in Equation (1.31) can be replaced by $\delta(E)$. Similarly, $|\mathbf{k}_i| = |\mathbf{k}_f|$ and the double differential cross section simplifies to

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N} \sum_{\tau_i} p_{\tau_i} \sum_{\tau_f} \left| \sum_j b_j \langle \tau_f | e^{i\mathbf{Q}_0 \cdot \mathbf{R}_j} | \tau_i \rangle \right|^2 \delta(E), \quad (1.118)$$

where $\mathbf{Q}_0$ is the value of $\mathbf{Q}$ when $k_f = k_i$ for a given k_i and scattering angle ϕ . Integration over the final energy E_f gives

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \int_{-\infty}^{E_i} \frac{d^2\sigma}{d\Omega dE_f} dE_f = \frac{1}{N} \sum_{\tau_i} p_{\tau_i} \sum_{j \in d, j' \in d'} \overline{b_j^*} \overline{b_{j'}} \left\langle e^{-i\mathbf{Q}_0 \cdot (\mathbf{R}_{j'} - \mathbf{R}_j)} \right\rangle \\ &= \sum_{dd'} c_d^{1/2} c_{d'}^{1/2} \overline{b_d^*} \overline{b_{d'}} S^{dd'}(\mathbf{Q}_0) + \sum_d c_d \frac{\sigma_i^d}{4\pi}. \end{aligned} \quad (1.119)$$

It is noteworthy that Equation (1.119) is equivalent to Equation (1.105), the approximate result for the total scattering at constant $\mathbf{Q}$, with $\mathbf{Q} = \mathbf{Q}_0$. Diffraction experiments from liquids and glasses are often interpreted on the assumption that Equation (1.119) is valid in the case of measurements at constant E_i and ϕ . However, there is no inelastic scattering in this model and the first moment of the scattering function, for example, is zero instead of the exact result given by Equation (1.115). We may therefore expect this approximation to be valid when $E_i \gg \hbar^2 Q^2 / 2M$ or $k_i \gg Q/A^{1/2}$. For diffraction measurements out to $Q \approx 10 \text{ \AA}^{-1}$, this condition may imply $k_i \approx 50 \text{ \AA}^{-1}$ and neutron energies of several eV, readily available at pulsed spallation neutron sources. For measurements with thermal neutrons, attempts are usually made to correct for the errors in the treatment of elastic scattering using the incoherent approximation and an expansion in powers of $(1/A)$, as originally introduced by Placzek [105–107] and further discussed in Ref. [108].

1.4.3.2 Free Particles and the Impulse Approximation

The scattering function for identical (noninteracting and structureless) particles can be calculated analytically and provides important insights into the expected behavior of dynamic structure factors. To this end, we consider an ensemble of N such particles in a box of volume V with momentum distribution $n(\mathbf{p})$. The associated translational wave function for the j th particle in the ensemble is simply given by

$$|\tau\rangle = \frac{1}{V^{1/2}} e^{i\mathbf{p}\cdot\mathbf{R}/\hbar}. \quad (1.120)$$

Since the particles are independent, it suffices to consider the single-particle $S(\mathbf{Q}, E)$. Using

$$\langle \tau_f | e^{i\mathbf{Q}\cdot\mathbf{R}} | \tau_i \rangle = \delta_{\mathbf{Q}, (\mathbf{p}_f - \mathbf{p}_i)/\hbar} \quad (1.121)$$

and

$$E_{\tau_f} - E_{\tau_i} = \frac{\hbar^2}{2M} (p_f^2 - p_i^2) = \frac{\hbar^2}{2M} \left(Q^2 + \frac{2\mathbf{Q}\cdot\mathbf{p}_i}{\hbar} \right). \quad (1.122)$$

The dynamic structure factor is then given by

$$\begin{aligned} S(\mathbf{Q}, E) &= \frac{1}{N} \sum_{\tau_i} p_{\tau_i} \sum_{\tau_f} \langle \tau_f | e^{i\mathbf{Q}\cdot\mathbf{R}} | \tau_i \rangle^2 \delta(E + E_{\tau_i} - E_{\tau_f}) \\ &= \sum_{\mathbf{p}_i} n(\mathbf{p}_i) \delta\left(E - \frac{\hbar^2 Q^2}{2M} - \frac{\hbar \mathbf{Q}\cdot\mathbf{p}_i}{M}\right). \end{aligned} \quad (1.123)$$

This is the well-known *impulse approximation* for scattering from a collection of independent particles. The scattering function is centered at the recoil energy

$$E_R = \frac{\hbar^2 Q^2}{2M} \quad (1.124)$$

and its zeroth moment is $\sum_{\mathbf{p}_i} n(\mathbf{p}_i) = 1$. Furthermore, its first moment is E_R and the second moment is given by

$$E_R^2 + 2E_R \sum_{\mathbf{p}_i} n(\mathbf{p}_i) \frac{(\mathbf{p}_i \cdot \hat{\mathbf{Q}})^2}{M} = E_R^2 + \frac{2E_R \langle p_i^2 \rangle}{3M}. \quad (1.125)$$

In this particular case, we see that first three moment relations are fully satisfied. Furthermore, the coherent and incoherent scattering functions are identical since the particles are assumed to be uncorrelated. The impulse approximation is generally considered to represent the high- Q limit of the scattering functions for *any* system. When the recoil energy is much larger than the energies of the natural excitations in the system, Equation (1.125) should be a good approximation. Systematic deviations from this exact result have been the subject of much theoretical and experimental work to date [81,109–111].

The precise shape of the dynamic structure factor depends on the type of statistics obeyed by the particle ensemble. For a Boltzmann distribution at temperature T , it can be written as

$$S(\mathbf{Q}, E) = \frac{1}{\sqrt{2\pi}\sigma_Q} e^{-(E-E_R)^2/2\sigma_Q^2}, \quad (1.126)$$

where

$$\sigma_Q^2 = 2E_R k_B T = \hbar^2 Q^2 k_B T / M. \quad (1.127)$$

The intermediate scattering function is given by

$$I(\mathbf{Q}, t) = e^{-\sigma_t^2 Q^2 / 2} \quad (1.128)$$

and the space–time correlation function by

$$G(\mathbf{r}, t) = \frac{1}{(2\pi)^{3/2} \sigma_t^3} e^{-r^2/2\sigma_t^2}, \quad (1.129)$$

where

$$\sigma_t^2 = t(t - i\hbar/k_B T) k_B T / M. \quad (1.130)$$

We see that $G(\mathbf{r}, t)$ is given by a Gaussian whose spatial width increases at a rate given by

$$\frac{\sigma_t}{t} = \left\langle \frac{v^2}{3} \right\rangle^{1/2} \left(1 - \frac{i\hbar}{k_B T t} \right)^{1/2}. \quad (1.131)$$

For a classical system, we would clearly expect a rate given simply by $\langle v^2/3 \rangle^{1/2}$. Even though we have used classical statistics to describe the dynamics of the scattering system, there is an additional factor involving $\hbar$ that describes the quantum mechanics of the interaction process. This leads to a scattering function centered at E_R instead of $E=0$, which satisfies the detailed balance condition. The double differential cross section corresponding to Equation (1.126) can be integrated explicitly to give a total scattering cross section (see pp. 93–95 in Vol. I of Ref. [63])

$$\sigma_{\text{tot}} = \frac{\sigma}{(1 + m_n/M)^2} \left[\left(1 + \frac{1}{2\varepsilon} \right) \Phi(\varepsilon^{1/2}) + \frac{e^{-\varepsilon}}{(\pi\varepsilon)^{1/2}} \right], \quad (1.132)$$

where $\varepsilon = E_i M / m_n k_B T$ and $\Phi(x) = 2\pi^{-1/2} \int_0^x e^{-t^2} dt$. For $\varepsilon \ll 1$ this gives

$$\sigma_{\text{tot}} \propto \frac{1}{E^{1/2}}, \quad (1.133)$$

due to the kinematic factor (k_f/k_i) in the inelastic cross section, and for $\varepsilon \gg 1$ we get

$$\sigma_{\text{tot}} \approx \sigma \left(\frac{A}{1+A} \right)^2, \quad (1.134)$$

which is the free-atom cross section, as required.

For quantum systems, $n(\mathbf{p})$ will no longer have the Maxwellian form of Equations (1.126) and (1.127) since it is affected by delocalization and particle-exchange effects. In quantum solids and nondegenerate quantum liquids like ^4He and ^3He , early neutron experiments [112,113] and computer simulations [114] have established that $n(\mathbf{p})$ (as well as $S(\mathbf{Q}, E)$ by virtue of Equation 1.123) is, to a first-order approximation, Gaussian but with the actual temperature in Equation (1.127) replaced by an effective temperature representing the kinetic energy associated with zero-point motions. Figure 1.13 shows a series of recoil-scattering inelastic spectra for solid ^4He using neutrons with energies in the electron-volt range and available at spallation sources (cf. Chapter 2). In this high-energy regime, it is customary to represent the neutron data in terms of the so-called *neutron Compton profile* $J(\mathbf{Q}, y)$ [111,116], where

$$y = \frac{M}{\hbar^2 Q} (E - E_R) \quad (1.135)$$

is a scaling variable which reflects the relationship between E and Q in the impulse limit. Within the impulse approximation, it is related to the dynamic structure factor $S_{\text{IA}}(\mathbf{Q}, E)$ by

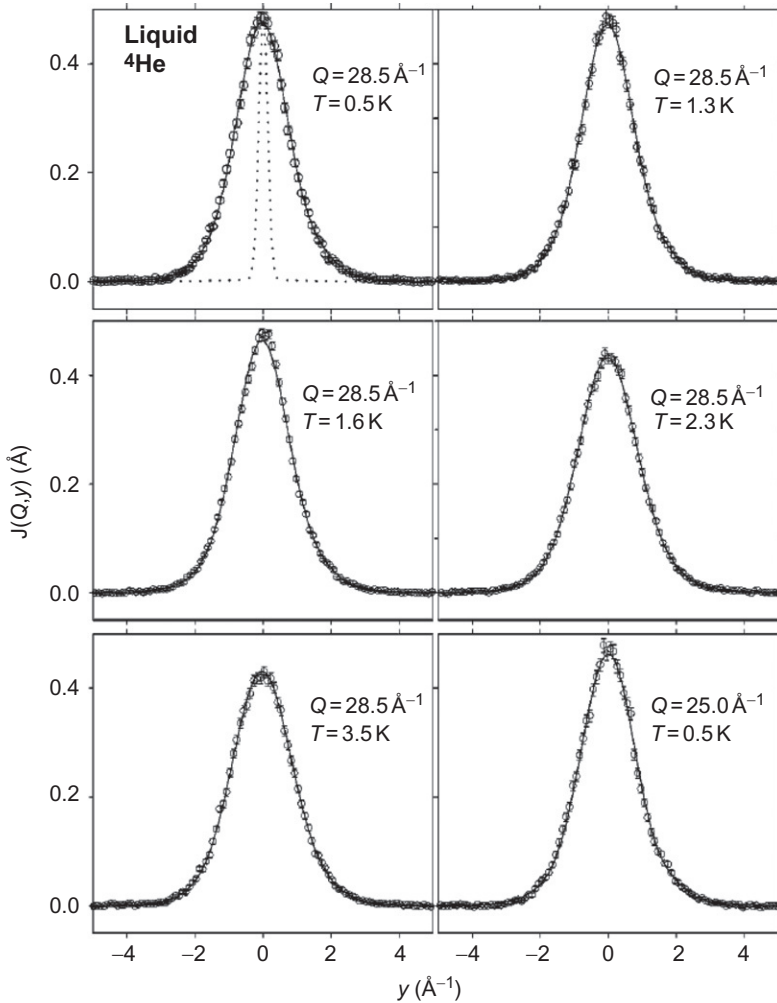


FIGURE 1.13 Recoil scattering of 0.75-eV neutrons from liquid ^4He above and below $T_\lambda = 2.17$ K. The dotted line denotes the instrumental resolution and the solid lines are fits to the experimental data using a model that accounts for the presence of a Bose–Einstein condensate fraction and departures from the impulse approximation. Reprinted with permission from Ref. [115].

$$J(\hat{\mathbf{Q}}, y) = \frac{\hbar Q}{M} S_{\text{IA}}(\mathbf{Q}, E). \quad (1.136)$$

Use of Equation (1.136) to represent experimental results necessarily assumes that it is possible to isolate the response from a given mass M in the neutron data. Also, measurements of this observable at different absolute Q values can be used to ascertain the validity and deviations from this

limit – we also recall that at high momentum transfers Q , the static structure factor $S(Q)$ tends to unity and, therefore, the observed (total) dynamic structure factor is entirely incoherent. Conversely, the different recoil lines followed by different masses M provides a means of isolating the response of different elements in the sample (particularly light atoms). This task requires a judicious choice of the kinematic trajectories to “cut through” the mass-recoil lines in (Q, E) space with sufficient mass discrimination, as recently demonstrated in simultaneous measurements of the mean kinetic energies of several nuclides in lithium-containing materials [117,118].

More generally, the neutron Compton profile is related to the momentum distribution $n(\mathbf{p})$ via the integral relation

$$J(\hat{\mathbf{Q}}, y) = \hbar \int n(\mathbf{p}) \delta(\hbar y - \mathbf{p} \cdot \hat{\mathbf{Q}}) d\mathbf{p}. \quad (1.137)$$

Formally speaking, Equation (1.137) corresponds to the Radon transform of the momentum distribution $n(\mathbf{p})$ and it defines the probability for a given nucleus to have a momentum within an interval $\hbar(y, y + dy)$ parallel to $\hat{\mathbf{Q}}$. Within the impulse approximation, the neutron Compton profile should be independent of the magnitude of momentum transfer Q , a result which is experimentally confirmed by the data shown in Figure 1.13. From these experiments, it is possible to extract quantitative estimates of the condensate fraction in this fundamental benchmark system. This experimental strategy has been used recently to establish whether claims of supersolidity in ^4He obtained from torsional-oscillator measurements could be traced back to microscopic mechanisms [119,120]. These studies have not detected any changes in the $n(\mathbf{p})$ of ^4He upon crossing the critical temperature, suggesting that the phenomenon is of a very different nature to the well-known case of superfluidity.

The use of epithermal neutrons to measure momentum distributions also transcends a purely academic interest in exotic quantum phases of matter. In the case of the hydrogen-bonded ferroelectric potassium dihydrogen phosphate (KH_2PO_4 or KDP), neutron Compton scattering has been used to probe the momentum distribution $n(\mathbf{p})$ of protons above and below the ferroelectric transition at $T = 124 \text{ K}$ [121]. These momentum distributions are shown in Figure 1.14 for a single crystal, from which it is possible to extract an effective Born–Oppenheimer potential for the hydrogen atoms in the material. These data also show that the proton remains coherently delocalized across two sites in the high-temperature phase, strongly suggesting the inapplicability of the commonly accepted order–disorder character of this transition.

As in the case of inelastic neutron-scattering experiments probing phonon structure at lower energy and momentum transfers, first-principles calculations can be used to predict the outcome of neutron Compton scattering measurements. Within the harmonic approximation, the neutron Compton profile for a given nuclide n may be described by a Gaussian line shape along a given crystallographic direction i by writing

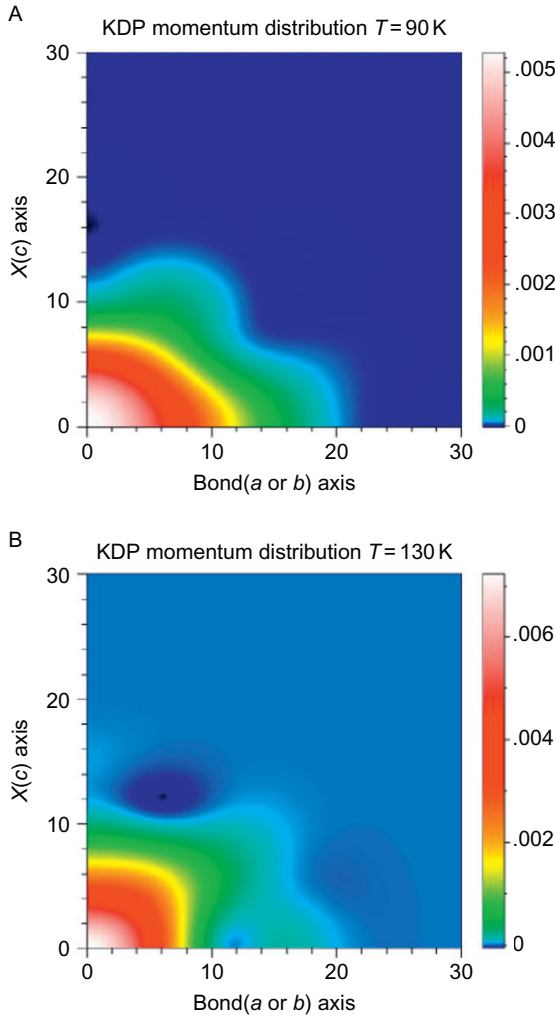


FIGURE 1.14 Proton momentum distributions in KDP below (a) and above (b) the ferroelectric transition. The green bulge in (b) ca. 30° away from the bond axis is ascribed to the repulsion experienced by the proton due to the neighboring phosphorous atoms in the lattice. *Reprinted with permission from Ref. [121].*

$$J_{n,i}(y) = \frac{1}{\sqrt{2\pi}\sigma_{n,i}} e^{-y^2/2\sigma_{n,i}^2}, \quad (1.138)$$

where we have assumed that the impulse approximation holds (i.e., no implicit Q dependence) and $\sigma_{n,i}$ describes the second moment of the distribution for nucleus n along direction i . Using the formalism presented in

Section 1.3, the neutron Compton profile widths $\sigma_n(\hat{\mathbf{q}})$ along a given direction $\hat{\mathbf{q}}$ can be computed from the phonon-dispersion relations using [122]

$$\sigma_n^2(\hat{\mathbf{q}}) = \frac{M_n}{\hbar^2 N} \sum_{\mathbf{q} \in \text{BZ}} \sum_k [\hat{\mathbf{q}} \cdot \mathbf{e}_n^k(\mathbf{q})]^2 \frac{E_k(\mathbf{q})}{2} \coth \left[\frac{E_k(\mathbf{q})}{2k_B T} \right], \quad (1.139)$$

where the sums run over all q -points sampled within the first Brillouin zone and over all phonon branches k . The *quasiharmonic* approximation [68] has been assumed in order to allow predictions at finite temperatures. Equation (1.139) provides a convenient starting point to discuss the various contributions to the Compton profile using the harmonic Born–Oppenheimer approximation, and has been shown to provide reliable estimates for a number of materials, including light hydrides [118,122], fluorides [117], and molecular hydrogen [123]. In the latter case, such an analysis of computational results has allowed the dissection of the experimental neutron data into various intra- and intermolecular contributions associated with the adsorbed state of this molecule, including the contributions from adsorbate–adsorbate interactions.

Beyond the harmonic approximation, path-integral molecular dynamics calculations allow for an exact evaluation of the nuclear momentum distribution $n(\mathbf{p})$ in terms of the density matrix $\rho(\mathbf{r}, \mathbf{r}') = \langle \mathbf{r} | \exp(-\hat{H}/k_B T) | \mathbf{r}' \rangle$ [124]:

$$n(\mathbf{p}) = \frac{1}{2\pi\hbar^3} \int d\mathbf{r} d\mathbf{r}' e^{i/\hbar(\mathbf{p} \cdot (\mathbf{r} - \mathbf{r}'))} \frac{\rho(\mathbf{r}, \mathbf{r}')}{Z}, \quad (1.140)$$

where Z is the partition function defined as

$$Z = \int d\mathbf{r} \rho(\mathbf{r}, \mathbf{r}). \quad (1.141)$$

Whereas the partition function Z is an integral over diagonal terms of the particle density matrix, the momentum distribution $n(\mathbf{p})$ requires the inclusion of off-diagonal elements. Its accurate evaluation is, therefore, far from being a trivial task. In the context of neutron Compton experiments, Lin *et al.* [125] have recently proposed an efficient and at the same time insightful method to evaluate the open-path distribution represented by Equation (1.140). In their formulation, free-particle and environmental contributions factorize and the latter can be likened to the free-energy surface, therefore facilitating the interpretation of experimental data. The calculation of these observables using *ab initio* computational methods also provides solid ground for the understanding of nuclear quantum effects in condensed matter beyond those captured by the harmonic approximation. These include tunneling and particle delocalization [126], the only exception at present being the full inclusion of particle exchange.

Recent examples of the use of epithermal neutrons and state-of-the-art computational techniques in a growing number of materials include the study

of proton momentum distributions and mean kinetic energies in supercritical and supercooled water [127,128], relaxation mechanisms in proton glasses [129], the potential energy landscape of protons in the superprotonic conductor $\text{Rb}_3\text{H}(\text{SO}_4)_2$ [130], nuclear quantum effects in the hydrogen-storage material lithium imide [131], and the determination of the spatial orientation of molecular hydrogen adsorbed in metal-graphite intercalates [123].

1.4.3.3 Classical Limit

Our final aim in this section is to establish a connection between the actual scattering function and that of a classical system interacting with the same potential. This is an important exercise because computer simulations using the molecular-dynamics method [132,133] play an invaluable role in furthering our understanding of classical liquids and dense fluids, and generally these calculations are done within the framework of classical mechanics. What is calculated is a classical intermediate scattering function (see Equation 1.96),

$$I_{\text{cl}}(\mathbf{Q}, t) = \frac{1}{N} \langle \rho_d(-\mathbf{Q}, 0) \rho_d(-\mathbf{Q}, 0) \rangle_{\text{cl}}, \quad (1.142)$$

where the averages are now classical thermal averages over the history of the model system. In this case $I_{\text{cl}}(\mathbf{Q}, t)$, $G_{\text{cl}}(\mathbf{r}, t)$, and $S_{\text{cl}}(\mathbf{Q}, E)$ are all symmetric with respect to simultaneous inversion of both variables, and the detailed balance condition, Equation (1.108), is not satisfied. A solution to this problem was suggested by Schofield [134] who noted that substitution of

$$\tau = t - \frac{i\hbar}{2k_{\text{B}}T} \quad (1.143)$$

in the classically derived correlation functions, yields functions

$$I(\mathbf{Q}, t) = I_{\text{cl}}(\mathbf{Q}, \tau) \quad (1.144)$$

in accord with the detailed-balance condition. For example, the classical solution of the noninteracting gas problem discussed in the previous section is

$$I_{\text{cl}}(\mathbf{Q}, t) = e^{-t^2 k_{\text{B}} T Q^2 / 2M}. \quad (1.145)$$

The above procedure leads to the intermediate scattering function

$$I(\mathbf{Q}, t) = e^{-\sigma_I^2 Q^2 / 2} \quad (1.146)$$

with

$$\sigma_I^2 = \frac{k_{\text{B}}T}{M} \tau^2 = \frac{k_{\text{B}}T}{M} \left(t - \frac{i\hbar}{2k_{\text{B}}T} \right)^2. \quad (1.147)$$

This result is identical to Equation (1.128) apart from a constant factor $\exp(E_{\text{R}}/4k_{\text{B}}T)$, which is generally close to unity for conditions under which measurements on classical liquids are performed. Similarly, the reverse

procedure can be used to obtain “symmetrized” dynamic structure factors from experimental data:

$$\tilde{S}(\mathbf{Q}, E) = e^{-E/2k_B T} S(\mathbf{Q}, E), \quad (1.148)$$

corresponding to

$$\tilde{I}(\mathbf{Q}, t) = I\left(\mathbf{Q}, t + \frac{i\hbar}{2k_B T}\right), \quad (1.149)$$

which can then be directly compared with molecular dynamics results.

1.4.4 From Order to Disorder: Diffuse Scattering

Most materials of practical interest display some form of disorder, with profound consequences on their physical and chemical properties [135]. In certain cases, these properties are intentionally modified by introducing controlled amounts of disorder, as in the case of the alloying of metals for technological applications. In other cases, disorder arises as a result of the material’s function, as in the cases of radiation damage of structural materials in nuclear reactors, or aging and corrosion in industrial applications. To gain an appreciation for the importance and technological relevance of disorder in our daily lives, the estimated global cost associated with corrosion prevention and mitigation *just* for the oil and gas sector has been valued at ca. \$19 billion in 2011 [136].

Macroscopic features of disorder, such as dislocation patterns and grain structures, and their relationship to the material’s properties are often understood at an empirical and phenomenological level. To advance beyond this level, static (within the timescale of the measurement) and dynamic disorder phenomena must be studied at the atomic scale. On this scale, disorder can be characterized in terms of vacancies and interstitials in the ideal lattice or by local fluctuations in chemical composition, all of which might also display a characteristic time dependence. The strains induced by these defects and the interaction and dynamical phenomena associated with these defects through their associated strain fields are important ingredients for a sound microscopic picture of disordered solids.

To proceed, we recall our preliminary discussion of diffuse scattering in Section 1.3.3. In a static picture (still conforming to our definition of a canonical solid), the loss of translational invariance in a solid gave way to diffuse scattering away from reciprocal lattice points satisfying the Bragg condition. In the extreme case of complete loss of translational disorder, all scattering becomes diffuse and one must abandon all together a description of the material in terms of well-defined Bragg reflections. This section extends these ideas and presents an outline of the use of diffuse nuclear scattering to study the emergence of disorder in solids. Magnetic diffuse scattering is discussed in Section 1.5.4. Beyond the classic text by Krivoglaz [137], more recent

monographs [95,103,138–140] discuss the subject well beyond the scope of this chapter, including diffuse X-ray scattering.

For the intermediate case of solids with long-range order but with various types of local disorder, one observes a combination of Bragg scattering and diffuse scattering away from Bragg points. A detailed analysis of Bragg versus diffuse intensities provides a wealth of information about the underlying (static and dynamic) disorder. To illustrate the methodology on more quantitative grounds, let us consider a solid in which a periodic lattice can be used as a basis for indexing the atomic positions. In this case, the unit cells in the crystal are positioned at $\mathbf{l}^0$ and atomic positions within the unit cell are denoted by $\mathbf{d}^0$. The superscript “0” is used to denote the ideal (undistorted) lattice points. Deviations from the lattice points are given by the vectors $\mathbf{u}$. We also define an occupational probability $P_{nj,k}$, which is equal to one if lattice site n,j is occupied by an atom of type k and 0 otherwise. The elastic coherent differential neutron cross section is then given by

$$\left(\frac{d\sigma}{d\Omega}\right)_{c,el} = \frac{1}{N_c} \left| \sum_{n=0}^{N_c-1} \sum_{j=0}^{J-1} \sum_{k=0}^{K-1} P_{nj,k} \overline{b}_k e^{-W_{nj,k}} e^{i\mathbf{Q} \cdot (\mathbf{l}_n^0 + \mathbf{d}_j^0 + \mathbf{u}_{nj,k})} \right|^2, \quad (1.150)$$

where N_c is the number of unit cells, J is the number of atomic positions within the unit cells, K is the number of different atom types, $\overline{b}_k$ is the coherent scattering length for atoms of type k , and W is the Debye–Waller exponent. The displacement $\mathbf{u}_{ijk}$ represents a linear superposition of displacements of atom n, j, k induced by all other atoms:

$$\mathbf{u}_{nj,k} = \sum_{n'j'k'} P_{n'j'k'} \mathbf{u}_{nj,k}^{n'j'k'}. \quad (1.151)$$

If we separate the occupation numbers into the average concentration of species k on site j , c_{jk} , and local fluctuations $\Delta P_{nj,k}$, the displacement can be written as

$$\mathbf{u}_{nj,k} = \sum_{n'j'k'} c_{j'k'} \mathbf{u}_{nj,k}^{n'j'k'} + \sum_{n'j'k'} \Delta P_{n'j'k'} \mathbf{u}_{nj,k}^{n'j'k'}. \quad (1.152)$$

The first sum in Equation (1.152) can be seen to define a new reference lattice,

$$\mathbf{l}_n + \mathbf{d}_j = \mathbf{l}_n^0 + \mathbf{d}_j^0 + \overline{\sum_{n'j'k'} c_{j'k'} \mathbf{u}_{nj,k}^{n'j'k'}} \quad (1.153)$$

where the average of the sum is over atomic species.

The second sum in Equation (1.152) accounts for the fluctuations in the occupation number about the average. As such, it describes distortions relative to the new reference lattice. From Equations (1.152) and (1.153) we obtain

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{c,el}} = \frac{1}{N_c} \left| \sum_{nj\mathbf{k}} P_{nj\mathbf{k}} \langle \bar{b}_{\mathbf{k}} \rangle e^{i\mathbf{Q} \cdot (\mathbf{l}_n + \mathbf{d}_j)} \exp \left(i\mathbf{Q} \cdot \sum_{n'j'k'} \Delta P_{n'j'k'} \mathbf{u}_{nj\mathbf{k}}^{n'j'k'} \right) \right|^2, \quad (1.154)$$

where $\langle \bar{b}_{\mathbf{k}} \rangle$ is the average of $\bar{b}_{\mathbf{k}} \exp(-W_{ijk})$ over the lattice indices n, j .

Further insight into the physical meaning of Equation (1.154) comes from considering the limiting case of small distortions. In this case, the second exponential in Equation (1.154) can be expanded as a power series. Keeping only the linear term in this expansion, the differential cross section becomes

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{c,el}} = \frac{1}{N_c} \left| \sum_{nj\mathbf{k}} [c_{jk} + P_{nj\mathbf{k}}] \langle \bar{b}_{\mathbf{k}} \rangle e^{i\mathbf{Q} \cdot (\mathbf{l}_n + \mathbf{d}_j)} \left[1 + i\mathbf{Q} \cdot \sum_{n'j'k'} \Delta P_{n'j'k'} \mathbf{u}_{nj\mathbf{k}}^{n'j'k'} \right] \right|^2. \quad (1.155)$$

By considering the expression resulting from the multiplication of the two square-bracketed terms in Equation (1.155), we can gain a first-order picture of the information content of the coherent elastic cross section for a solid with compositional disorder and small distortions:

- The term proportional to c_{jk} describes the Bragg conditions of the new reference lattice described by the vectors $\mathbf{l}$ and $\mathbf{d}$.
- The mixed terms involving c_{jk} times the sum of the other three terms in the square brackets will also affect the intensity in the Bragg peaks.
- The remaining terms give rise to diffuse scattering between the Bragg peaks.

The average distortion of the lattice due to disorder thus leads to a shift in the position of the Bragg peaks, while local distortions and compositional fluctuations are manifested both in the intensity of Bragg reflections and in the diffuse scattering between them. We also note that in our derivation of the differential cross section given by Equation (1.155), we have not assumed any implicit time dependence of the displacement terms. To treat this case, it becomes necessary to follow a similar methodology using the double differential cross section, followed by integration over final neutron energy. We shall return to this case in Section 1.4.5.

As a specific example of the information obtained from the results of a diffuse scattering experiment, we consider the case in which lattice distortions can be neglected. The differential cross section for diffuse scattering is then obtained from Equation (1.155) to read

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{c,el}}^{\text{diffuse}} = \frac{1}{N} \left| \sum_{nj\mathbf{k}} P_{nj\mathbf{k}} \langle \bar{b}_{\mathbf{k}} \rangle e^{i\mathbf{Q} \cdot (\mathbf{l}_n + \mathbf{d}_j)} \right|^2. \quad (1.156)$$

Representing compositional fluctuations by their Fourier transform in reciprocal space

$$\Delta P_{jk}(\mathbf{Q}) = \frac{1}{N_c} \sum_n \Delta P_{nj} \mathbf{e}^{i\mathbf{Q} \cdot \mathbf{r}_n} \quad (1.157)$$

leads to

$$\left(\frac{d\sigma}{d\Omega} \right)_{c,el}^{\text{diffuse}} = \frac{N_c}{J} \left| \sum_{jk} \langle \bar{b}_k \rangle \Delta P_{jk}(\mathbf{q}) \mathbf{e}^{i\mathbf{Q} \cdot \mathbf{d}_j} \right|^2, \quad (1.158)$$

where ΔP is given as function of the reduced wave vector $\mathbf{q} = \mathbf{Q} - \boldsymbol{\tau}$, where $\boldsymbol{\tau}$ is a reciprocal vector of the lattice. Thus, the diffuse scattering in this case is entirely defined by the spectrum of compositional fluctuations in reciprocal space.

On a practical front, it becomes clear that the study of diffuse scattering is best accomplished using single-crystal specimens. Discerning static versus dynamic contributions to the observed scattering patterns may be accomplished by performing experiments at different incident wavelengths and temperatures. The former task is easily achievable at pulsed spallation sources using a broad range of incident energies. Figure 1.15 shows neutron-scattering data for deuterated benzil, and illustrates the wealth of information afforded by diffuse neutron-scattering experiments on hydrogen-containing molecular crystals. Similar to what is observed with X-rays and indicative of strong longitudinal-displacement correlations in this crystal, the middle image in this figure contains a diffuse hexagon connecting the set of intense reflections (400), (040), ($\bar{4}\bar{4}0$), ($0\bar{4}0$), ($\bar{4}00$), and ($\bar{4}40$) arising from Bragg scattering by 1.169 Å (59.8 meV) neutrons. As shown in the top image, this feature disappears from the data when the incident neutron wavelength is increased to 3.445 Å (6.9 meV), signaling a strong contribution of inelastic events to the former dataset. The bottom image shows broad regions of diffuse scattering, quite prominent at high momentum transfers and indicative of intramolecular motions—cf. Figure 1.10 and our previous discussion of inelastic form factors for one- and multiphonon scattering in Section 1.3.5. This particular case therefore represents a situation where analysis of scattered intensities away from Bragg reflections gives partial access to the double differential cross section. In some cases like polycrystalline MgO, it has been shown that detailed information on lattice dynamics for harmonic solids can also be obtained from total scattering data in a model-independent manner [142,143].

1.4.5 Stochastic Diffusion

Our treatment of the canonical solid presented in Section 1.3 made the tacit assumption that all atoms occupy well-defined equilibrium positions. In this case, the double differential neutron-scattering cross section is conveniently discussed in terms of *inelastic* scattering (i.e., exchange of energy and

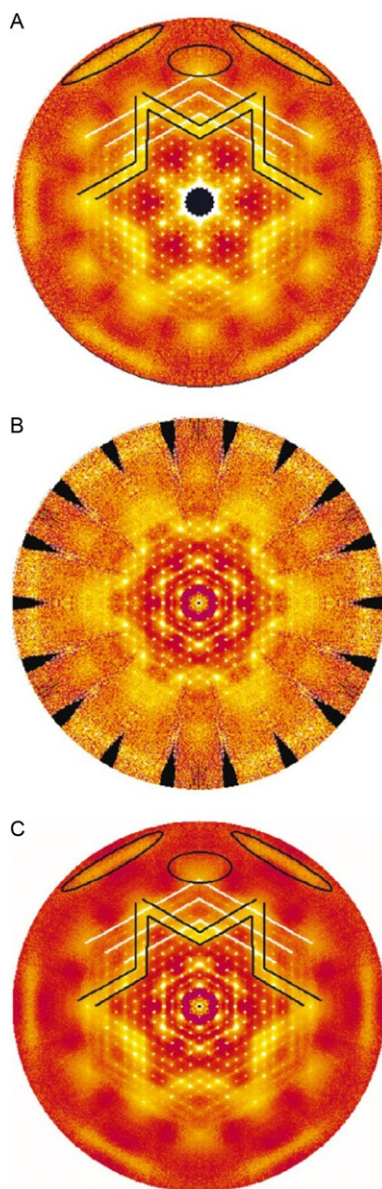


FIGURE 1.15 $(hk0)$ reciprocal sections of deuterated benzil (1,2-diphenylethane-1,2-dione; $C_{14}D_{10}O_2$) measured on a pulsed-source single-crystal diffractometer at several scattering angles: (a) $+142.5^\circ$; (b) $+37.5^\circ$; and the sum over detectors located at $\pm 37.5^\circ$, $\pm 90^\circ$, and $\pm 142.5^\circ$. In (c), only incident neutron energies above 27 meV have been considered in the data analysis. The dark lines in (a) and (c) emphasize prominent regions of diffuse intensity. Reprinted with permission from Ref. [141].

momentum with well-defined excitations in the solid), and *elastic* scattering (exchange of momentum with the entire solid recoiling as a rigid unit). Physically speaking, elastic scattering describes the response from particles at their equilibrium sites over the duration of the experiment (usually taken to be infinitely long relative to other characteristic timescales intrinsic to the material under investigation). In real solids, this is not generally the case. In addition to thermal motion around equilibrium sites (giving rise to a Debye–Waller factor, see Equation (1.59) in Section 1.3.2), atoms may occasionally move or diffuse between such sites. The ability to define a characteristic timescale for these diffusive motions immediately translates into the emergence of a spectral broadening of the elastic line in the energy domain. The energy widths of these *quasielastic* features constitute a direct measure of diffusion phenomena. If these processes are sufficiently fast, quasielastic spectral widths can be resolved experimentally and analyzed in terms of its details at the atomic scale. Recalling the space–time diagram shown by Figure 1.8 in Section 1.1.5.1, neutron scattering is quite unique as it provides simultaneous access to the spatial and temporal characteristics of diffusive motions via a study of the momentum and energy-transfer dependence of the double differential cross section.

Illustrations of this phenomenon in the solid state include metal hydrides, characterized by rapid translational diffusion of hydrogen atoms within the lattice, or fast-ion conductors, where the conducting ions are highly mobile. Lithium diffusion in LiC_6 and LiCoO_2 (materials sitting at the heart of modern lithium-ion-battery technology) are good examples of the latter. Likewise, in certain crystals displaying orientational disorder (so-called “plastic crystals”) [144], molecules undergo rapid reorientations between a finite number of quasi-equilibrium configurations. This phenomenon corresponds to the rapid transport of individual atoms between quasi-equilibrium sites and, thus, also gives rise to quasielastic features in the neutron-scattering response. On a more fundamental front, molecular plastic crystals represent an interesting intermediate situation between liquids and canonical solids, where specific intermolecular normal modes of vibration become unstable and stochastic in nature prior to the onset of the liquid state *per se* [135,145,146]. In this case, quasielastic features in the neutron-scattering response are best understood as the transition between well-defined (underdamped) oscillatory motions characterized by an energy E much larger than its associated spectral width Γ such that $E \gg \Gamma$, to overdamped modes with $E \ll \Gamma$.

Liquids represent an extreme case where the concept of equilibrium sites breaks down altogether. In this situation, translational diffusion of the center of mass of a given particle happens in a more or less continuous manner, and quasielastic scattering is always observed. Moreover, vibrational excitations are strongly damped and the quasielastic and the inelastic regions of the dynamic structure factor can no longer be separated unambiguously. In

certain cases and, in particular, at small values of the neutron momentum transfer, the quasielastic component can be isolated and used to derive distinct information about the diffusion process.

Chapter 6 provides a series of recent examples on the use of quasielastic neutron-scattering techniques to investigate stochastic dynamics and relaxation in solids and liquids. Detailed treatments can also be found in the monographs by Bee [147] and Hempelmann [148].

1.4.5.1 Liquids and Dense Gases

To describe particle diffusion, it is convenient to use the space-time formalism outlined in Section 1.4, that is, the space-time (van Hove) correlation functions [94]. In this case, the solution to the diffusion equation is obtained in $(\mathbf{r}, t)$ -space and the dynamics of the diffusing particle is treated as a one-particle process. The corresponding Van Hove self-correlation function is often written within the so-called Gaussian approximation [149]

$$G_s(\mathbf{r}, t) = [4\pi\gamma(t)]^{-3/2} \exp[-r^2/4\gamma(t)], \quad (1.159)$$

where, for a classical system, $\gamma(t)$ is the mean-square displacement of a particle relative to its position at $t=0$:

$$\gamma(t) = \frac{1}{6} \langle r^2(t) \rangle = \frac{1}{6} \int r^2 G_s(\mathbf{r}, t) d\mathbf{r}. \quad (1.160)$$

Recalling the properties of Gaussian distributions, the intermediate scattering function corresponding to Equation (1.159) is also a Gaussian (this time in Q space):

$$I_s(\mathbf{Q}, t) = \exp[-Q^2\gamma(t)]. \quad (1.161)$$

In this approximation, $I_s(\mathbf{Q}, t)$, and thus $S_i(\mathbf{Q}, E)$, are completely specified by the time dependence of the mean-square displacement. We note that the Gaussian approximation is rigorously valid in certain cases only, namely, for a harmonic solid, for an ideal gas, and for the *long-time* behavior of an atom diffusing in a liquid [149,150]. It is, nonetheless, used extensively for other cases as well. To illustrate the formalism presented above, we consider two simple cases, namely, the free-flight motion of a particle in a dilute gas and the Brownian motion of an atom in a dense fluid.

For particles of mass M in a dilute gas, the mean-square displacement is given by

$$\gamma(t) = \frac{1}{6} \langle v^2 \rangle t^2 = \frac{k_B T}{2M} t^2, \quad (1.162)$$

and the scattering function is

$$S_i(\mathbf{Q}, E) = \left[\frac{M}{4\pi k_B T \hbar^2 Q^2} \right]^{1/2} \exp \left[-\frac{ME^2}{2k_B T \hbar^2 Q^2} \right], \quad (1.163)$$

that is, a Gaussian profile in energy transfer with a full-width-at-half-maximum given by

$$\Gamma(Q) = 2\sqrt{2 \ln 2} \hbar Q \left(\frac{k_B T}{M} \right)^{1/2}. \quad (1.164)$$

In dense fluids, such as in the liquid state, the atoms are in continuous interaction with their neighbors. As such, these interactions can be represented as an average fluctuating force on the particle. In the classical Langevin theory of Brownian motion, the corresponding mean-square displacement reads:

$$\gamma(t) = D \{ t - \tau [1 - \exp(-t/\tau)] \}, \quad (1.165)$$

where

$$D = \frac{k_B T}{\eta M} = \frac{k_B T}{M} \tau \quad (1.166)$$

is the self-diffusion coefficient, η is the viscosity, and $\tau = 1/\eta$ is a measure of the rate of relaxation of velocity correlations in the fluid. From an expansion of the exponential function in Equation (1.165) and using Equation (1.166), it can be shown that Equation (1.165) is equal to the corresponding result for a particle in a dilute gas in the limit $t \ll \tau$. In the other limit, $t \gg \tau$, we obtain

$$\gamma(t) = D(t - \tau), \quad (1.167)$$

which should be compared to the result obtained from the solution of the macroscopic diffusion equation, namely

$$\gamma(t) = Dt, \quad (1.168)$$

that is, the inclusion of the friction term introduces a delay in the onset of purely diffusive behavior.

When inserted into Equation (1.161), $\gamma(t)$ in Equations (1.167) and (1.168) both yield the same result for the scattering function, namely

$$S_i(\mathbf{Q}, E) = \frac{1}{\pi} \frac{\hbar D Q^2}{E^2 + (\hbar D Q^2)^2}, \quad (1.169)$$

which is a Lorentzian with a half-width-at-half-maximum

$$\Gamma(Q) = \hbar D Q^2. \quad (1.170)$$

At small values of Q , where the above models are valid, the width of the quasielastic line is thus proportional to the diffusion coefficient. Typical diffusion coefficients in liquids lie in the range $\sim 0.1 \text{ \AA}^2 \text{ ps}^{-1}$, resulting in a spectral half-width-at-half-maximum Γ of $\sim 60 \text{ \mu eV}$ for $Q \approx 1 \text{ \AA}^{-1}$. These spectral

widths are routinely (and quite uniquely) accessible in scattering experiments using cold-neutron spectrometers.

Strictly speaking, the attentive reader will note that the incoherent scattering function shown in Equation (1.169) does not have a well-defined second moment and, therefore, is in clear violation of the moment rule given by Equation (1.117) in Section 1.4.3. Although this deficiency has little consequence in most practical applications, we note that it arises from considering only the dynamics at long times. To illustrate the origin of this behavior, we recall that the diffusion coefficient D can be expressed in terms of a time integral of the velocity autocorrelation function [151]

$$D = \int_0^\infty \langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle dt = \int_0^\infty Z(t) dt, \quad (1.171)$$

where for times much longer than the correlation time of stochastic forces we can write

$$Z(t) = D \left(\frac{\Gamma_v}{\hbar} \right) e^{-t(\Gamma_v/\hbar)}, \quad (1.172)$$

where Γ_v is a friction or damping constant expressed in energy units. The corresponding spectral density in the energy domain is then given by

$$Z(E) = D \frac{1}{\pi} \frac{\Gamma_v}{E^2 + \Gamma_v^2}. \quad (1.173)$$

As in the case of the dynamic structure factor, this spectral density is clearly ill-defined in terms of its second moment and therefore can only be valid for small energies E . One way to overcome this deficiency is to allow for an energy-dependent damping constant $\Gamma_v(E)$ that satisfies the fluctuation-dissipation theorem [152]. Using analytic-continuation techniques onto the complex plane, a commonly used functional form with a well-defined second (but not fourth!) moment reads

$$\Gamma_v(E) = \Gamma_{v0} \frac{\Gamma_{v1}}{\Gamma_{v1} + iE}, \quad (1.174)$$

where we note that $\Gamma_v(E) \rightarrow \Gamma_{v0}$ as $E/\Gamma_{v1} \rightarrow 0$. Therefore, only experiments seeking to probe *both* short- and long-time regimes will require such a generalization of the damping term. One such case is the extraction of the spectral density of the velocity autocorrelation function $Z(E)$ from neutron data using the relation [153]

$$E^2 \left[\frac{S_i(Q, E)}{Q^2} \right]_{Q \rightarrow 0} \rightarrow Z(E) \quad (1.175)$$

which, in combination with Equation (1.174), leads to a spectral density of the form

$$\left[\frac{E^2}{Q^2} S_i(Q, E) \right]_{Q \rightarrow 0} \rightarrow \frac{\Gamma_{v0} \Gamma_{v1}^2}{(E^2 + \Gamma_{v1}^2) E^2 + \Gamma_0^2 \Gamma_{v1}^2} \quad (1.176)$$

More elaborate treatments of diffusion in liquids than what is captured by Equations (1.169) and (1.170) take into account a stepwise transport between quasi-equilibrium sites, including a description of pseudo-oscillatory behavior and residence time of a given atom temporarily trapped in a solvation cage [102,154]. In general, these models lead to Lorentzian-shaped scattering functions with a model-dependent $\Gamma(Q)$.

For bulk liquids, a commonly used extension of the Fickian limit described by Equation (1.170) considers a first-order correction to the high- Q limit describing a residence time τ_0 between translational jumps. In this jump-diffusion model of translational stochastic motions $\Gamma_T(Q)$ is given by [153]

$$\Gamma_T(Q) = \hbar \left[\frac{D_T Q^2}{1 + D_T Q^2 \tau_0} \right] \quad (1.177)$$

where it is easily verified that $\Gamma_T(Q)$ tends asymptotically to either $\hbar D_T Q^2$ or $\hbar/\tau_0$ at low and high momentum transfers Q , respectively. An experimental determination of energy widths as a function of Q can then be used to test the validity of these theoretical models for translational diffusion. Figure 1.16 shows the momentum-transfer dependence of quasielastic spectral widths for incoherent (proton-dominated) scattering from liquid hydrogen fluoride as a function of temperature. The spectroscopic data are represented by recourse to a linearized form of Equation (1.177) reading

$$\Gamma_T^{-1}(Q) = \frac{1}{\hbar} [D_T^{-1} Q^{-2} + \tau_0] \quad (1.178)$$

In this form, the jump-diffusion correction to Fick's law becomes an additive constant, and the slope as a function of Q^{-2} is inversely proportional to the *absolute value* of the microscopic self-diffusion coefficient D_T .

In molecular liquids such as in the example given above, in addition to the diffusive transport of the centre of mass, molecules can also rotate—the specific case of stochastic rotational motions in solids is also discussed in the next section. In the liquid phase, both translational and rotational motions determine the observed broadening. A first-order theoretical treatment of the resulting dynamic structure factor under the assumption of no couplings between translational and rotational modes in the time domain leads to the successive convolution (or folding) of the corresponding spectral functions in the energy domain. We can illustrate the approach by considering in more detail the case of hydrogen fluoride presented above.

As a heteronuclear diatomic molecule containing a single hydrogen atom, the incoherent self-intermediate scattering function in the time domain must include the contribution from center-of-mass translations of the whole molecule (assumed rigid) and rotation of the hydrogen atom about the molecular

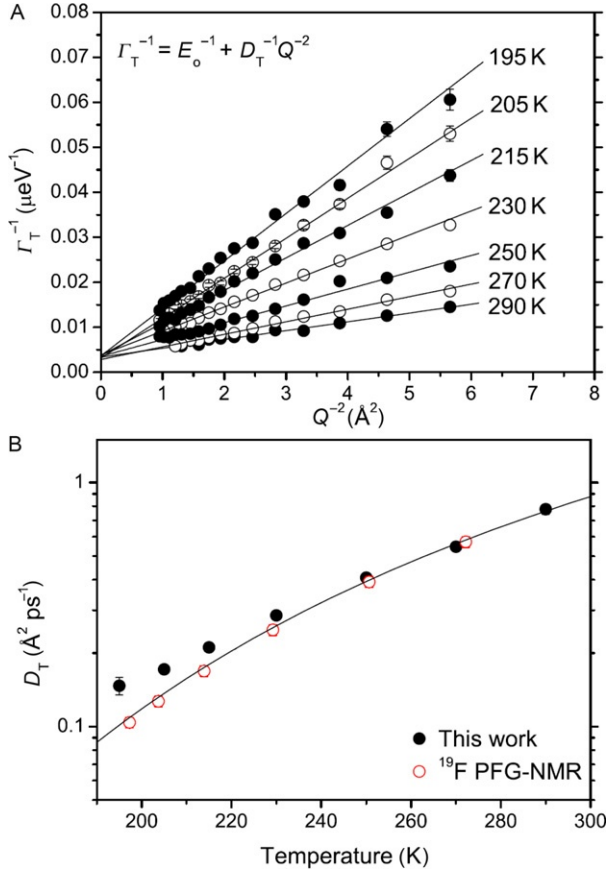


FIGURE 1.16 Quasielastic neutron-scattering data for liquid hydrogen fluoride as a function of temperature: (a) inverse spectral widths $\Gamma_T^{-1}(Q)$ for translational diffusion as a function of Q^{-2} , as described in the text; (b) derived translational self-diffusion coefficients and comparison with ^{19}F PFG-NMR results. Reprinted with permission from Ref. [155].

center of mass. If these two motions are decoupled from each other, we can then write the following self-intermediate scattering function for hydrogen:

$$I_S(Q, t) = I_T(Q, t) I_R(Q, t) \quad (1.179)$$

or in the energy domain

$$S_i(Q, E) = S_T(Q, E) \otimes S_R(Q, E), \quad (1.180)$$

where $\otimes$ denotes a convolution product. $S_T(Q, E)$ is given by Equation (1.177), and describes translational motions of the *entire* molecule, as well as the possibility of high-frequency vibrational motions as introduced in Equation (1.63)

$$S_T(Q, E) = e^{-Q^2 \langle u^2 \rangle / 3} \left[\frac{1}{\pi E^2 + \Gamma_T(Q)^2} \right]. \quad (1.181)$$

We note that both $\langle u^2 \rangle$ and $\Gamma_T(Q)$ are accessible to experimental scrutiny via a measurement of intensities and spectral widths as a function of momentum transfer Q , respectively.

To describe stochastic rotational diffusion about the molecular centre of mass, a common expression to describe the isotropic case in terms of partial (angular momentum) waves is [156–158]

$$S_R(Q, E) = j_0^2(QR) \delta(E) + \sum_{l>0} (2l+1) j_l^2(QR) \left[\frac{1}{\pi E^2 + (l(l+1)\Gamma_R)^2} \right]. \quad (1.182)$$

where R is radius of gyration and j_l is a spherical Bessel function of order l . The rotational diffusion constant D_R is defined such that $\Gamma_R = \hbar D_R$. Inserting Equations (1.181) and (1.182) into (1.180), the incoherent dynamic structure factor becomes

$$S_i(Q, E) = e^{-Q^2 \langle u^2 \rangle / 3} \left\{ j_0^2(QR) \left[\frac{1}{\pi E^2 + \Gamma_T(Q)^2} \right] + \sum_{l>0} (2l+1) j_l^2(QR) \left[\frac{1}{\pi E^2 + \Gamma_{RT}^2(l; Q)} \right] \right\} \quad (1.183)$$

with $\Gamma_{RT}(l; Q) = \Gamma_T(Q) + l(l+1)\Gamma_R$. The first term in this expression corresponds to a purely translational contribution to the incoherent scattering function, and its form factor contains a $j_0^2(QR)$ term. In the particular case of hydrogen fluoride, it leads to the spectral widths shown in Figure 1.16a. We note, however, that *all* spectral components in Equation (1.183) have an intrinsic momentum-transfer dependence. For simple molecular liquids $\Gamma_R \gg \Gamma_T(Q)$ (particularly at low Q), and this dependence is far less marked in $\Gamma_{RT}(Q; l)$, usually leading to the approximation that it is Q independent. Moreover, the contribution of higher-order partial-wave terms beyond $l=1$ over typical momentum transfers is sufficiently small and it is generally omitted in the data analysis.

Equation (1.183) is also a convenient starting point to explore the effects of freezing translational motions, naturally leading to the formation of a plastic crystal (or an orientationally disordered glass). As $\Gamma_T(Q) \rightarrow 0$, the energy-dependent term in Equation (1.181) can be replaced by a delta function in E

$$S_T(Q, E) \rightarrow e^{-Q^2 \langle u^2 \rangle / 3} \delta(E) \quad (1.184)$$

and the total incoherent scattering law reduces to

$$S_i(Q, E) = e^{-Q^2 \langle u^2 \rangle / 3} \left\{ j_0^2(QR) \delta(E) + \sum_{l>0} (2l+1) j_l^2(QR) \left[\frac{1}{\pi E^2 + (l(l+1)\Gamma_R)^2} \right] \right\}. \quad (1.185)$$

The first term within the curly brackets represents a purely elastic term reflecting a vibrating molecule about its center of mass, that is, translationally speaking, the system behaves as a canonical solid, as introduced in [Section 1.3](#). Higher-order terms in the sum shown by Equation (1.185) describe rotational stochastic motions, and their associated spectral widths are now Q independent. A multitude of models for the dynamic structure factor are discussed in Refs. [147,148]. Diffusion in bound and confined media lead to important changes to the momentum-transfer dependence of quasielastic neutron-scattering spectra particularly at low energy transfers, as originally discussed in Refs. [159,160]. For a recent treatment of this particular case, see Refs. [161,162]. As covered in detail in [Chapter 6](#), computer simulation has become a key tool to interpret quasielastic neutron-scattering experiments, particularly for complex systems. Nonetheless, we emphasize that a physical interpretation of either experimental or computational data ultimately requires recourse to simplified models in order to gain the requisite level of insight.

In the preceding discussion, only the incoherent component to the scattering is considered and the results are directly related to self-correlation functions. Coherent scattering, on the other hand, is related to the full space-time correlation function. In this case, the theory is much less developed [153,154], yet still quasielastic neutron-scattering experiments have provided much-needed input to competing theories of the liquid state, see, for example, the elegant neutron experiments of Morkel *et al.* [163] to test the predictions of mode-coupling theories on liquid sodium. In certain cases, it is possible to separate the coherent and the incoherent scattering functions from measurements on samples with different isotopic compositions or via the use of polarization techniques (see [Section 1.5.2](#)). This has, for example, been done for liquid argon [97] as shown in [Figure 1.12](#) in [Section 1.4.2](#). In other cases, the Vineyard convolution approximation [149],

$$S_c(\mathbf{Q}, E) = S(\mathbf{Q})S_i(\mathbf{Q}, E), \quad (1.186)$$

or an *ad hoc* modification of the convolution approximation [164]

$$S_c(\mathbf{Q}, E) = S(\mathbf{Q})S_i(\mathbf{Q}', E), \quad (1.187)$$

with $\mathbf{Q}' = \mathbf{Q}\sqrt{S(\mathbf{Q})}$ is used to express the coherent scattering function in terms of the incoherent scattering function and the liquid structure factor. The observed scattering function can then again be analyzed for information about single-particle motions.

1.4.5.2 Disordered Solids

In certain solids, particle-diffusion rates are sufficiently large to enable neutron-scattering studies. For conventional spectrometers the condition is that $D \geq 0.01 \text{ \AA}^2 \text{ ps}^{-1}$. However, backscattering or spin-echo spectrometers can offer spectral resolutions below $1 \text{ \mu eV}$, and measurable diffusion

coefficients can be as low as $10^{-3} \text{ \AA}^2 \text{ ps}^{-1}$. Diffusion rates in these ranges are observed in fast-ion conductors and in certain metal hydrides, to name a few.

In analogy with some models for the liquid state, diffusion in the solid state may be regarded as taking place via discontinuous jumps between available quasi-equilibrium sites in the lattice. While residing at a given site, the particle undergoes high-frequency vibrations (describable via an appropriate Debye–Waller factor), and occasionally undergoes a transition to a nearby site. The appropriate rate equation for stepwise translational diffusion was first formulated by Chudley and Elliott [165]. Assuming instantaneous and statistically independent jumps across sites, the probability $P(\mathbf{r}, t)$ that the particle is at site $\mathbf{r}$ at time t is obtained from the following differential equation:

$$\frac{\partial}{\partial t} P(\mathbf{r}, t) = \frac{1}{n\tau} \sum_{k=1}^n [P(\mathbf{r} + \mathbf{d}_k, t) - P(\mathbf{r}, t)], \quad (1.188)$$

where τ is the residence time at a given site and $\mathbf{d}_k$ is the distance to the k th site in the lattice. It is assumed that the available sites form a Bravais lattice so that the set of jump vectors $\mathbf{d}_k$ is the same for all sites. With the condition that the particle starts at the origin at time zero

$$P(\mathbf{r}, 0) = \delta(\mathbf{r}), \quad (1.189)$$

we obtain

$$G_s^D(\mathbf{r}, t) = P(\mathbf{r}, t), \quad (1.190)$$

where $G_s^D(\mathbf{r}, t)$ is the Van Hove correlation function that describes the diffusion process. We will further assume that the diffusion process is decoupled from the thermal vibrations so that the scattering function corresponding to G_s^D may be derived independently.

Using an intermediate scattering function of the form

$$I_s^D(\mathbf{Q}, t) = \exp \left[-\frac{\Gamma(\mathbf{Q})}{\hbar} t \right], \quad (1.191)$$

and the integral relationship between I_s and G_s given in Section 1.4.1, the incoherent dynamic structure factor is of the form

$$S_i^D(\mathbf{Q}, E) = \frac{1}{\pi} \frac{\Gamma(\mathbf{Q})}{E^2 + \Gamma^2(\mathbf{Q})}, \quad (1.192)$$

with

$$\Gamma(\mathbf{Q}) = \frac{\hbar}{\tau n} \sum_{k=1}^n [1 - e^{-i\mathbf{Q} \cdot \mathbf{d}_k}]. \quad (1.193)$$

Equations (1.192) and (1.193) render themselves to a simple physical interpretation, as the scattering function is a Lorentzian with a width that is

determined by the jump distance $\mathbf{d}$ and jump rate τ . Nonetheless, it is at first sight somewhat surprising that the incoherent scattering should contain structural information. However, we note that this information is not obtained from wave interference from different particles (as in the case of coherent scattering), but rather from the interference of waves scattered from the same particle at different times. To illustrate the use of Equation (1.193), the width function for a simple cubic lattice with lattice parameter a is given by

$$\Gamma(\mathbf{Q}) = \frac{\hbar}{3\tau} (3 - \cos Q_x a - \cos Q_y a - \cos Q_z a), \quad (1.194)$$

where we note that for $|\mathbf{Q}| \rightarrow 0$, the half-width-at-half-maximum becomes

$$\Gamma(Q) = \hbar \frac{Q^2 a^2}{6\tau} \equiv \hbar D Q^2, \quad (1.195)$$

that is, the result derived above for continuous (Fickian) diffusion in a fluid is recovered in this limit. For systems in which the sites do not form a Bravais lattice, the scattering function is obtained from a coupled set of differential equations, each one representing a Bravais subset of lattice points, and the scattering function is then a sum of Lorentzians. An in-depth discussion of the mathematical formalism and physical models applicable to a wide range of solid-state systems can be found in the monograph by Hempelmann [148].

To illustrate the above concepts, Figure 1.17 shows neutron-spin-echo for the hydrated perovskite $\text{BaZr}_{0.90}\text{Y}_{0.10}\text{O}_{2.95}$, a proton-conducting material [166,167]. As mentioned in Section 1.2.6, spin-echo methods probe the intermediate scattering functions and can extend the dynamic range of neutron spectroscopy well into the nanosecond time domain. Notwithstanding the experimental challenges associated with the isolation of the spectral response associated with proton motions using spin-echo techniques, these data provide a direct measurement of stochastic diffusion over timescales associated with atomic transport at the mesoscale. Ongoing developments in neutron sources and instrumentation aim to make these challenging measurements routine in next-generation neutron facilities.

The theoretical approach outlined thus far for solid-state diffusion has only considered the incoherent scattering function. In the case of scattering from hydrogen (e.g., hydrides, proton conductors), the incoherent scattering dominates and only $S_i(\mathbf{Q}, E)$ is of practical interest. In the case of other fast-ion conductors, the mobile atoms often display significant coherent scattering and the treatment above is then not applicable. In some cases, notably Ag diffusion in the well-studied fast-ion conductor AgI [168], the Vineyard convolution approximation [149], or an *ad hoc* modification of the Vineyard approximation [164], has been used to evaluate the coherent scattering function [169]. When possible, comparison of incoherent and coherent experiments also provide information on the differences between self and collective diffusion coefficients and the underlying short- and long-range

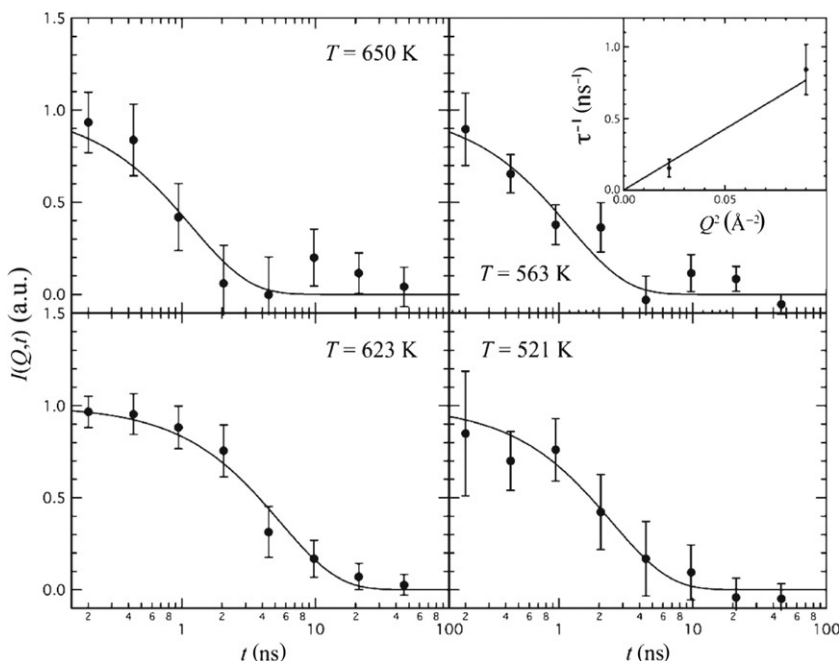


FIGURE 1.17 Intermediate scattering functions for hydrated $\text{BaZr}_{0.90}\text{Y}_{0.10}\text{O}_{2.95}$ at several temperatures and momentum transfers: $T = 650$ K ($0.20 \text{ \AA}^{-1}$), 623 K ($0.20 \text{ \AA}^{-1}$), 563 K ($0.30 \text{ \AA}^{-1}$), and 521 K ($0.30 \text{ \AA}^{-1}$). Solid lines are fits to an exponentially decay with a Q -dependent relaxation time τ . The inset illustrates the Q^2 dependence of $\tau(Q)$ at $T = 563$ K. Reprinted with permission from Ref. [166].

order associated with inter-species (also called “chemical”) diffusion. For a detailed discussion, see Chapter 6 in Ref. [148]. As example, quasielastic scattering data from diffusing ions in yttria-stabilized zirconia at elevated temperatures are shown in Figure 1.18. In this case, the observed spectral broadenings are related to the diffusion of O^{2-} ions in the lattice. Studies of this sort at elevated temperatures have been greatly facilitated by ongoing developments in levitated (containerless) techniques [171].

There is one more important class of materials that show quasielastic scattering, namely, certain molecular solids in which specific functional groups (particularly methyl groups) are free to reorient on the time scale of picoseconds. As reorientations take place, individual atoms will move about a given molecular axis occupying a sequence of quasi-equilibrium orientations. Rate equations such as the one given in Equation (1.188) can again be used to solve the dynamics of these reorientational jumps. In this case, however, there is a finite probability that the particle will visit a given site more than once and, therefore, the total probability density will remain finite in the limit $t \rightarrow \infty$. As shown by Equation (1.185), this feature gives rise to a truly elastic component to the scattering, in addition to a broad quasielastic line corresponding to

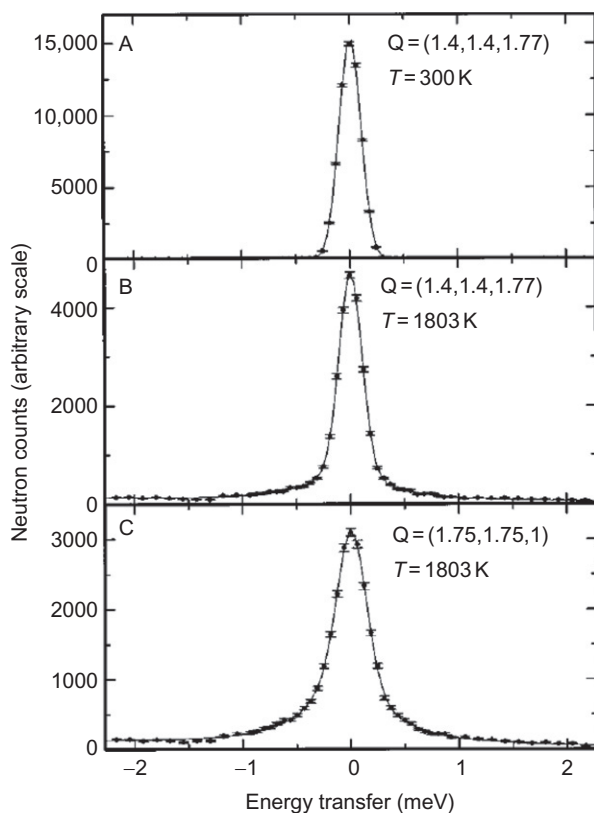


FIGURE 1.18 Quasielastic scattering spectra at constant Q from a single crystal of $(\text{ZrO}_2)_{0.87}(\text{Y}_2\text{O}_3)_{0.13}$ at room temperature and 1800 K. Reprinted with permission from Ref. [170].

the temporal redistribution of probability density between the sites. At sufficiently low temperatures, these reorientational motions can give rise to well-defined spectroscopic transitions associated with quantum-mechanical particle delocalization across sites, also called “tunneling transitions” [172,173]. These interesting dynamical effects are not only present in simple molecular solids involving a high-symmetry species such as solid methane or ammonium salts [173] but have also been observed in complex materials like polymers [174]. Interest in understanding low-energy excitations (uniquely accessible with neutron scattering) transcends a purely academic interest in solid-state spectroscopy, as it constitutes the basis for a growing number of neutron studies probing the interaction of technologically relevant molecules for gas storage and catalysis. These include methane [175], molecular hydrogen [176,177], or carbon dioxide [178] with surfaces and nanoporous media. Chapter 6 covers these in more detail in the context of the dynamics of atoms and molecules.

1.4.6 Beyond Atoms and Molecules: Large-Scale Structures

Neutron-scattering techniques can also be used to probe the properties of so-called “large-scale structures,” supramolecular collections of atoms of sizes up to several thousand times interatomic distances. Examples of such systems are found in a plethora of scientific and technological applications including precipitates and voids in solids, macro, and biomolecules in solution, polymer chains in solid and liquid solutions and thin films, food, and consumer products, etc. In principle, scattering experiments yield information on both size and shape of the relevant structural units (form factor), as well as on the relative arrangement of these units (structure factor).

The use of neutron scattering to study large-scale structures has experienced an impressive growth over the past three decades. The origin of the discipline dates back to the 1970s, when position-sensitive detectors became generally available for small-angle neutron-scattering (SANS) experiments [179]. Further progress in instrumentation in the 1980s led to the development of neutron-reflectometry (NR) techniques for the study of interfaces and layered materials. In both cases, the increased availability of cold moderators at major neutron sources (both reactor- and spallation-based) has greatly stimulated this area of neutron science. As a brief complement to Chapter 5, we limit ourselves to a general overview of the overarching principles, as well as to how these can be related to our earlier discussion of neutron scattering by atomic and molecular systems.

In ordinary Bragg diffraction (Section 1.1.4), intense and coherent elastic scattering is observed whenever the Bragg condition is satisfied:

$$\lambda_i = 2d \sin \frac{\phi}{2}. \quad (1.196)$$

This condition can also be written

$$Q = 2\pi/d, \quad (1.197)$$

where Q is the wave vector at the Bragg peak, namely,

$$Q = \frac{4\pi}{\lambda_i} \sin \frac{\phi}{2}. \quad (1.198)$$

We also recall that the Bragg condition results from the constructive interference of neutron waves scattered from individual atomic planes. These planes define a periodically modulated neutron-scattering-length density. Operationally speaking, a minimum scattering angle corresponding to the maximum plane spacing can always be derived from Equation (1.196). In principle, there are no physical restrictions on the length scales associated with variations in scattering-length density, and these may very well vary over distances that are significantly larger than typical interatomic distances. As implied by Equation (1.197), the scattering vector Q is inversely proportional

to the characteristic distance over which the scattering-length density fluctuates. Large objects thus give rise to scattering at small wave vectors and hence, in general, to scattering at small angles.

1.4.6.1 Small-Angle Scattering

The differential cross section for neutron scattering from individual atoms was introduced in [Section 1.3](#) and reads

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \left| \sum_{\mathbf{R}} \overline{b_{\mathbf{R}}} e^{i\mathbf{Q} \cdot \mathbf{R}} \right|^2. \quad (1.199)$$

For scattering at small wave vectors, $Q \ll \pi/d$, individual atoms are not resolved and the scattering process is effectively governed by the interference of neutron waves scattered from regions of linear dimensions l such that

$$l \approx \pi/Q. \quad (1.200)$$

In this case, the summation over discrete atoms in Equation (1.199) can be written as a spatial integral of the form

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \left| \int_V \rho_b(\mathbf{r}) e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r} \right|^2, \quad (1.201)$$

where $\rho_b(\mathbf{r})$ is the local scattering-length density, that is, a continuum approximation to the actual scattering-length density at a particular point. The spatial integration is over the entire volume of the target. We also express $\rho_b(\mathbf{r})$ in terms of its average value plus fluctuations around the average, namely,

$$\rho_b(\mathbf{r}) = \overline{\rho_b} + \delta\rho(\mathbf{r}). \quad (1.202)$$

The constant term $\overline{\rho_b}$ gives a contribution to the differential cross section at $Q=0$ only. For $Q>0$, the scattering is given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \left| \int_V \delta\rho(\mathbf{r}) e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r} \right|^2. \quad (1.203)$$

For the case of N particles of homogeneous scattering-length density ρ_p imbedded in a matrix of homogeneous scattering length density ρ_m , Equation (1.203) gives

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \left(\rho_p - \rho_m \right)^2 \left| \int_{V_p} e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r} \right|^2, \quad (1.204)$$

where the integral is over all particles in the system. If all the N_p particles are identical, the differential cross section becomes

$$\frac{d\sigma}{d\Omega} = \frac{V_p^2 N_p}{N} (\rho_p - \rho_m)^2 |F_p(\mathbf{Q})|^2, \quad (1.205)$$

where the last term is the single-particle form factor given by

$$P(\mathbf{Q}) = |F_p(\mathbf{Q})|^2 = \left| \frac{1}{V_p} \int_{V_p} e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r} \right|^2, \quad (1.206)$$

Since $F_p(0) = 1$ in the limit $Q \rightarrow 0$, we can write:

$$\left(\frac{d\sigma}{d\Omega} \right)_{Q \rightarrow 0} = \frac{V_p^2 N_p}{N} (\rho_p - \rho_m)^2. \quad (1.207)$$

For a sphere of radius R , the single-particle form factor $P(\mathbf{Q})$ displays the following momentum-transfer dependence:

$$P(\mathbf{Q}) = \{3[\sin QR - QR \cos QR] / Q^3 R^3\}^2. \quad (1.208)$$

The differential cross section thus shows characteristic maxima and minima, a result which is not only restricted to spherically shaped objects, as shown in Figure 1.19.

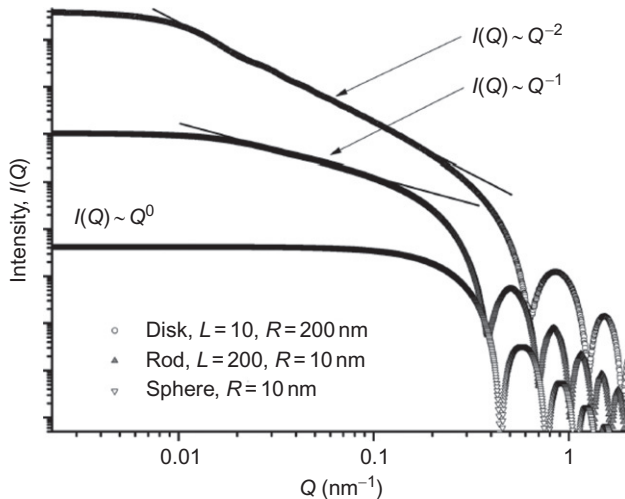


FIGURE 1.19 Simulated SANS profiles for monodisperse and dilute ensembles of disks (top trace), rods (middle), and spheres (bottom). The strong decay of the oscillations at high momentum transfers is the result of Porod's law. Disks (rods) exhibit a region where the integrated intensity $I(Q)$ scales as Q^{-2} (Q^{-1}). The spherical case (the smallest object of the trio) most clearly shows a flat plateau extending to low momentum transfers—this feature is actually present in all cases as long as $RQ < 1$. Characteristic radii of gyration can be obtained from a Guinier analysis of these data, as described in the main text. *Reprinted with permission from Ref. [180].*

In the general case of a single particle of arbitrary shape and orientation, it is not possible to express the scattering function in a simple closed form. Notwithstanding these methodological limitations, there are two limiting cases which can still provide useful information from the general behavior of the scattering function: the *Guinier approximation* [181], which applies at small values of Q such that $Ql \ll 1$, where l is the characteristic length over which the scattering-length density varies (the particle size); and the *Porod approximation* [182], which is valid at sufficiently large Q values such that $Ql \gg 1$.

When the Guinier approximation is applied to randomly oriented particles, the form factor is given by

$$P(\mathbf{Q}) = \exp(-Q^2 R_G^2/3), \quad (1.209)$$

where R_G , the radius of gyration, is obtained from

$$R_G^2 = \frac{1}{V_p} \int_{V_p} r^2 d\mathbf{r}, \quad (1.210)$$

and the integration is over the entire volume of the particle. Spherical particles of radius R give $R_G = \sqrt{3/5}R$ and the Guinier approximation is valid for $QR_G \leq 1.2$.

For particles of surface area A_p and homogeneous scattering length density, the Porod approximation predicts a scattering law proportional to Q^{-4} :

$$S(\mathbf{Q}) = 2\pi \left(\frac{A_p}{V_p^2} \right) \frac{1}{Q^4}. \quad (1.211)$$

Additional information can be obtained from integrated intensities. From Equations (1.202) and (1.203) we obtain

$$I_{\text{TOT}} = \frac{N}{V_p} \int \frac{d\sigma}{d\Omega} d\mathbf{Q} = (2\pi)^3 [\delta\rho^2(\mathbf{r})]_{\text{av}}, \quad (1.212)$$

where the average is taken over the whole system. For homogeneous particles imbedded in a homogeneous matrix, the integrated intensity is given by a slightly modified expression

$$I_{\text{TOT}} = (2\pi)^3 c_p (1 - c_p) (\rho_p - \rho_m)^2, \quad (1.213)$$

where $c_p = N_p V_p / V$. Thus, total scattered intensities provide a direct measure of volume fractions. Extrapolation to $Q=0$ (see Equation 1.207) gives $V_p^2 N_p / N = V_p c_p$. In combination with c_p obtained from the integrated intensity, it is then possible to determine the average particle volume V_p . An example of the determination of particle volumes by SANS in ionic-liquid-in-oil microemulsions is shown in Figure 1.20.

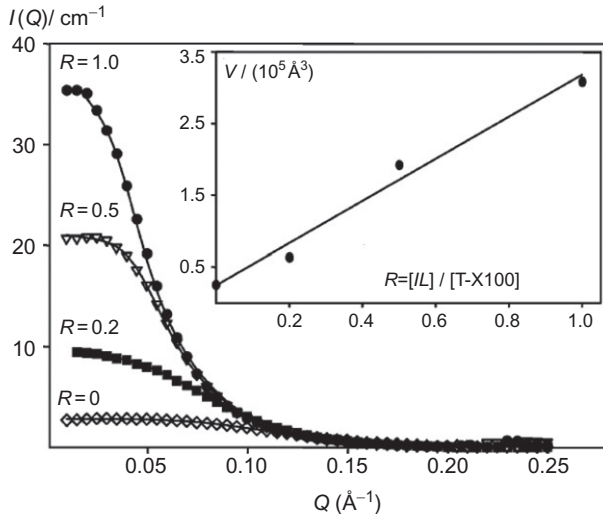


FIGURE 1.20 SANS from microemulsions of the ionic liquid $[h\text{-bmin}]\text{BF}_4$ in cyclohexane- d_{12} , stabilized by the nonionic surfactant T-X100 at 55°C . R is defined as the ratio of ionic liquid to nonionic surfactant, as shown in the inset. Fits to the SANS data have been performed using a form factor for homogeneous ellipsoids. The inset shows the particle volume as a function of R , indicating that substantial micelle swelling takes place as R is increased. Adapted with permission from Ref. [183].

Up until now, we have explicitly assumed low particle concentrations so that interparticle correlations can be safely neglected. In many cases, the relative arrangement of large structural units is important—for example, in colloidal solutions at high concentrations and in biological molecules where the arrangement of the subunits is often of vital importance for biological function (e.g., quaternary protein structure or DNA pairing). To obtain the differential cross section in the more general case, the system can be subdivided into N_p cells such that each cell contains exactly one particle. The position of the centre of the j th cell is given by $\mathbf{R}_j$ and the position of the k th particle in the cell relative to the centre of the cell is denoted by $\mathbf{d}_k$. With these definitions, the macroscopic differential cross section becomes

$$\frac{d\Sigma}{d\Omega} = \frac{1}{V} \left\langle \left| \sum_{j=1}^{N_p} e^{i\mathbf{Q}\cdot\mathbf{R}_j} \sum_{k=1}^{N_i} \overline{b_{jk}} e^{i\mathbf{Q}\cdot\mathbf{d}_k} \right|^2 \right\rangle, \quad (1.214)$$

where $\overline{b_{ij}}$ is the scattering length for particle j in cell i , and N_i is the number of atoms in the cell. Defining form factors for the individual cells as

$$F_j(\mathbf{Q}) = \sum_{k=1}^{N_i} \overline{b_{jk}} e^{i\mathbf{Q}\cdot\mathbf{d}_k}, \quad (1.215)$$

allows us to rewrite Equation (1.214) in the following form

$$\frac{d\Sigma}{d\Omega} = \frac{1}{V} \left\langle \sum_{j=1}^{N_p} \sum_{k=1}^{N_p} F_j(\mathbf{Q}) F_k(\mathbf{Q}) e^{i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_k)} \right\rangle. \quad (1.216)$$

For a system of identical particles with the same orientation, or spherical particles, the form factors are the same for all particles and Equation (1.216) can be written as

$$\frac{d\Sigma}{d\Omega} = \frac{N_p}{V} |F(\mathbf{Q})|^2 S(\mathbf{Q}), \quad (1.217)$$

where $S(\mathbf{Q})$ is now used to denote the *interparticle structure factor*

$$S(\mathbf{Q}) = \frac{1}{N_p} \left\langle \sum_{j=1}^{N_p} \sum_{k=1}^{N_p} e^{i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_k)} \right\rangle. \quad (1.218)$$

Equation (1.217) forms the basis for the determination of the relative arrangement of large structural units and it is used extensively in the study of the architecture underlying supramolecular chemical and biological systems. By selective deuteration, any chosen subunit can be made to match the scattering-length density of the matrix, which often is an aqueous solvent. If all but two subunits in a structure are made to match the solvent, $S(\mathbf{Q})$ will contain a term $S'(Q)$ describing the interference between scattering-length density centers of the two subunits separated by a distance l . This term is approximately given by

$$S'(Q) \approx j_0(Ql), \quad (1.219)$$

where $j_0(Ql) = \sin(Ql)/Ql$ is the already-familiar spherical Bessel function of the first kind with $l=0$. The zeros of this function occur at $Q = 2\pi n/l$ for any nonzero integer n , from which it is possible to obtain the distance between subunits. This approach has been used to determine the mesoscale structure of ribosomes in the seminal studies of Engelman *et al.* [184].

SANS techniques for the study of biological structure have evolved significantly in the past two decades [185]. Owing to the inherent complexity of these systems (e.g., presence of tertiary and quaternary structure in proteins, etc.), a common strategy involves the simultaneous analysis of neutron and X-ray small-angle data, as illustrated in Figure 1.21.

1.4.6.2 Reflection from Surfaces and Interfaces

Neutron reflectivity (NR) has become a powerful technique for the study of density fluctuations in surfaces and interfaces. As in optics, we can define a neutron refractive index n describing the change of neutron wave vector k_n in a medium relative to that *in vacuo*:

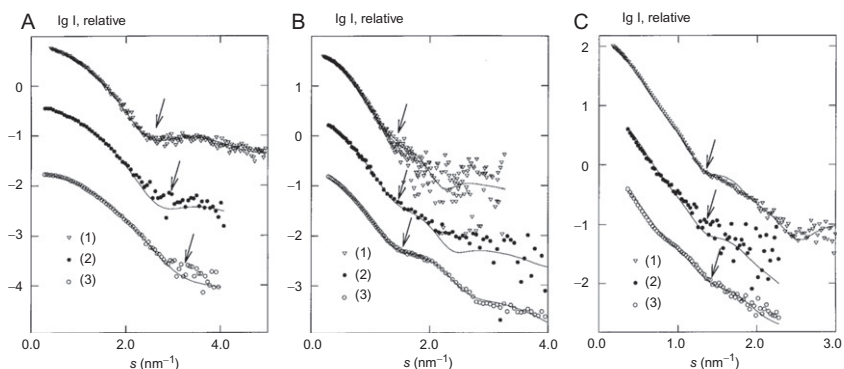


FIGURE 1.21 Small-angle scattering from proteins: (a) lysozyme; (b) *Escherichia coli* thioredoxin reductase; (c) and *Escherichia coli* ribonucleotide reductase protein R1. Solid lines are fits using the CRYSON and CRYSON programs. In each figure, labels (1–3) correspond to X-rays, neutrons with H₂O, and neutrons with D₂O, respectively. Reprinted with permission from Ref. [186].

$$n = 1 - \frac{1}{2\pi} \lambda_n^2 \rho \bar{b}, \quad (1.220)$$

where ρ is the number of scatterers per unit volume. As the coherent neutron-scattering length $\bar{b}$ for most materials is positive (hydrogen being an important exception), the associated neutron refractive index n is generally smaller than one and neutrons will be externally reflected by most materials (see also Section 1.5.5). This property is exploited in neutron guides to provide long experimental flight paths without reducing the acceptance angle and hence the outgoing intensity, as discussed in Chapter 3.

Total reflection from a surface or interface will take place at grazing angles of incidence θ below the critical angle θ_c . Values of θ_c are, however, generally small. In the case of total reflection, diffraction measurements can also be made by varying the azimuthal angle, giving information about the in-plane structure of the atoms at the surface. These *grazing-* or *glancing-incidence* measurements (sometimes called GISANS) open up the possibility of investigating the surface properties of nanostructured materials, including information about surface roughness and lateral correlations [187].

For values of $\theta > \theta_c$, information about the surface or interface structure can be obtained, including the study of buried interfaces not accessible using optical or electron microscopies. The specular component to the reflectivity is characterized by equal incident and reflected angles relative to the surface normal z and can be conveniently described within the kinematic approximation [188]. As in photon-based optics, this component to the reflectivity profile is confined to the plane defined by incident and reflected neutron beams. As such, it is related to the square of the Fourier transform of the scattering length density profile, $\rho(z)$, along z :

$$R(Q_z) = \frac{16\pi^2}{Q_z^2} \left| \int \rho(z) e^{-iQ_z z} dz \right|^2, \quad (1.221)$$

where Q_z is the wave-vector transfer perpendicular to the surface ($Q_z = (4\pi/\lambda) \sin \theta$, where θ is the angle of incidence).

The vast majority of NR studies performed to date have focused on an analysis of the specular reflectivity described by Equation (1.221), and many examples are given in [Chapter 5](#). Gaining information about in-plane structure (on the xy plane) requires the detection of scattered intensities away from the specular plane so as to access nonzero momentum transfers away from the surface normal. So-called off-specular NR can also be of a diffuse character when it is associated with disorder and surface roughness [189,190]. A comprehensive account of the formalism to describe neutron and X-ray reflectivity is given in Ref. [191], and more recent applications of nonspecular NR in soft matter have been summarized by Dalglish [192]. The study of magnetic large-scale structures is also an active and growing area of research, and it is discussed below in [Section 1.5.5](#).

In closing our discussion on large-scale structures, we emphasize that, in reality, there is no sharp distinction between SANS and the total-scattering techniques to probe shorter length scales ([Section 1.4.2](#)). Profiting from the availability of a wide range of incident neutron energies at spallation sources, a number of last-generation instruments [193–195] have been purposely designed to bridge the gap between microscopic and mesoscopic length scales, all within a single measurement.

1.5 MAGNETIC STRUCTURE AND POLARIZED NEUTRONS

This section covers the use of neutron scattering to study the spatial distribution of unpaired electron density in magnetic materials. This task is primarily achieved by means of neutron diffraction experiments. Such studies not only tell us about magnetism *per se* but also provide insights into the electronic structure of the material. Polarized neutrons are also included in this discussion, as they have been used extensively to discriminate between nuclear and magnetic neutron scattering.

1.5.1 Basic Principles

We outline the theory of neutron–magnetic interactions and associated scattering cross sections, so as to provide a suitable framework for the discussion of specific results in the sections to follow.

1.5.1.1 Magnetic Interactions and Cross Sections

In analogy to our discussion of the master formula and nuclear scattering in [Section 1.2](#), we now consider the interaction of a neutron (in spin state σ) with

a moving electron of momentum $\mathbf{p}$ and spin state $\mathbf{s}$ —note that Pauli operators $\boldsymbol{\sigma}$ are used to define the spin state of the neutron [196]. As introduced in Section 1.1.1, their corresponding magnetic moments are $\gamma\mu_N\boldsymbol{\sigma}$ and $g_e\mu_B\mathbf{s}$, where $\gamma = -1.913$, $g_e = -2.002319$ is the electron g -factor, and μ_N and μ_B are the nuclear and Bohr magnetons, respectively. In our treatment below, we shall make the implicit assumption that $g_e = -2$, an approximation of a similar nature to assuming m_n to be equal to the unified atomic-mass unit for nuclear scattering. While not conventional choices in the neutron literature, these definitions of the neutron-magnetic-moment-to-nuclear-magneton ratio γ and electron g -factor g_e conform to the latest recommendations on the use of units and physical constants [2]. Convincing physical arguments behind this sign convention have been presented in Ref. [197] for both atomic and molecular systems.

With these definitions in mind, the neutron–electron interaction potential is a distance- and momentum-dependent function of the form

$$V(\mathbf{r}) = -(2\gamma\mu_N\mu_B)\boldsymbol{\sigma} \cdot \left(\nabla \times \left(\frac{\mathbf{s} \times \hat{\mathbf{r}}}{r^2} \right) + \frac{\mathbf{p} \times \hat{\mathbf{r}}}{\hbar r^2} \right). \quad (1.222)$$

Physically speaking, this interaction potential is sensibly more intricate than the corresponding one for nuclear scattering involving the Fermi pseudo-potential, and given explicitly by Equation (1.27).

The double differential scattering cross section for a target containing a collection of unpaired electrons is obtained by substituting Equation (1.222) in the master formula, Equation (1.26). In the evaluation of matrix elements of the form $\langle \mathbf{k}_f \sigma_f \tau_f | V | \mathbf{k}_i \sigma_i \tau_i \rangle$, we first consider expectation values associated with the neutron plane-wave states $|\mathbf{k}_i\rangle$ and $|\mathbf{k}_f\rangle$, namely,

$$\langle \mathbf{k}_f | V | \mathbf{k}_i \rangle = 8\pi(\gamma\mu_N\mu_B)\boldsymbol{\sigma} \cdot \mathbf{D}_\perp(\mathbf{k}_i - \mathbf{k}_f), \quad (1.223)$$

where $\mathbf{D}_\perp(\mathbf{Q})$ is the *magnetic interaction operator*

$$\mathbf{D}_\perp(\mathbf{Q}) = \sum_j \left(\hat{\mathbf{Q}} \times (\mathbf{s}_j \times \hat{\mathbf{Q}}) + \frac{i}{\hbar Q} \mathbf{p} \times \hat{\mathbf{Q}} \right) e^{i\mathbf{Q} \cdot \mathbf{r}_j}. \quad (1.224)$$

To evaluate the matrix elements over neutron spin states $|\sigma_i\rangle$ and $|\sigma_f\rangle$, we first consider the case of *unpolarized* neutrons, where the corresponding matrix elements of products involving the neutron-spin operator $\boldsymbol{\sigma}$ satisfy the conditions

$$\sum_{\sigma_i} p_{\sigma_i} \sum_{\sigma_f} |\langle \sigma_f | \sigma_\alpha \sigma_\beta | \sigma_i \rangle|^2 = \delta_{\alpha\beta}, \quad (1.225)$$

and so the master formula becomes

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_\alpha \sum_{\tau_i} p_{\tau_i} \sum_{\tau_f} \langle \tau_i | \mathbf{D}_{\perp\alpha}^+ | \tau_f \rangle \langle \tau_f | \mathbf{D}_{\perp\alpha} | \tau_i \rangle \delta(E + E_{\tau_i} - E_{\tau_f}), \quad (1.226)$$

where

$$r_0 = \frac{e^2}{4\pi\epsilon_0 m_e c^2} = 2.8179 \text{ fm} \quad (1.227)$$

is the classical electron radius (of a similar magnitude as the range of nuclear interactions), and N_m represents the total number of magnetic centers. As in [Section 1.2.3](#), it is convenient to switch to a real-time representation of the double differential cross section

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_{\alpha} \frac{1}{2\pi\hbar} \int \langle \mathbf{D}_{\perp\alpha}(-\mathbf{Q}, 0) \mathbf{D}_{\perp\alpha}(\mathbf{Q}, t) \rangle e^{-iEt/\hbar} dt, \quad (1.228)$$

where the $\mathbf{D}_{\perp\alpha}(\mathbf{Q}, t)$ terms are magnetic interaction operations in the Heisenberg representation. Comparison of Equation (1.228) with the nuclear-scattering counterpart (cf. Equation 1.37) shows that the characteristic magnetic scattering cross section per electron is $4\pi(\gamma r_0)^2 = 3.652 \text{ b}$, and under suitable conditions magnetic scattering can therefore be of the same order of magnitude as nuclear scattering.

The magnetic interaction operators $\mathbf{D}_{\perp}(\mathbf{Q}, t)$ are inextricably linked to the magnetic properties of real systems, as they are projections of a generalized vector operator $\mathbf{D}(\mathbf{Q}, t)$ on a plane perpendicular to the neutron-momentum transfer $\mathbf{Q}$. We can always write:

$$\mathbf{D}_{\perp} = \hat{\mathbf{Q}} \times \mathbf{D} \times \hat{\mathbf{Q}} = \mathbf{D} - (\mathbf{D} \cdot \hat{\mathbf{Q}}) \hat{\mathbf{Q}}, \quad (1.229)$$

where $\mathbf{D}$ is related to the Fourier transform of the magnetization operator $\mathbf{M}$ via:

$$\mathbf{D}(\mathbf{Q}, t) = -\frac{\mathbf{M}(\mathbf{Q}, t)}{2\mu_B} = -\frac{1}{2\mu_B} \int \mathbf{M}(\mathbf{r}, t) e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r}. \quad (1.230)$$

Using Equation (1.229), Equation (1.230) can be reexpressed in terms of $\mathbf{D}(\mathbf{Q}, t)$ as

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{Q}_{\alpha} \hat{Q}_{\beta} \right) \frac{1}{2\pi\hbar} \int \langle D_{\alpha}(-\mathbf{Q}, 0) D_{\beta}(\mathbf{Q}, t) \rangle e^{-iEt/\hbar} dt. \quad (1.231)$$

Therefore, scattering probabilities depend on the magnetic fluctuations in the system, an entirely analogous situation to the way in which nuclear scattering depends on density fluctuations and discussed in [Section 1.4.1](#).

The directional dependence through the tensor $(\delta_{\alpha\beta} - \hat{Q}_{\alpha} \hat{Q}_{\beta})$ becomes a crucial and distinct feature of magnetic neutron scattering, as it “picks out”

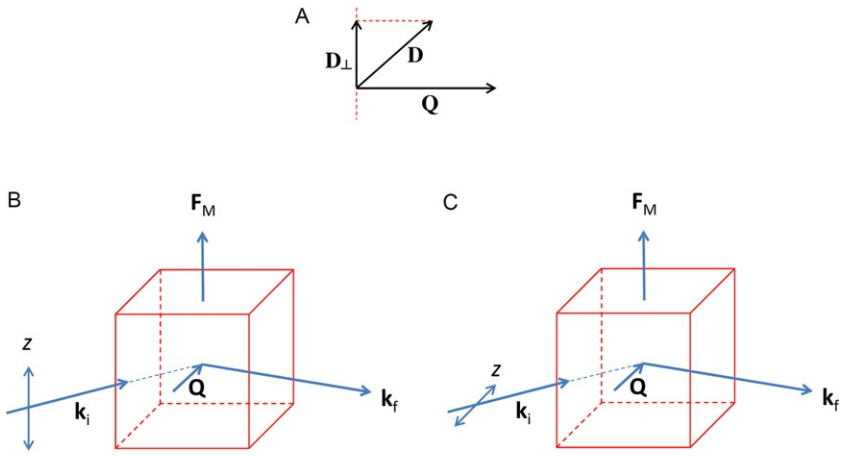


FIGURE 1.22 (a) Definition of the magnetic interaction operator $\mathbf{D}_\perp$. (b) Scattering geometry with the direction of neutron polarization z orthogonal to $\mathbf{Q}$. (c) Scattering geometry with z parallel to $\mathbf{Q}$.

the components of the magnetization orthogonal to the momentum transfer $\mathbf{Q}$, as shown in Figure 1.22a. For a spatially uniform magnetization, it is convenient to use a unit vector $\hat{\eta}$ ($\hat{\eta}$ is a vector quantity) in which case the directional dependence implied in Equation (1.231) is described by the factor

$$1 - (\hat{\mathbf{Q}} \cdot \hat{\eta})^2. \quad (1.232)$$

This factor makes it possible to deduce the orientation of the magnetic moments in a crystalline material relative to a given crystallographic direction.

The evaluation of $\mathbf{D}$ in terms of microscopic properties depends entirely on the nature of the magnetic electrons—for a detailed discussion, see Chapter 6 in Ref. [108]. For electrons of intrinsic spin and orbital angular momentum $\mathbf{s}$ and $\mathbf{l}$, respectively, localized on atoms at positions $\mathbf{R}_j$, and with positions $\mathbf{r}_n$ relative to their corresponding atomic centers,

$$\mathbf{D}(\mathbf{Q}, t) = \sum_j e^{i\mathbf{Q} \cdot \mathbf{R}_j} \sum_{n \in j} e^{i\mathbf{Q} \cdot \mathbf{r}_n} (\mathbf{s}_n + \mathbf{l}_n). \quad (1.233)$$

For electrons obeying Russell–Saunders (or LS) coupling, we can write

$$\mathbf{D}(\mathbf{Q}, t) = \sum_j f_j(\mathbf{Q}) \boldsymbol{\mu}_j(t) e^{i\mathbf{Q} \cdot \mathbf{R}_j(t)}, \quad (1.234)$$

where we define $\boldsymbol{\mu}_j = \frac{1}{2} g_j \mathbf{S}_j$ as the magnetic moment of the j th atom (in Bohr magnetons μ_B), and $\mathbf{S}_j$ its atomic-spin operator. Use of the symbol $\mathbf{S}$ above

adheres to convention, yet we note that it can be used (rather interchangeably!) to denote intrinsic spin, total electronic angular momentum $\mathbf{J}$, or (more generally) an effective angular-momentum operator for the case of partially quenched orbital angular momenta, the latter case requiring magnetization measurements on the system under investigation. With these definitions, $f_j(\mathbf{Q})$ is the corresponding form factor defined by

$$f(\mathbf{Q}) = \frac{g_S}{g} \bar{j}_0(\mathbf{Q}) + \frac{g_L}{g} [\bar{j}_0(\mathbf{Q}) + \bar{j}_2(\mathbf{Q})] \quad (1.235)$$

with

$$\bar{j}_n(\mathbf{Q}) = \int j_n(Qr) |\psi(r)|^2 dr, \quad (1.236)$$

where $\psi(r)$ is the electronic wave function, j_n 's are spherical Bessel functions, and g , g_L , and g_S correspond to g -factors associated with total, orbital, and intrinsic electronic spins, respectively. We also note that $f_j(\mathbf{Q})$ is normalized so that $f_j(0) = 1$. Thus, $\boldsymbol{\mu}_j(t)$ represents the magnitude and direction of the magnetic moment and $f_j(\mathbf{Q})$ the Fourier transform of its spatial distribution about the atom center.

Assuming that the magnetic moments associated with the electrons have a negligible effect on interatomic forces, the sums over electronic and nuclear coordinates of $\mathbf{D}$ in Equation (1.234) can be carried out separately and Equation (1.231) becomes

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE_f} &= \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta \right) \sum_{jj'} f_j^*(\mathbf{Q}) f_{j'}(\mathbf{Q}) \times \\ &\times \frac{1}{2\pi\hbar} \int \langle \mu_{j\alpha}(0) \mu_{j'\beta}(t) \rangle \left\langle e^{-i\mathbf{Q}\cdot\mathbf{R}_j(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt, \end{aligned} \quad (1.237)$$

which is the magnetic analog of Equation (1.38). It is important to note that the double differential scattering cross section shown in Equation (1.237) still contains correlation functions for the atomic positions, so that magnetic scattering is still sensitive to the structure and dynamics of these, as well as to magnetic properties.

1.5.1.2 Random Spins

The simplest case to apply the formalism introduced in the previous section corresponds to a paramagnetic solid with localized electrons at zero magnetic field. In this case, there are no correlations between spin orientations on different atoms, and therefore

$$\langle \mu_{j\alpha}(0) \mu_{j'\beta}(t) \rangle = \langle \mu_{j\alpha}(0)^2 \rangle \delta_{jj'} \delta_{\alpha\beta} = \frac{1}{12} g_j^2 S_j(S_j + 1) \delta_{jj'} \delta_{\alpha\beta}. \quad (1.238)$$

Since $\sum_{\alpha\beta} (\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta) = 2$, Equation (1.237) gives

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_j |f_j(Q)|^2 \frac{1}{6} g_j^2 S_j(S_j+1) \frac{1}{2\pi\hbar} \int \left\langle e^{-i\mathbf{Q}\cdot\mathbf{R}_j(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_j(t)} \right\rangle e^{-iE_f t/\hbar} dt, \quad (1.239)$$

which is equivalent to Equation (1.48) for incoherent nuclear scattering with the cross section σ_i replaced by

$$4\pi(\gamma r_0)^2 |f_j(Q)|^2 \frac{1}{6} g_j^2 S_j(S_j+1). \quad (1.240)$$

Apart from the magnetic form factors, there are no new features associated with the magnetic scattering from such a system. Equation (1.235) no longer applies in the presence of exchange interactions between different ions, or if there is an electrostatic interaction between magnetic and neighboring ions giving rise to a *crystal field* that lifts the M -sublevel degeneracy of the electronic ground state. Another way to lift this angular-momentum degeneracy is via the application of an external magnetic field B leading to characteristic Zeeman splitting of the magnetic sublevels. If B is applied in the z -direction, it will induce a nonzero average spin S_z , leading to coherent scattering with an effective scattering length of

$$(\gamma r_0) f_j(Q) \frac{1}{2} g_j \bar{S}_z. \quad (1.241)$$

We note that this quantity depends on the magnitude of $\mathbf{Q}$, not its direction.

1.5.1.3 Elastic Scattering from Spin Order

Spin-ordering phenomena imply nonzero correlations between magnetic moments on different atoms in the material. If these persist for long times (relative to the duration of the measurement), they necessarily lead to purely elastic scattering, just as the static order in nuclear positions discussed in Section 1.3. In this case the correlation function in Equation (1.228) becomes a time-independent quantity and the differential scattering cross section reads

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{el}} = \frac{1}{N_m} (\gamma r_0)^2 |\langle \mathbf{D}_\perp(\mathbf{Q}) \rangle|^2. \quad (1.242)$$

For a periodic structure, we define a time-dependent position vector for the j th spin as $\mathbf{R}_j(t) = \mathbf{l} + \mathbf{d} + \mathbf{u}_j(t)$, where the $\mathbf{l}$ vectors now have a periodicity corresponding to the *magnetic structure*. After averaging the electronic and nuclear parts independently, Equation (1.234) becomes

$$\langle \mathbf{D}(\mathbf{Q}) \rangle = \sum_l \sum_d f_d(\mathbf{Q}) \langle \mu_d \rangle e^{i\mathbf{Q}\cdot(\mathbf{l}+\mathbf{d})} e^{-W_d(\mathbf{Q})} = \frac{1}{|\gamma r_0|} \sum_l \mathbf{F}_M(\mathbf{Q}) e^{i\mathbf{Q}\cdot\mathbf{l}}, \quad (1.243)$$

with the *magnetic unit-cell structure factor* $\mathbf{F}_M(\mathbf{Q})$ is defined as

$$\mathbf{F}_M(\mathbf{Q}) = |\gamma r_0| \sum_d f_d(\mathbf{Q}) \langle \mu_d \rangle e^{i\mathbf{Q} \cdot \mathbf{d}} e^{-W_d(\mathbf{Q})}. \quad (1.244)$$

Summation over lattice vectors gives

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{el}} = \frac{1}{N_m} \frac{(2\pi)^3}{v_0} \sum_{\boldsymbol{\tau}_M} \delta(\mathbf{Q} - \boldsymbol{\tau}_M) |\mathbf{F}_{M\perp}(\boldsymbol{\tau}_M)|^2, \quad (1.245)$$

where

$$\mathbf{F}_{M\perp} = \hat{\mathbf{Q}} \times \mathbf{F}_M \times \hat{\mathbf{Q}} \quad (1.246)$$

is sometimes called the *magnetic structure factor*, $\boldsymbol{\tau}_M$ represent the reciprocal lattice vectors of the magnetic structure, and v_0 is the magnetic unit-cell volume. For the special case of itinerant electrons (e.g., magnetic metals), Equation (1.245) still holds provided that the unit-cell structure factor is replaced by

$$\mathbf{F}_M(\mathbf{Q}) = \hat{\boldsymbol{\eta}} |\gamma r_0| \left(\int_{V_0} |\varphi(\mathbf{r})|^2 e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r} \right) \left(\frac{n_{\uparrow} - n_{\downarrow}}{2} \right) e^{-W(\mathbf{Q})}, \quad (1.247)$$

where $|\varphi(\mathbf{r})|^2$ is the spin density in each cell and $n_{\uparrow}$ and $n_{\downarrow}$ are the average numbers of up and down spins. In general, $\hat{\boldsymbol{\eta}}$ will itself be a function of $\mathbf{Q}$ if the moments do not all point in the same direction. A detailed discussion of the physics behind Equation (1.247) is given in Sections 5.5 and 7.2 of Ref. [108].

Comparison of the nuclear-scattering structure factor with Equation (1.237) or (1.244) indicates that the magnetic analog provides information on both the magnitude and direction of magnetic moments on each site, and the form factors reflect the distribution of unpaired electron density in the material. Both of these features provide a strong motivation for carrying out magnetic diffraction studies.

1.5.2 Polarized Neutrons

In our discussion up until this point, the spin state (or polarization) of both incident and scattered neutron beams has been assumed to be evenly distributed between the two possible $\pm 1/2$ eigenstates for a spin-1/2 particle. Furthermore, Section 1.2.2 and 1.5.1 have presented differential scattering cross sections by performing an explicit average over all possible spin states. For many problems involving magnetic scattering (and some involving nuclear scattering), it is nonetheless advantageous to have the additional variable of neutron polarization at one's disposal so as to distinguish between different types of scattering. To this end, it becomes necessary to retrace the formalism of the previous sections taking explicit account of initial and final polarizations.

1.5.2.1 Nuclear Scattering

The dependence of nuclear scattering on neutron spin requires defining scattering-length operators of the form:

$$\hat{b}_j = \bar{b}_j + \frac{1}{2} b_{Nj} (\boldsymbol{\sigma} \cdot \mathbf{I}_j) \quad (1.248)$$

where $\mathbf{I}$ and $1/2\boldsymbol{\sigma}$ are the spin angular momentum operators for the j th nucleus and neutron, respectively. If the eigenvalues of $\hat{b}_j$ are $\hat{b}_j^\pm$ for the compound neutron–nucleus system of total angular momentum $I_j \pm 1/2$, respectively, the corresponding (spin-dependent) scattering lengths are given by

$$\begin{aligned} \bar{b}_j &= \frac{1}{2I_j + 1} \left[(I_j + 1) b_j^+ + I_j b_j^- \right], \\ b_{Nj} &= \frac{2}{2I_j + 1} \left[b_j^+ - b_j^- \right]. \end{aligned} \quad (1.249)$$

The *spin-dependent scattering length* b_N defined above is proportional to the so-called *incoherent scattering length* (see the [Appendix](#)). Expectation values of the scattering-length operators over the possible polarization states of incoming and outgoing neutrons can be obtained by recourse to angular-momentum algebra:

$$\begin{aligned} \langle \uparrow | \hat{b}_j | \uparrow \rangle &= \bar{b}_j + \frac{1}{2} b_{Nj} I_{jz}, \\ \langle \downarrow | \hat{b}_j | \downarrow \rangle &= \bar{b}_j - \frac{1}{2} b_{Nj} I_{jz}, \\ \langle \downarrow | \hat{b}_j | \uparrow \rangle &= \frac{1}{2} b_{Nj} (I_{jx} + i I_{jy}), \\ \langle \uparrow | \hat{b}_j | \downarrow \rangle &= \frac{1}{2} b_{Nj} (I_{jx} - i I_{jy}), \end{aligned} \quad (1.250)$$

where down/up arrows denote the state with spin anti/parallel to the quantization (z) axis. The upper two cases correspond to non-spin-flip (nsf) and the lower two to spin-flip (sf) events.

For a single-element specimen consisting of randomly mixed isotopes a with concentrations c_a and randomly oriented spins I_a , the corresponding coherent scattering lengths are given by

$$\begin{aligned} \bar{b}_{\text{nsf}} &= \sum_a c_a \bar{b}_a = \sum_a \frac{c_a}{2I_a + 1} \left[(I_a + 1) b_a^+ + I_a b_a^- \right] = \bar{b}, \\ \bar{b}_{\text{sf}} &= 0, \end{aligned} \quad (1.251)$$

where $\bar{b}$ is the mean scattering length for unpolarized neutrons (see Equation 1.44). Thus, coherent scattering takes place without a neutron-spin flip. The corresponding incoherent cross sections are given by

$$\sigma_{i,\text{nsf}} = 4\pi \sum_a c_a \bar{b}_a^2 - 4\pi \left(\sum_a c_a \bar{b}_a \right)^2 + \frac{\pi}{3} \sum_a c_a I_a (I_a + 1) b_{\text{Na}}^2 = \sigma_i - \sigma_{i,\text{sf}},$$

$$\text{with } \sigma_{i,\text{sf}} = \frac{2\pi}{3} \sum_a c_a I_a (I_a + 1) b_{\text{Na}}^2 = \frac{8\pi}{3} \sum_a c_a \frac{I_a (I_a + 1)}{(2I_a + 1)^2} [\bar{b}_a^+ - \bar{b}_a^-]^2.$$
(1.252)

Thus, if there are isotopes with $b_a^+ \neq b_a^-$, there is a nonzero spin-flip scattering component equal to two-thirds of the spin-incoherent scattering. In this case, polarization analysis can be used to change the ratio of coherent to incoherent scattering and thus separate out both scattering functions. With the appropriate substitutions of Equations (1.251) and (1.252), the formalism in Section 1.2 can be taken over.

Figure 1.23 shows a recent example where neutron-polarization analysis has been used to separate the coherent and incoherent response in hemoglobin and myoglobin solutions. The data above $Q = 1 \text{ \AA}^{-1}$ shown in Figure 1.23c display a similar behavior to that discussed earlier in relation to the incoherent and coherent response from simple liquids (cf. Section 1.4.2 and Figure 1.12), with a prominent peak at $2.0 \text{ \AA}^{-1}$ associated with solvent-solvent correlations. At lower energy transfers, coherent-scattering features are ascribed to intramolecular helix-helix separations within the protein ($0.6\text{--}0.7 \text{ \AA}^{-1}$) and protein-protein distances ($0.10\text{--}0.15 \text{ \AA}^{-1}$). Also note the predominance of coherent scattering in the low-angle region below $0.05 \text{ \AA}^{-1}$.

1.5.2.2 Magnetic Scattering

For magnetic scattering, it is necessary to evaluate the neutron-spin dependence of the matrix elements associated with the interaction potential (cf. Equation 1.223). In analogy with Equation (1.250) for spin-dependent nuclear scattering, we define

$$\begin{aligned} \langle \uparrow | \sigma \cdot \mathbf{D}_\perp | \uparrow \rangle &= D_{\perp z}, \\ \langle \downarrow | \sigma \cdot \mathbf{D}_\perp | \downarrow \rangle &= -D_{\perp z}, \\ \langle \downarrow | \sigma \cdot \mathbf{D}_\perp | \uparrow \rangle &= D_{\perp x} + iD_{\perp y}, \\ \langle \uparrow | \sigma \cdot \mathbf{D}_\perp | \downarrow \rangle &= D_{\perp x} - iD_{\perp y}. \end{aligned}$$
(1.253)

The matrix elements can then be used to compute differential cross sections using the master formula.

1.5.2.3 The Localized Paramagnet

As a first illustration of polarization analysis in magnetic systems, we use the double differential cross section for non-spin-flip magnetic scattering given by

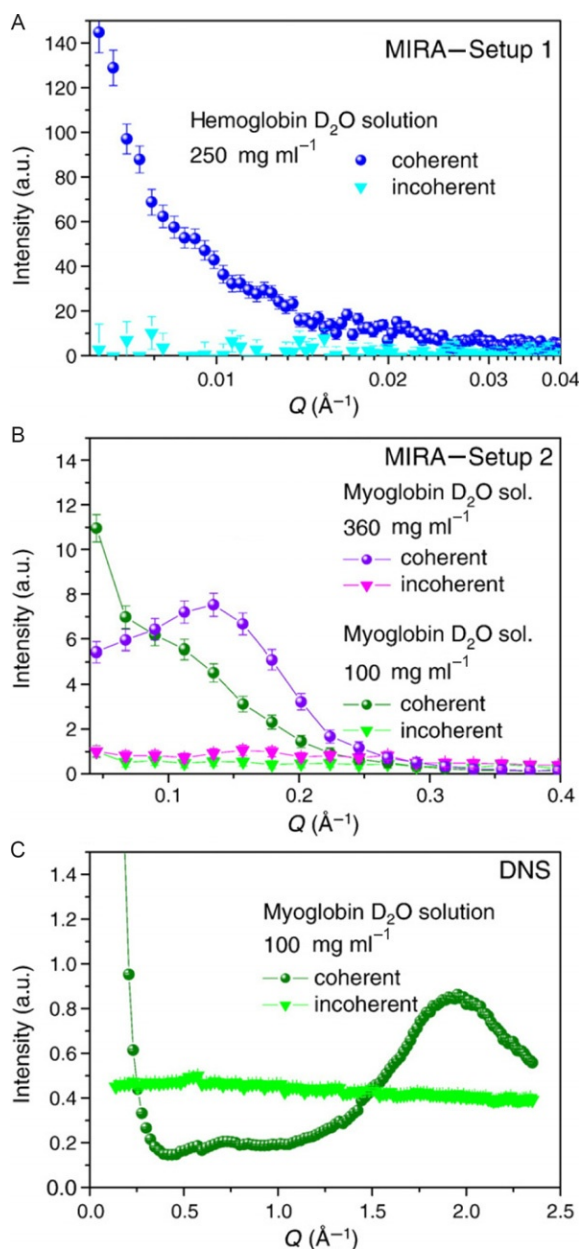


FIGURE 1.23 Incoherent- and coherent-scattering intensities of a series of protein samples: (a) low; (b) intermediate; and (c) intermediate and high momentum transfers Q . Reprinted with permission from Ref. [198].

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_{\tau_i} p_{\tau_i} \sum_{\tau_f} |\langle \tau_f | D_{\perp z} | \tau_i \rangle|^2 \delta(E + E_{\tau_i} - E_{\tau_f}). \quad (1.254)$$

For a paramagnetic solid with localized electrons in zero magnetic field, we may integrate over energy, and sum over initial (τ_i) and final target states (τ_f), to obtain

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{nsf}} = \frac{1}{N_m} (\gamma r_0)^2 \langle D_{\perp z}^+ D_{\perp z} \rangle. \quad (1.255)$$

Using the results of Equations (1.234) and (1.248), this becomes

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{nsf}} = \frac{1}{N_m} (\gamma r_0)^2 \sum_j |f_j(Q)|^2 \frac{g_j^2}{12} S_j(S_j + 1) (1 - \hat{Q}_z^2). \quad (1.256)$$

Comparison with Equation (1.240) shows that the spin-flip scattering must be

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{sf}} = \frac{1}{N_m} (\gamma r_0)^2 \sum_j |f_j(Q)|^2 \frac{g_j^2}{12} S_j(S_j + 1) (1 + \hat{Q}_z^2). \quad (1.257)$$

Thus, if $\mathbf{Q}$ is parallel to the direction of neutron polarization, $(1 - \hat{Q}_z^2) = 0$ and all the scattering is spin-flip. Alternatively, if $\mathbf{Q}$ is perpendicular to the polarization direction, $\hat{Q}_z = 0$ and both spin-flip and non-spin-flip scattering cross sections are equal. The use of polarized neutrons makes it possible to separate paramagnetic from nuclear scattering by means of measurements using two distinct polarizations.

1.5.2.4 Probing Magnetic Order

In a second illustration of polarization analysis, we examine coherent and elastic neutron scattering from a system of magnetic atoms or ions, allowing for the presence of both nuclear and magnetic scattering. The interaction potential for nuclear scattering has matrix elements given by Equation (1.29) with the generalization to polarized neutrons (cf. Equation 1.248):

$$\langle \mathbf{k}_f | V | \mathbf{k}_i \rangle_{\text{nuc}} = -\frac{2\pi\hbar^2}{m_n} \sum_j \hat{b}_j e^{i\mathbf{Q} \cdot \mathbf{R}_j}. \quad (1.258)$$

For magnetic scattering, we use Equations (1.223), (1.229), and (1.234) to write

$$\langle \mathbf{k}_f | V | \mathbf{k}_i \rangle_{\text{mag}} = \frac{2\pi\hbar^2}{m_n} (\gamma r_0) \sigma \cdot \mathbf{D}_{\perp} = \frac{2\pi\hbar^2}{m_n} \sum_j \sigma \cdot \mathbf{C}_j e^{i\mathbf{Q} \cdot \mathbf{R}_j}, \quad (1.259)$$

where the coefficient $\mathbf{C}_j$ is defined as

$$\mathbf{C}_j = (\gamma r_0) f_j(\mathbf{Q}) (\hat{\mathbf{Q}} \times \boldsymbol{\mu}_j \times \hat{\mathbf{Q}}). \quad (1.260)$$

The total matrix element is therefore

$$\langle \mathbf{k}_f | V | \mathbf{k}_i \rangle_{\text{mag}} = \frac{2\pi\hbar^2}{m_n} \sum_j (\hat{b}_j + \boldsymbol{\sigma} \cdot \mathbf{C}_j) e^{i\mathbf{Q} \cdot \mathbf{R}_j}, \quad (1.261)$$

and from Equation (1.25) the scattering cross section for an event with initial and final neutron spin states σ_i and σ_f and initial and final states of the system τ_i and τ_f is

$$\left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{\sigma_i \tau_i \rightarrow \sigma_f \tau_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) \left| \left\langle \sigma_f \tau_f \left| \sum_j (\hat{b}_j + \boldsymbol{\sigma} \cdot \mathbf{C}_j) e^{i\mathbf{Q} \cdot \mathbf{R}_j} \right| \sigma_i \tau_i \right\rangle \right|^2 \delta(E + E_{\tau_i} - E_{\tau_f}). \quad (1.262)$$

For coherent elastic scattering, we may use the results from Section 1.2.3 and generalize Equation (1.70) to the case of polarized-neutron analysis of combined magnetic and nuclear scattering:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\sigma_i \rightarrow \sigma_f} = \frac{(2\pi)^3}{v_0} \sum_{\boldsymbol{\tau}} \left| \langle \sigma_f | \hat{F}(\boldsymbol{\tau}) | \sigma_i \rangle \right|^2 \delta(\mathbf{Q} - \boldsymbol{\tau}) \quad (1.263)$$

with a combined nuclear-magnetic form factor of the form

$$\hat{F}(\boldsymbol{\tau}) = \sum_d (\bar{b}_d + \boldsymbol{\sigma} \cdot \mathbf{C}_d) e^{-W_d} e^{i\boldsymbol{\tau} \cdot \mathbf{d}}. \quad (1.264)$$

For the case of coherent scattering, the neutron-spin-dependent matrix elements of $\hat{F}(\boldsymbol{\tau})$ can be obtained from Equations (1.250) and (1.253), and are given by

$$\begin{aligned} \langle \uparrow | \hat{F} | \uparrow \rangle &= F_N - F_{M\perp z}, \\ \langle \downarrow | \hat{F} | \downarrow \rangle &= F_N + F_{M\perp z}, \\ \langle \downarrow | \hat{F} | \uparrow \rangle &= -(F_{M\perp x} + iF_{M\perp y}), \\ \langle \uparrow | \hat{F} | \downarrow \rangle &= -(F_{M\perp x} - iF_{M\perp y}), \end{aligned} \quad (1.265)$$

where $F_N(\boldsymbol{\tau})$ is the unit-cell structure factor for nuclear scattering given by Equation (1.71), and $F_{M\perp}(\boldsymbol{\tau})$ is the magnetic structure factor given by Equations (1.244) and (1.246). In essence, Equation (1.263) with the matrix elements shown in Equation (1.265) replaces the expressions for unpolarized neutrons given earlier by Equations (1.70) and (1.245). The matrix elements given by Equation (1.265) also serve to illustrate that the scattering depends on the relative orientation of three vectors, namely: the neutron-spin quantization axis z , the magnetization of the system described by $\mathbf{F}_M$, and the

wave-vector $\mathbf{Q}$ (governing the orientation and magnitude of $\mathbf{F}_{M\perp}$). The participation of these three distinct vector quantities suggests that there are several distinct ways of conducting magnetic-structure experiments with polarized neutrons.

When z is orthogonal to $\mathbf{Q}$ and parallel to $\mathbf{F}_M$ (see Figure 1.22b), Equation (1.265) shows that the spin-flip-scattering channel is zero and that the *flipping ratio* (ratio of cross sections for neutrons parallel and antiparallel to $\mathbf{F}_{M\perp}$) is

$$r = \left(\frac{F_N - F_{M\perp}}{F_N + F_{M\perp}} \right)^2 \approx 1 - \frac{4F_{M\perp}}{F_N}. \quad (1.266)$$

When $F_{M\perp}$ is small compared with F_N , it is preferable to measure it through the flipping ratio, not the total scattering cross section as the latter is increased only by a fraction $(F_{M\perp}/F_N)^2$ arising from the presence of magnetic scattering. Furthermore, flipping ratios are amenable to accurate measurements owing to the cancellation of common systematic errors. From these considerations, it should come as little surprise that much of the study of magnetic structures with polarized neutrons has involved extensive use of Equation (1.266).

The flipping ratio as defined by Equation (1.266) also provides the most commonly used method for producing and analyzing polarized beams, arranging for $\mathbf{F}_M$ in a magnetic crystal to be perpendicular to $\mathbf{Q}$, so as to fulfill the condition $\mathbf{F}_{M\perp} = \mathbf{F}_N$, and choosing crystals such that $\mathbf{F}_N \approx \mathbf{F}_{M\perp}$. In this case, the outgoing neutron beam becomes polarized either parallel or antiparallel to $\mathbf{F}_M$. For a Bravais lattice, the required condition is equivalent to

$$|b| = f(\mathbf{Q})|\gamma r_0 \mu|, \quad (1.267)$$

a requirement which holds reasonably well, for example, for the (1.200) reflection of the Heussler alloy $\text{Co}_{0.92}\text{Fe}_{0.08}$. The output beam in this case is both monochromatic and polarized. For the production of broadband polarized beams, other methods must be used such as: neutron guides (see Refs. [199–201] and Section 1.5.5 below); ^4He -based spin filters [202,203]; the spin dependence of nuclear absorption resonances such as that from ^{149}Sm [204]; or the scattering from polarized nuclei (see Ref. [205] and Section 1.7).

An alternative geometry to the one described above involves arranging $\mathbf{Q}$ parallel to the neutron quantization axis z , so that $\mathbf{F}_{M\perp}$ becomes perpendicular to z (see Figure 1.22c). As shown by Equation (1.265), magnetic scattering becomes all spin-flip, while nuclear scattering remains non-spin-flip. For a ferromagnet, a magnetic field must be applied to align the domains, and this will define the z direction, so that normally there will be no component of $\mathbf{F}_{M\perp}$ perpendicular to z . For noncollinear magnetism (both ferro and antiferro), however, such components are nonvanishing and can be identified by measuring the spin-flip and non-spin-flip cross sections separately.

1.5.3 Magnetic Bragg Scattering

1.5.3.1 Ferromagnets

Ferromagnetic crystals are characterized by identical magnetic and nuclear unit cells and, therefore, magnetic and nuclear scattering take place at identical positions in reciprocal space. The simplest (and most widely used) method to separate these two contributions involves taking the difference between datasets above and below the ordering (Curie) temperature. Another approach involves applying a saturating magnetic field along the momentum-transfer vector $\mathbf{Q}$. The effect of the field is to remove magnetic scattering since in this case $\mathbf{F}_{M\perp} \equiv 0$. At zero field, on the other hand, the domains will be randomly oriented with a scattered intensity proportional to

$$1 - \overline{(\hat{\tau} \cdot \hat{\eta})^2}, \quad (1.268)$$

where the second term represents an average over domains. For randomly oriented powders and cubic crystals this term is equal to $2/3$, and in the general case it is greater than zero. For uniaxial or orthorhombic structures, the average will depend on the specific orientation of the reciprocal-lattice vector $\hat{\tau}$. As such, it can be used to deduce the inclination of the spin direction relative to the principal axes. For powders, it is generally hard to saturate the sample parallel to $\mathbf{Q}$, and the temperature dependence is frequently used to separate magnetic and nuclear scattering.

The most powerful and least ambiguous method for measuring ferromagnetic structures is diffraction from single crystals with polarized neutrons using the configuration $\mathbf{F}_M \parallel \mathbf{z} \perp \boldsymbol{\tau}$ (Figure 1.22b). In this case $\mathbf{F}_{M\perp} \approx \mathbf{F}_M$ and the flipping ratios give $\mathbf{F}_M(\boldsymbol{\tau})$ in terms of $\mathbf{F}_N(\boldsymbol{\tau})$. The results may be interpreted in terms of magnetic moments on individual atoms as in Equation (1.244), or $\mathbf{F}_M(\boldsymbol{\tau})$ can be Fourier transformed to give the spin density as a function of position in the unit cell. Figure 1.24 shows the spin density obtained for the (001) plane in iron by Shull and Yamada in the early days of polarized neutron scattering [13]. The higher densities along the direction of the cube edges show that the 3d electronic orbitals are preferentially aligned along these directions—see also Figure 1.4 in Section 1.1.

1.5.3.2 Antiferromagnets

Antiferromagnets represent a departure from the previous case, as they possess distinctly different nuclear and magnetic lattices leading to purely magnetic Bragg reflections. In this case, it is therefore not necessary to use polarized neutrons to isolate the magnetic response—most antiferromagnetic structures have been determined with unpolarized beams, and often with powder samples. If the structure has sufficiently low symmetry, it may be possible to obtain the orientation of the moments in the magnetic structure, as already discussed in Section 1.5.3.1.

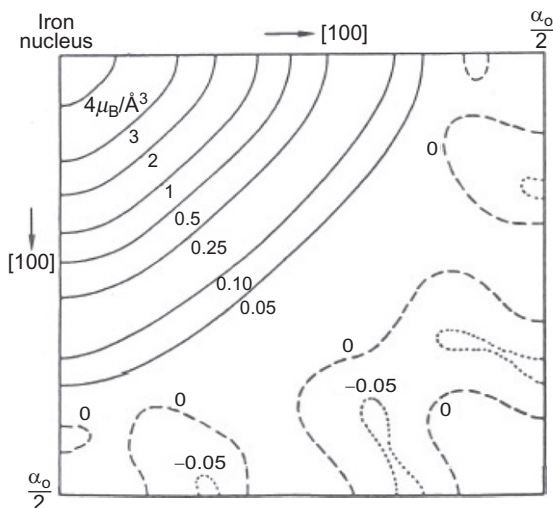


FIGURE 1.24 Spin density for iron on the (001) plane. Reprinted with permission from Ref. [13].

An instructive illustration of antiferromagnetic ordering is provided by compounds whose magnetic structure is characterized by a spin direction that changes with position in the lattice. When the repeat distance is not an exact multiple of the repeat distance of the nuclear structure (a rather common situation), the magnetic structure is then called incommensurate. If the magnitude and direction of the moments on each magnetic atom are described by a function $\boldsymbol{\mu}(\mathbf{r})$, the periodicity of $\boldsymbol{\mu}(\mathbf{r})$ can be described with propagation vectors $\boldsymbol{\lambda}$:

$$\boldsymbol{\mu}(\mathbf{r}) = \sum_n \boldsymbol{\mu}_n e^{i n \boldsymbol{\lambda} \cdot \mathbf{r}}. \quad (1.269)$$

And the magnetic unit-cell structure factor, Equation (1.244), for the n th component is

$$\mathbf{F}_{Mn}(\mathbf{Q}) = -(\gamma r_0) \sum_d \boldsymbol{\mu}_n f_d(\mathbf{Q}) e^{-W_d(\mathbf{Q})} e^{i(n\boldsymbol{\lambda} + \mathbf{Q}) \cdot \mathbf{d}}. \quad (1.270)$$

The differential cross section in Equation (1.245) becomes

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{el}} = \frac{(2\pi)^3}{v_0} \sum_n \sum_{\boldsymbol{\tau}} \delta(\mathbf{Q} + n\boldsymbol{\lambda} - \boldsymbol{\tau}) |\mathbf{F}_{Mn\perp}(\boldsymbol{\tau})|^2. \quad (1.271)$$

The most salient feature of Equation (1.271) is the emergence of satellite peaks in the magnetic diffraction pattern, displaced from nuclear Bragg peaks by multiples of the propagation vector $\boldsymbol{\lambda}$. The intensity of these satellites depends on the Fourier components $\boldsymbol{\mu}_n$.

The experiments on holmium by Koehler *et al.* [206] provide a beautiful and legendary example of satellite reflections in a magnetic structure. In the

temperature region below the Néel temperature (133 K), the magnetic moments form a simple helical structure

$$\mu_x(\mathbf{r}) = \mu \cos(\boldsymbol{\lambda} \cdot \mathbf{r}), \quad \mu_y(\mathbf{r}) = \mu \sin(\boldsymbol{\lambda} \cdot \mathbf{r}), \quad \mu_z(\mathbf{r}) = 0, \quad (1.272)$$

which has just one Fourier component, so there is a pair of satellites $\boldsymbol{\tau} + \boldsymbol{\lambda}$ about each Bragg reflection. Figure 1.25a shows the crystal rocking curve about the (100) reflection at 77 K; at this temperature λ is about $0.46\pi/c$, where c is the lattice parameter. As the temperature is lowered, λ decreases and at the same time new peaks arise in the diffraction pattern. At 4.2 K (Figure 1.25b), $\lambda = \pi/(1.3c)$, that is, the helical structure is now commensurate with the crystal lattice, and peaks are now observed at $\boldsymbol{\tau} \pm 5\boldsymbol{\lambda}$ (labeled 5, 7 in the figure) and also at $\boldsymbol{\tau} \pm \boldsymbol{\lambda}$ (labeled 1, 11). This is due to slight changes in the orientation of the moments, in opposite directions on successive planes, which produce a lower energy configuration on account of crystal-anisotropy effects. The new magnetic structure is thus described by

$$\mu_x(\mathbf{r}^\pm) = \mu \sin \gamma \cos(\boldsymbol{\lambda} \cdot \mathbf{r} \pm \delta), \quad \mu_y(\mathbf{r}^\pm) = \mu \sin \gamma \sin(\boldsymbol{\lambda} \cdot \mathbf{r} \pm \delta), \quad \mu_z(\mathbf{r}^\pm) = \mu \cos \gamma, \quad (1.273)$$

where the plus and minus signs refer to alternating planes normal to the c axis, and the z component represents a small ferromagnetic contribution. The occurrence of the fifth- and seventh-harmonic satellites can be qualitatively understood by considering the structure of Equation (1.273) as a doubling of the magnetic unit cell structure in the c direction. This structure leads to new reciprocal points $\boldsymbol{\tau}'$, such as (001), leading to new satellites at $\boldsymbol{\tau}' + \boldsymbol{\lambda}$, for example, (005/6) and (007/6). Comparison of the measured intensities of these satellites with those calculated from the magnetic unit-cell structure factor (Equation 1.270) shows that δ is about 6° at 4.2 K.

1.5.4 Diffuse Scattering from Magnetic Disorder

As in the nuclear case, when the magnetic moments do not follow a periodic pattern, the elastic magnetic scattering becomes a continuous function of $\mathbf{Q}$ —diffuse scattering. Assuming that the moments lie along a unique direction $\hat{\mathbf{n}}$, the differential cross section given by Equation (1.242) is

$$\frac{d\sigma}{d\Omega} = \frac{1}{N_m} (\gamma r_0)^2 \left(1 - \hat{\mathbf{Q}} \cdot \hat{\mathbf{n}}\right)^2 |\langle \mathbf{D}(\mathbf{Q}) \rangle|^2 \quad (1.274)$$

with

$$|\langle \mathbf{D}(\mathbf{Q}) \rangle|^2 = \sum_{j,j'} f_j(\mathbf{Q}) f_{j'}(\mathbf{Q}) \langle \mu_j \rangle \langle \mu_{j'} \rangle e^{i\mathbf{Q} \cdot (\mathbf{r}^j - \mathbf{r}^{j'})} e^{-W_j - W_{j'}}. \quad (1.275)$$

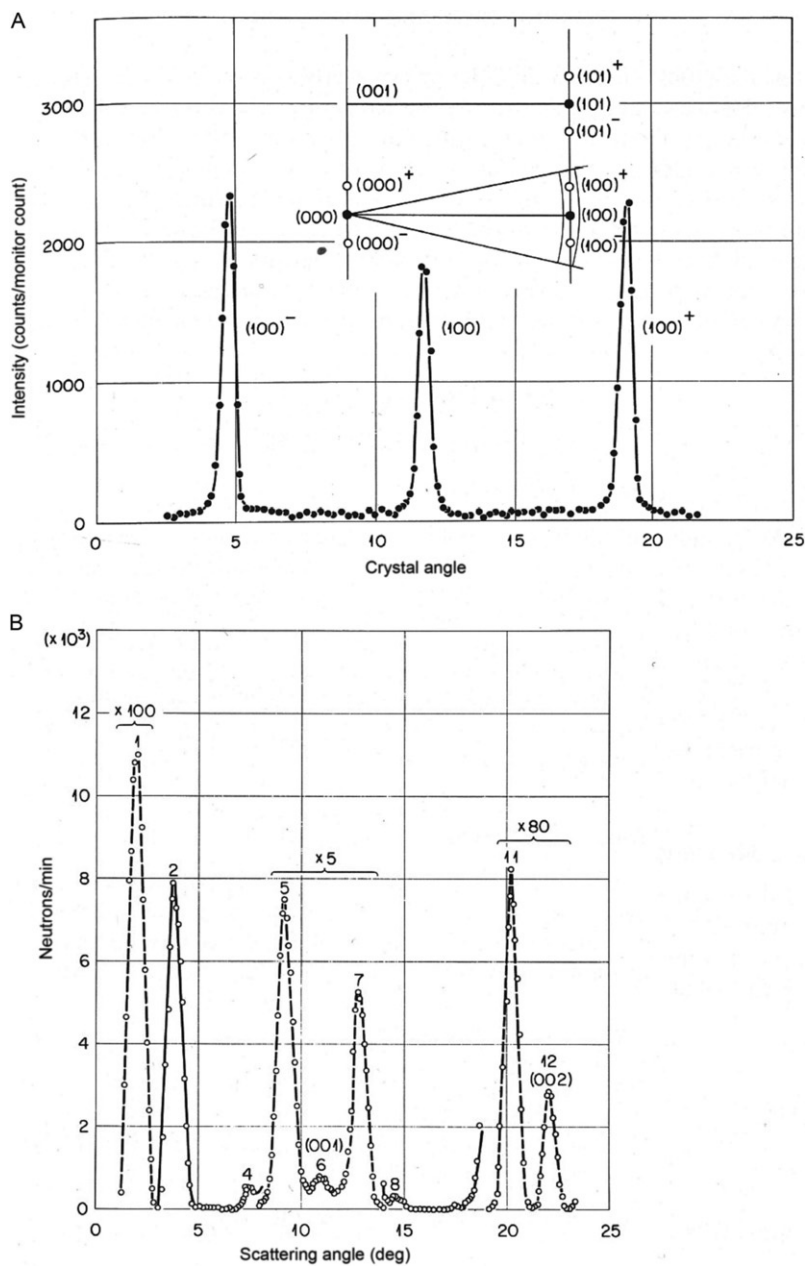


FIGURE 1.25 (a) Crystal rocking curve about the (100) reflection in holmium at 77 K and (b) at 4.2 K. Reprinted with permission from Ref. [206].

Considering an average spin of atoms of type d as

$$\bar{\mu}_d = \frac{1}{N_d} \sum_{j \in d} \langle \mu_j \rangle, \quad (1.276)$$

this average value in combination with Equation (1.275) leads to Bragg scattering as defined in the previous section. Deviations from the average, given by

$$\langle \mu_j \rangle \approx \bar{\mu}_d + \Delta\mu_j, \quad (1.277)$$

give rise to diffuse scattering proportional to

$$|\langle D(\mathbf{Q}) \rangle|_{\text{diff}}^2 = \sum_{\substack{j \in d \\ j' \in d'}} f_d(\mathbf{Q}) f_{d'}(\mathbf{Q}) \Delta\mu_j \Delta\mu_{j'} e^{i\mathbf{Q} \cdot (\mathbf{j}' - \mathbf{j})} e^{-W_d - W_{d'}}. \quad (1.278)$$

These considerations are entirely analogous to the diffuse nuclear scattering discussed in Section 1.4.4. As illustration of this formalism, a disordered alloy with concentration c of impurity atoms is characterized by deviations of the form

$$\Delta\mu_j = \Delta\mu \cdot p_j + \sum_{j' \neq j} \phi(\mathbf{j} - \mathbf{j}') p_{j'}, \quad (1.279)$$

where $\Delta\mu$ is the difference between the magnetic moment of the impurity and that of the host atoms. Furthermore, $p_j = 1$ if the site is occupied by an impurity atom and 0 otherwise, and $\phi(\mathbf{j} - \mathbf{j}')$ represents the effect of an impurity atom on the spin of the host. Ignoring the effect of the impurity atom on $f(\mathbf{Q})$, one gets

$$|\langle D(\mathbf{Q}) \rangle|_{\text{diff}}^2 = f^2(\mathbf{Q}) c(1-c) [\Delta\mu + \phi(\mathbf{Q})]^2 e^{-2W}, \quad (1.280)$$

where $\phi(\mathbf{Q})$ is the Fourier transform of $\phi(\mathbf{j} - \mathbf{j}')$. This equation assumes that the magnetic defects associated with each impurity atom are independent of each other, a reasonable assumption in the dilute limit. Also, $\phi(\mathbf{Q}=0)$ represents the total magnetic-defect moment per impurity atom other than that at the impurity site itself.

A classic set of measurements of dilute magnetic alloys was carried out by Low and Collins in the 1960s [207]. They used a simple time-of-flight diffractometer with a saturating magnetic field that could be rotated parallel and perpendicular to the scattering vector $\mathbf{Q}$ in order to isolate the magnetic scattering. Some results on iron alloys are shown in Figure 1.26. An interesting feature is the behavior shown by several alloys in which the cross section passes through zero near $Q \approx 0.5 \text{ \AA}^{-1}$. This result implies that $\phi(\mathbf{Q}=0)$ is opposite in sign to $\Delta\mu$ so that the sum of the two crosses zero when $|\phi| = |\Delta\mu|$. Thus, the change in moment at the impurity site itself is opposite in sign to the net moment due to the impurity. The fact that the scattering goes

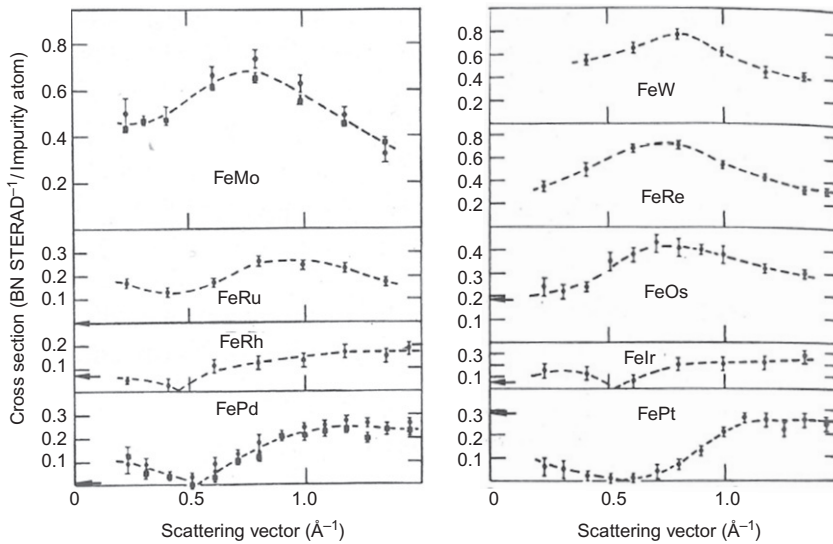


FIGURE 1.26 Diffuse magnetic scattering from iron alloys. Reprinted with permission from Ref. [207].

precisely to zero at a certain value of Q implies that the distribution of $\phi(\mathbf{Q})$ with respect to direction must be nearly isotropic. More precise treatments must take account of the effects of short-range order (which can be inferred from the *nuclear* diffuse scattering) and interactions between the defects ignored in the treatment leading to Equation (1.280). In this case, departures from the dilute limit in defect concentration can be examined. A comprehensive review is given in Ref. [208].

In certain classes of materials, there are static moments which do not exhibit long-range order either due to disorder or spatial constraints (e.g., nanoparticles). In this case, the magnetic scattering is diffuse in character. One class is the spin glasses, where the moments exhibit short-range correlations and there is no net magnetic moment. Another is that of amorphous magnetic materials and magnetic liquids, where the atomic positions as well as the moments only have short-range order. Relevant reviews on magnetically disordered systems can be found in Refs. [209–211].

To conclude this section, we return to a more fundamental example of the use of polarized neutron diffraction to probe diffuse magnetic scattering in the spin ice $\text{Ho}_2\text{Ti}_2\text{O}_7$, a material that has been predicted to support magnetic-monopole excitations [212]. Figure 1.27 shows diffuse-scattering data using an experimental setup configured to measure two distinct scattering functions associated with two-spin correlation functions along the y (spin-flip) and z (non-spin-flip) directions. The characteristic pinch points observed in the

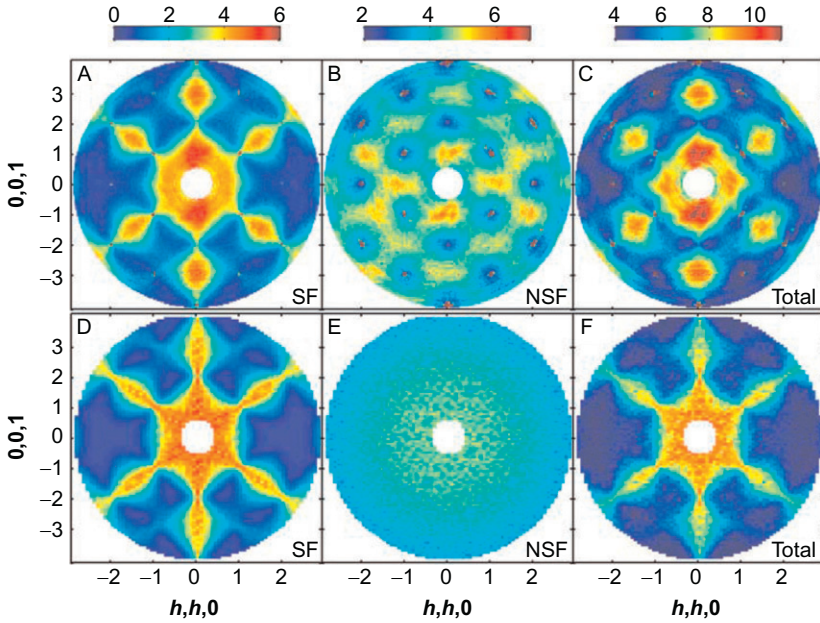


FIGURE 1.27 Diffuse scattering using polarized neutron diffraction techniques (a–c) compared against theoretical predictions using a near-neighbor model (d–f) for a single crystal of the spin-ice compound $\text{Ho}_2\text{Ti}_2\text{O}_7$: (a) spin-flip scattering at $T = 1.7$ K; (b) non-spin-flip scattering at the same temperature; (c) sum of spin-flip and non-spin-flip scattering intensities. *Reprinted with permission from Ref. [212].*

spin-flip data at the Brillouin zone centers $(0,0,2)$, $(1,1,1)$, and $(2,2,2)$ in [Figure 1.27a](#) provide an unambiguous signature of the dipolar form of the spin correlation function expected for a Coulomb phase, the quasiparticle vacuum for magnetic monopoles. These characteristic features in the spin-flip data are not at all evident in the unpolarized data shown in [Figure 1.27c](#).

1.5.5 Large-Scale Magnetic Structures

We recall the expression for the differential cross section for elastic magnetic scattering (cf. Equation 1.242)

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{el}} = \frac{1}{N_m} (\gamma r_0)^2 |\langle \mathbf{D}_\perp(\mathbf{Q}) \rangle|^2, \quad (1.281)$$

where $\mathbf{D}(\mathbf{Q}) = -\mathbf{M}(\mathbf{Q})/2\mu_B$, and $\mathbf{M}(\mathbf{Q})$ is the Fourier transform of the magnetization density

$$\mathbf{M}(\mathbf{Q}) = \int \mathbf{M}(\mathbf{r}) e^{i\mathbf{Q} \cdot \mathbf{r}} d\mathbf{r}. \quad (1.282)$$

An elegant use of small-angle magnetic neutron scattering in the study of emergent properties and exotic phases of matter is shown in [Figure 1.28](#). At

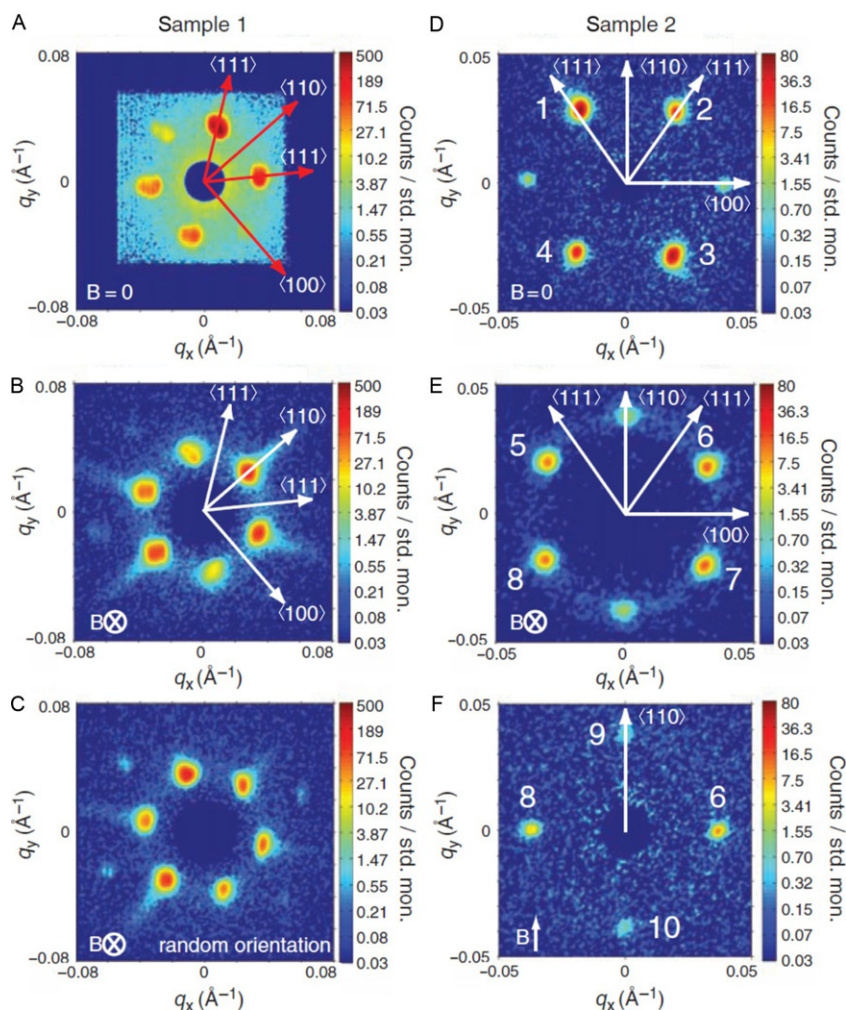


FIGURE 1.28 Small-angle neutron-scattering data showing the formation of a skyrmion lattice in two separate MnSi specimens, as labeled in the figure. *Reprinted with permission from Ref. [213].*

zero applied magnetic field, the small-angle scattering pattern for the chiral itinerant-electron magnet MnSi evinces the presence of helical order in the zero-field-cooled state. Sixfold symmetry emerges when a small field ($B \sim 0.15$ T) is applied to the sample, as shown in (b) of this figure. Remarkably, the same structure is still present for a randomly oriented sample (c). The observed magnetic structure is orthogonal to the applied magnetic field and entirely independent of the underlying crystalline order. These beautiful neutron-scattering data have been interpreted in terms of the spontaneous formation of a two-dimensional lattice of skyrmion lines—a topologically stable field configuration with particle-like properties, in this case involving electron-spin states.

As discussed earlier in the context of NR (Section 1.4.6.2), neutron transport through materials can be described in terms of a refractive index:

$$n = 1 - \frac{\rho}{2\pi} \lambda_n^2 \bar{b}, \quad (1.283)$$

where ρ is the number density of scattering centers characterized by a mean scattering length $\bar{b}$. In a magnetic medium, the latter necessarily contains magnetic contributions. If the spin direction is along the direction of magnetization, Equation (1.283) reads

$$n^\pm = 1 - \frac{\rho}{2\pi} \lambda_n^2 (\bar{b} \pm C), \quad (1.284)$$

for the magnetization parallel (+) and antiparallel (−) to the neutron-spin direction. For a neutron travelling in air and encountering the surface of such a medium, total reflection takes place at glancing angles less than the critical angles defined by

$$\theta_c^\mp = \lambda_n \left[\frac{\rho}{\pi} (\bar{b} \mp C) \right]^{1/2}. \quad (1.285)$$

For angles larger than θ_c , reflection is still possible but falls off over an angular range $\sim \theta_c$. This feature constitutes the basis for a magnetic neutron mirror. For $b \approx C$, it will reflect neutrons of the down-spin state only. Mirrors are used as polarizing devices where relatively long-wavelength neutrons are required and small angular divergences can be tolerated.

Equation (1.284) also provides a means for studying domain distributions in magnetically ordered materials, since refraction at a domain boundary is also governed by the refractive index n . An early experiment in this area was carried out by Hughes *et al.* [214] who measured the angular divergence of a neutron beam passed through an iron slab in magnetized and unmagnetized states. The average deflection at each boundary inferred from the estimated domain size and sample thickness agreed with that calculated from Equation (1.284) within 10%.

An important application of neutron-optical effects in magnetic scattering is in the field of surface magnetism, where the profile of the magnetization

near the surface can be derived from measurements of the reflected intensity as a function of grazing angle and neutron polarization. In a pioneering experiment by Felcher *et al.* [215], the penetration of magnetic field into the surface of superconducting niobium was measured from the reflection of polarized neutrons. In this case Equation (1.284) takes the form

$$n^{\pm}(z) = 1 - \frac{1}{2\pi}[\rho\bar{b} \mp cH(z)], \quad (1.286)$$

where c is a material-dependent constant. It is also assumed that the magnetic-field penetration into the material is of the form

$$H(z) = He^{-z/\Lambda}, \quad (1.287)$$

where Λ is a characteristic penetration depth. An external field H is applied parallel to the surface and it is less than the critical entry field for magnetic flux, H_{c1} . Using a polarized beam with a broadband wavelength spectrum (from a pulsed neutron source), and polarized by a mirror of the type described above in conjunction with a spin flipper in front of the sample, the flipping ratio of the reflectivities for the two spin states, R^+/R^- , as a function of neutron wavelength can be measured. Once again, measurement of a flipping ratio provides a sensitive measurement since most sources of systematic error are eliminated. The expression for the reflectivities is more complicated than Equation (1.284) for the refractive indices, but an approximate expression can be written down explicitly for the field dependence given above. From the flipping ratio and an extrapolation of the temperature dependence to $T=0$, a value of $\Lambda(0)=410\pm40$ Å can be derived. This figure agrees well with estimates based on local Ginzburg–Landau theory and the measured temperature dependence of the critical fields.

Progress in the use of polarized neutrons to study magnetic interfaces and superlattices has moved well beyond these seminal studies. Figure 1.29 shows NR data for a heterostructure made of thin (nm) layers of the half-metallic ferromagnet $\text{La}_{2/3}\text{Ca}_{1/3}\text{MnO}_3$ and the superconductor $\text{YBa}_2\text{Cu}_3\text{O}_7$. In this figure, the line across $Q_x=0$ corresponds to specular reflectivity (both nuclear and magnetic) and remains the dominant contribution to the signal well below the onset of ferromagnetism at $T=180$ K. The absence of diffuse scattering under these conditions also tells us that the in-plane magnetic roughness in this multi-layered material is negligible. Below the structural transition at $T=105$ K, there is a broadening of the main Bragg feature along the off-specular direction, thought to be related to the distortion of the substrate surface. Below the superconducting temperature, the off-specular scattering shows a dramatic change in terms of in-plane structure, characteristic of the presence of stripe-like magnetic domains. From the total range and inter-peak separation of these Bragg features along Q_x , it becomes possible to quantify both domain size (d) and domain-wall width (w). These properties are quite

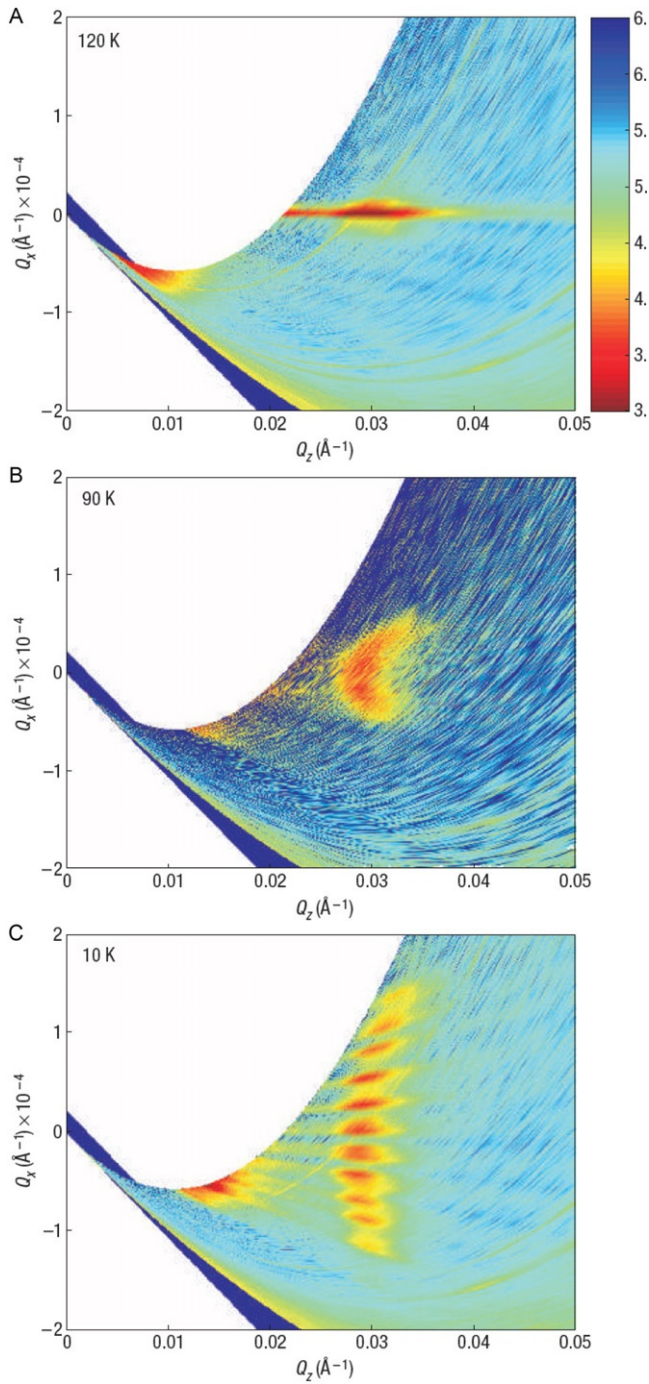


FIGURE 1.29 Temperature dependence of specular and off-specular neutron reflectivity around the first superlattice Bragg reflection in $\text{La}_{2/3}\text{Ca}_{1/3}\text{MnO}_3/\text{YBa}_2\text{Cu}_3\text{O}_7$ multilayers: (a) below the onset of ferromagnetism; (b) below the structural transition in the substrate at $T=105$ K; and (c) below the superconducting transition temperature at 78 K. Note the large difference in momentum-transfer scales along the specular (bottom) and off-specular directions (left). *Reprinted with permission from Ref. [216].*

sensitive to temperature, with $d=24(1.41)\mu\text{m}$ and $w=2(1.3)\mu\text{m}$ at $T=10$ (1.63)K. In addition to providing information on the spin and orbital electronic structure at the interfaces, these data thus demonstrate the possibility of extensive rearrangements of magnetic-domain structure at the superconducting transition temperature in artificial nanostructured materials.

1.6 SPIN DYNAMICS

1.6.1 Generalized Susceptibility

As in the case of inelastic nuclear scattering presented in Section 1.3, magnetic spins can sustain single-particle and collective excitations (spin waves). Their theoretical treatment requires going back to the double differential scattering cross section presented in Equation (1.237) of Section 1.5.1:

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE_f} = & \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta \right) \sum_{jj'} f_j^*(\mathbf{Q}) f_{j'}(\mathbf{Q}) \times \\ & \times \frac{1}{2\pi\hbar} \int \left\langle \mu_{j\alpha}(0) \mu_{j'\beta}(t) \right\rangle \left\langle e^{-i\mathbf{Q}\cdot\mathbf{R}_j(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (1.288)$$

The two correlation functions shown under the integrand above can be expressed as the sum of time-independent and time-dependent parts:

$$\begin{aligned} J_{jj'}^{\alpha\beta}(t) & \equiv \left\langle \mu_{j\alpha}(0) \mu_{j'\beta}(t) \right\rangle = J_{jj'}^{\alpha\beta}(\infty) + \tilde{J}_{jj'}^{\alpha\beta}(t), \\ I_{jj'}(\mathbf{Q}, t) & = \left\langle e^{-i\mathbf{Q}\cdot\mathbf{R}_j(0)} e^{i\mathbf{Q}\cdot\mathbf{R}_{j'}(t)} \right\rangle = I_{jj'}(\mathbf{Q}, \infty) + \tilde{I}_{jj'}(\mathbf{Q}, t). \end{aligned} \quad (1.289)$$

With these definitions, the Fourier time integral in Equation (1.288) can be written as

$$\int \left[J_{jj'}^{\alpha\beta}(\infty) + \tilde{J}_{jj'}^{\alpha\beta}(t) \right] \left[I_{jj'}(\mathbf{Q}, \infty) + \tilde{I}_{jj'}(\mathbf{Q}, t) \right] e^{-iEt/\hbar} dt. \quad (1.290)$$

From the four distinct terms arising from this expression, the time-independent term $J_{jj'}^{\alpha\beta}(\infty)I_{jj'}(\mathbf{Q}, \infty)$ corresponds to elastic scattering, as discussed in Section 1.5. The *magneto-vibrational* term $J_{jj'}^{\alpha\beta}(\infty)I_{jj'}(\mathbf{Q}, t)$ has the same energy dependence as the nuclear scattering but the interaction of the neutron with the system is through the magnetic potential. For ferromagnets with identical spins aligned parallel to the z direction, we can write

$$J_{jj'}^{\alpha\beta}(\infty) = \delta_{\alpha\beta} \delta_{\alpha z} \langle \mu^z \rangle^2 \quad (1.291)$$

and the double differential scattering cross section for elastic and magneto-vibrational scattering is

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 (1 - \hat{Q}_z^2) f^2(\mathbf{Q}) \langle \mu^z \rangle^2 \sum_{jj'} \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \left\langle e^{-i\mathbf{Q} \cdot \mathbf{R}_j(0)} e^{i\mathbf{Q} \cdot \mathbf{R}_{j'}(t)} \right\rangle e^{-iEt/\hbar} dt \quad (1.292)$$

Comparison with Equation (1.47) shows that this expression corresponds to the coherent nuclear scattering function with the factor

$$(\gamma r_0)^2 (1 - \hat{Q}_z^2) f^2(\mathbf{Q}) \langle \mu^z \rangle^2 \quad (1.293)$$

instead of $|\bar{b}|^2$ for nuclear scattering.

$J_{jj'}^{\alpha\beta}(t) I_{jj'}(\mathbf{Q}, \infty)$ in the product above is responsible for inelastic magnetic scattering. For a solid with long-range order we may write (similar to Equation 1.68)

$$\mathbf{j} = \mathbf{l} + \mathbf{d} \quad (1.294)$$

and the results of Section 1.2.3 can be used to give the double differential magnetic scattering cross section (elastic plus inelastic) as

$$\frac{d^2\sigma}{d\Omega dE_f} = \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) \frac{(\gamma r_0)^2}{4\pi\mu_B^2} \sum_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta \right) \left(1 - e^{-E/k_B T} \right)^{-1} \sum_{dd'} e^{-[W_d(\mathbf{Q}) + W_{d'}(\mathbf{Q})]} \left(\chi'' \right)_{\alpha\beta}^{dd'}(\mathbf{Q}, E), \quad (1.295)$$

where χ'' is the imaginary part of a *generalized susceptibility* function $\chi_{\alpha\beta}^{dd'}(\mathbf{Q}, E)$, corresponding to

$$\begin{aligned} \left(\chi'' \right)_{\alpha\beta}^{dd'}(\mathbf{Q}, E) &= 4\pi\mu_B^2 f_{\mathbf{d}}^*(\mathbf{Q}) f_{\mathbf{d}'}(\mathbf{Q}) \left(1 - e^{-E/k_B T} \right) \\ &\times \sum_{\mathbf{l}} e^{i\mathbf{Q} \cdot (\mathbf{l} + \mathbf{d}' - \mathbf{d})} \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \left\langle \mu_{0\mathbf{d}\alpha}(0) \mu_{\mathbf{l}\mathbf{d}'\beta}(t) \right\rangle e^{-iEt/\hbar} dt. \end{aligned} \quad (1.296)$$

The generalized susceptibility function is analogous to the dynamic structure factor $S(\mathbf{Q}, E)$ for nuclear inelastic scattering, as it contains detailed information about the dynamics of the magnetic system. The factor $(1 - e^{-E/k_B T})$ in Equation (1.296) ensures that detailed balance (see Section 1.4.3) is obeyed, also consistent with the condition

$$\chi''(\mathbf{Q}, -E) = -\chi''(\mathbf{Q}, E). \quad (1.297)$$

The function

$$\chi(\mathbf{Q}) \equiv \chi(\mathbf{Q}, 0) \quad (1.298)$$

is the *isothermal susceptibility* representing the $\mathbf{Q}$ dependence of the response to a static magnetic field. According to the Kramers–Kronig causality relations, $\chi(\mathbf{Q})$ is also given by

$$\chi(\mathbf{Q}) = \frac{1}{\pi} \int_{-\infty}^{\infty} \chi''(\mathbf{Q}, E) \frac{dE}{E}, \quad (1.299)$$

which represents a sum rule for $\chi''(\mathbf{Q}, E)$ analogous to the zeroth-moment sum rule for $S(\mathbf{Q}, E)$ (Equation 1.111). In what follows, we seek to examine $\chi''(\mathbf{Q}, E)$ in some specific situations.

1.6.2 Spin Waves

Long-range magnetic order can sustain collective excitations known as spin waves. In essence, these excitations arise from sinusoidal deviations of the spin components along a particular direction as one moves across magnetic centers. Like phonons, spin waves are quantized, and a single quantum is called a magnon. As such, much of the treatment of spin waves using a second-quantization picture is analogous to what we have discussed for excitations associated with nuclear motion. For a Bravais lattice, the Heisenberg operators for the spins $\mathbf{S}_I(t)$ can be written in terms of the spin waves as

$$\begin{aligned} S_I^x(t) + S_I^y(t) &= \left(\frac{2S}{N}\right)^{1/2} \sum_{\mathbf{q}} b(\mathbf{q}) e^{i[\mathbf{q} \cdot \mathbf{r} - \omega(\mathbf{q})t]}, \\ S_I^x(t) - S_I^y(t) &= \left(\frac{2S}{N}\right)^{1/2} \sum_{\mathbf{q}} b^+(\mathbf{q}) e^{-i[\mathbf{q} \cdot \mathbf{r} - \omega(\mathbf{q})t]}, \\ S_I^z(t) &= S - \frac{1}{N} \sum_{\mathbf{q}\mathbf{q}'} b(\mathbf{q}) b^+(\mathbf{q}') e^{i\{(\mathbf{q}' - \mathbf{q}) \cdot \mathbf{r} - [\omega(\mathbf{q}') - \omega(\mathbf{q})t]\}}, \end{aligned} \quad (1.300)$$

where $\omega(\mathbf{q})$ is the frequency of a spin wave of reduced wave vector $\mathbf{q}$ and the z axis is taken as the direction of spin quantization. The $\hbar\omega(\mathbf{q})$ terms represent eigenvalues of the spin Hamiltonian, which for a Heisenberg ferromagnet has the form

$$H = - \sum_{II'} J_{II'} \mathbf{S}_I \cdot \mathbf{S}_{I'}, \quad (1.301)$$

where the coefficients $J_{II'}$ are Heisenberg exchange integrals arising from electrostatic interactions between the electrons. The $b(\mathbf{q})$ and $b^+(\mathbf{q})$ are quantum-mechanical operators representing the annihilation or creation of a magnon, analogous to the phonon annihilation and creation operators introduced in Section 1.3.1. Under the so-called linear approximation (the magnetic analog of the harmonic approximation for phonons), deviations of the spin z -components from ground-state values are small relative to the S_I . In this case, the Hamiltonian in Equation (1.301) can be expressed in terms of these annihilation and creation operators

$$H = H^0 + \sum_{\mathbf{q}} \hbar \omega(\mathbf{q}) b^+(\mathbf{q}) b(\mathbf{q}), \quad (1.302)$$

where

$$H^0 = -S^2 N J(0), \quad (1.303)$$

$$\hbar \omega(\mathbf{q}) = 2S[J(0) - J(\mathbf{q})], \quad (1.304)$$

and

$$J(\mathbf{q}) = \sum_{l''} J_{ll''} e^{i\mathbf{q} \cdot (\mathbf{l} - \mathbf{l}')} \quad (1.305)$$

Spin-wave dispersion relations are explicitly given by Equation (1.304) in terms of Heisenberg exchange integrals. For a lattice with r atoms per unit cell, the dispersion relation has in general r branches with both acoustic and optic magnons. For reference, the case for a hexagonal-closed-packed ferromagnet is treated in detail in Section 9.5 of Ref. [108].

For a Bravais lattice, Equation (1.296) can be rewritten as

$$\chi''(\mathbf{Q}, E) = \pi g^2 \mu_B^2 \left(1 - e^{-E/k_B T}\right) \sum_{\mathbf{l}} e^{i\mathbf{Q} \cdot \mathbf{l}} \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \langle S_{0\alpha}(0) S_{\mathbf{l}\beta}(t) \rangle e^{-iEt/\hbar} dt. \quad (1.306)$$

From Equation (1.300), it can be shown that the only nonzero components of the time-correlation function $\langle S_{0\alpha}(0) S_{\mathbf{l}\beta}(t) \rangle$ are

$$\begin{aligned} \langle S_{0x}(0) S_{\mathbf{l}x}(t) \rangle &= \langle S_{0y}(0) S_{\mathbf{l}y}(t) \rangle \\ &= \frac{S}{2N} \sum_{\mathbf{q}} \left[e^{-i[\mathbf{q} \cdot \mathbf{l} - \omega(\mathbf{q})t]} \langle n_{\mathbf{q}} + 1 \rangle + e^{i[\mathbf{q} \cdot \mathbf{l} - \omega(\mathbf{q})t]} \langle n_{\mathbf{q}} \rangle \right], \end{aligned} \quad (1.307)$$

$$\begin{aligned} \langle S_{0x}(0) S_{\mathbf{l}y}(t) \rangle &= -\langle S_{0y}(0) S_{\mathbf{l}x}(t) \rangle \\ &= \frac{iS}{2N} \sum_{\mathbf{q}} \left[e^{-i[\mathbf{q} \cdot \mathbf{l} - \omega(\mathbf{q})t]} \langle n_{\mathbf{q}} + 1 \rangle + e^{i[\mathbf{q} \cdot \mathbf{l} - \omega(\mathbf{q})t]} \langle n_{\mathbf{q}} \rangle \right], \end{aligned} \quad (1.308)$$

and

$$\langle S_{0z}(0) S_{\mathbf{l}z}(t) \rangle = S^2 - \frac{S}{2N} \sum_{\mathbf{q}} \langle n_{\mathbf{q}} \rangle, \quad (1.309)$$

where

$$\langle n_{\mathbf{q}} \rangle = \frac{1}{\exp[\hbar \omega(\mathbf{q})/k_B T] - 1} \quad (1.310)$$

is the Bose population factor already encountered in our previous treatment of vibrations (both phonons and magnons are bosonic pseudoparticles). The time-independent correlation function shown in Equation (1.309) represents

the elastic scattering discussed in [Section 1.5](#). Furthermore, the xy and yx cross terms cancel in the expression for the cross section (cf. Equation 1.295), leaving only xx and yy contributions. The double differential inelastic scattering cross section is therefore reduced to a sum of terms for one-magnon creation and annihilation:

$$\begin{aligned} \left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{m,+1} &= \left(\frac{k_f}{k_i} \right) \frac{(2\pi)^3}{2N_m v_0} (\gamma r_0)^2 \left(\frac{g^2 S}{4} \right) \left(1 + \hat{Q}_z^2 \right) f^2(\mathbf{Q}) e^{-2W(\mathbf{Q})} \times \\ &\times \sum_{\tau\mathbf{q}} \langle n_{\mathbf{q}} + 1 \rangle \delta(\mathbf{Q} - \mathbf{q} - \boldsymbol{\tau}) \delta[E - \hbar\omega(\mathbf{q})], \end{aligned} \quad (1.311)$$

$$\begin{aligned} \left(\frac{d^2\sigma}{d\Omega dE_f} \right)_{m,-1} &= \left(\frac{k_f}{k_i} \right) \frac{(2\pi)^3}{2N_m v_0} (\gamma r_0)^2 \left(\frac{g^2 S}{4} \right) \left(1 + \hat{Q}_z^2 \right) f^2(\mathbf{Q}) e^{-2W(\mathbf{Q})} \\ &\times \sum_{\tau\mathbf{q}} \langle n_{\mathbf{q}} \rangle \delta(\mathbf{Q} + \mathbf{q} - \boldsymbol{\tau}) \delta[E + \hbar\omega(\mathbf{q})], \end{aligned} \quad (1.312)$$

These differential scattering cross sections have a similar structure as those encountered in our previous treatment phonon scattering, see, for example, Equation (1.81). As in the phonon case, strong scattering features ought to satisfy

$$\mathbf{Q} = \boldsymbol{\tau} \pm \mathbf{q}, \quad E = \hbar\omega(\mathbf{q}). \quad (1.313)$$

The geometrical factor $(1 + \hat{Q}_z^2)$ averages to unity for a multidomain cubic crystal. For an applied magnetic field strong enough to orient the spin direction, this factor can vary between 1 and 2 depending on the direction of the field.

The Hamiltonian of Equation (1.301) is based on a localized picture of electronic band structure, which might not be appropriate in cases such as, for example, transition-metal ferromagnets with wide d bands. Nonetheless, it has been found to provide a reasonable description of spin-wave structure, perhaps because of the strong couplings among d electrons on a given atom compared with (typically) weaker electronic correlations with other atoms.

[Figure 1.30](#) shows spin-wave maps for the magnetically ordered iron chalcogenide $\text{Fe}_{1.05}\text{Te}$. These data have been measured on state-of-the-art spallation-based time-of-flight neutron spectrometers. These instruments access high energy transfers up to the electron-volt range and have become workhorses in the study of strongly correlated electron systems. Dissection of these experimental data along well-defined crystallographic directions allows for the determination of the spin-wave Hamiltonian via the use of Equations (1.302)–(1.305). This particular example is of particular relevance to

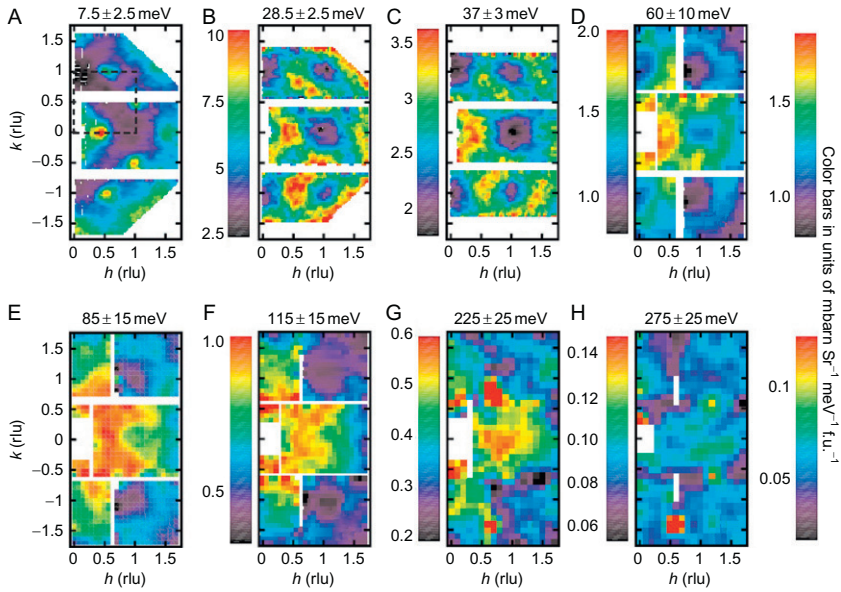


FIGURE 1.30 Constant energy-transfer slices showing the spin-wave structure as a function of increasing energy in $\text{Fe}_{1.05}\text{Te}$ at $T=10$ K. A crystallographic Brillouin zone is shown by the dashed line. Also note the absolute units of the color scale. *Reprinted with permission from Ref. [217].*

our understanding of superconductivity in the pnictides, a new class of materials with unusually high T_c 's. The data shown in Figure 1.30 show that these spin-wave relations are distinctly different from first-principles predictions and parallel observations in the closely related compound CaFe_2As_2 . Whereas nearest-neighbor exchange couplings are quite different in both systems, next-nearest-neighbor terms are remarkably similar. This finding suggests that superconductivity in the pnictides and chalcogenides share a common magnetic origin intimately related to next-nearest-neighbor energy couplings.

Antiferromagnetic materials with purely exchange interactions between ions on different sublattices lead to dispersion relations for the “acoustic” branches that are linear in q for small q . Anisotropic interactions, in which the spin direction is linked to the crystal lattice through the interaction of the orbital moment with the ligand field, usually introduce an extra term leading to a dispersion relation with finite frequency at $q=0$.

1.6.3 Crystal Fields and Magnetic Clusters

Couplings between nearby spins lead to collective spin-wave behavior. In the opposite limit, strong interactions with the electrostatic crystalline field of neighboring ions can lead to energy levels with zero or little dispersion.

Transitions between these levels are observed with inelastic neutron scattering provided that the symmetry is such so as to allow for magnetic-dipole-mediated transitions. These neutron experiments may be particularly useful when optical data are not available, for example, metallic systems.

The double differential cross section for such processes can be derived from Equations (1.226), (1.229), and (1.234):

$$\begin{aligned} \frac{d^2\sigma}{d\Omega dE_f} = & \frac{1}{N_m} \left(\frac{k_f}{k_i} \right) (\gamma r_0)^2 \sum_{\alpha\beta} \left(\delta_{\alpha\beta} - \hat{Q}_\alpha \hat{Q}_\beta \right) |f(\mathbf{Q})|^2 e^{-2W} \times \\ & \times \frac{1}{Z} \sum_{\tau_i \tau_f} e^{-E_{\tau_i}/k_B T} \langle \tau_i | \mu_\alpha^+ | \tau_f \rangle \langle \tau_f | \mu_\beta | \tau_i \rangle \delta(E + E_i - E_f), \end{aligned} \quad (1.314)$$

where Z is the partition function $\sum_{\tau_i} e^{-E_{\tau_i}/k_B T}$.

For rare-earth compounds, simple models for the crystal field based on charged point ions may provide a good description of experimental data if appropriate values are taken for the ionic charges and the moments of the radial wave functions of the magnetic ions. Figure 1.31 shows neutron spectra for PrBi for which these simple considerations give an excellent fit to the data [218]. Crystal-field levels in actinide compounds are usually less well defined because of strong exchange interactions—see the review by Pfeleiderer [219].

Magnetic clusters involving interactions between neighboring magnetic centers represent an intermediate case between the limits of long-range order and isolated spins discussed so far. Interest in this regime (typically involving tens of magnetic ions) comes from their ubiquity in nature, from artificial nanostructures to metalloproteins. Small clusters also offer the opportunity to study fundamental magnetic interactions in a controlled manner and can display novel quantum phenomena such as quantum tunneling of the magnetization. Figure 1.32 shows inelastic neutron data for the magnetic cluster Mn_{12} acetate [220]. This compound is characterized by a total spin of $S=10$ in its ground state, giving rise to an unusually rich neutron spectrum at low energy transfers associated with the splitting of this multiplet structure. In a simplified picture of this cluster, an external ring of eight Mn(III) ions with $S=2$ surrounds a tetrahedron of four Mn(IV) ions ($S=3/2$), and the molecule adopts a plate-like shape with local tetragonal symmetry. At low temperatures, this cluster exhibits regular steps in the hysteresis cycle associated with the reversal of its magnetization, a phenomenon associated with a quantum-tunneling mechanism governed by small (higher-order) spin interactions. The spin Hamiltonian is of the form

$$H = D \left[S_z - \frac{1}{3} S(S+1) \right] + B_4^0 O_4^0 + B_4^4 O_4^4 \quad (1.315)$$

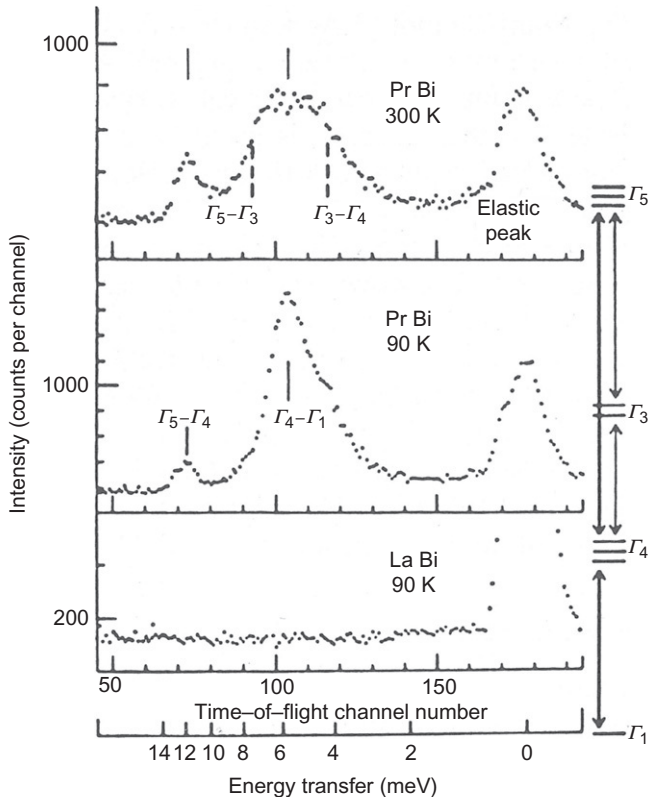


FIGURE 1.31 Neutron-scattering spectra of crystal-field excitations in PrBi. A schematic diagram of the underlying energy-level structure is shown on the right. *Reproduced with permission from Ref. [218].*

with

$$O_4^0 = 35S_z^4 - [30S(S+1) - 25]S_z^2 - 6S(S+1) + 3S^2(S+1)^2 \quad (1.316)$$

and

$$O_4^4 = \frac{1}{2} [S_+^4 + S_-^4], \quad (1.317)$$

where $S_{\pm}$ are raising and lowering operators. The inelastic neutron data are shown in Figure 1.32, where only $\Delta M = \pm 1$ transitions are allowed by virtue of the character of magnetic interactions. A decrease in energy spacings with decreasing energy transfer is primarily dominated by the B_4^0 term above. The best fit of the experimental data using the Hamiltonian given by Equation (1.315) yields $D = -56.7(2)\mu\text{eV}$ and $B_4^0 = -2.89(2) \times 10^{-3}\mu\text{eV}$. A nonzero B_4^4 will mix states with different M values and manifests itself most clearly for energy transfers below 0.3 meV, corresponding to energy levels near the top of the anisotropy barrier. Analysis of the inelastic neutron spectra

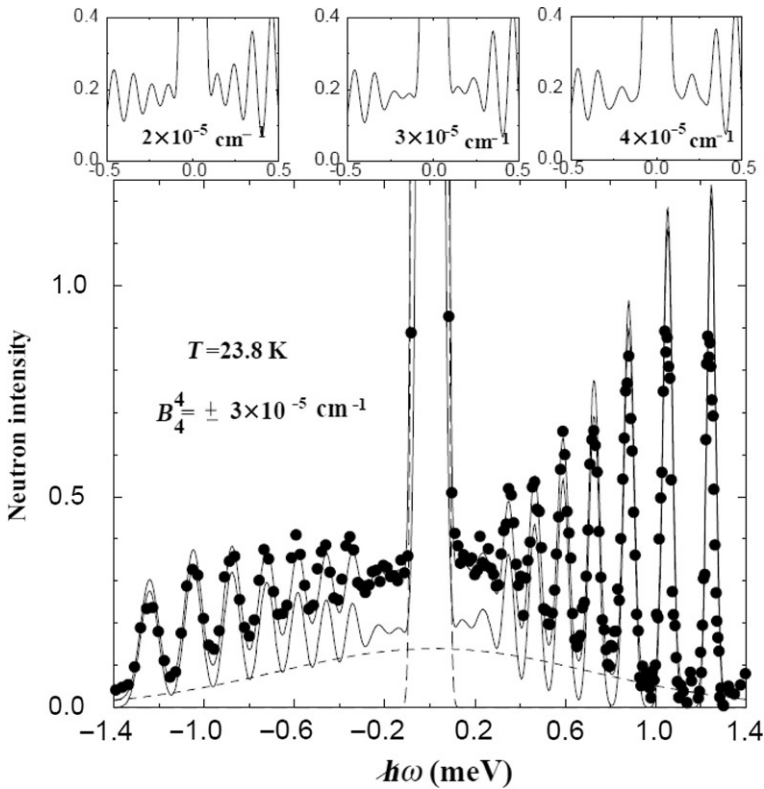


FIGURE 1.32 Inelastic neutron-scattering data for the magnetic cluster Mn_{12} acetate at cryogenic temperatures. The thin line in the main figure corresponds to a simulated spectrum. The top three figures show the sensitivity of the neutron spectra to different values of the coupling constant B_4^4 (see main text for details). *Reprinted with permission from Ref. [220].*

gives $|B_4^4| = 3.71(3) \times 10^{-3} \mu\text{eV}$, where the absolute sign of this parameter remains undetermined owing to the symmetry of the Hamiltonian. This term allows for tunneling subject to the selection rule $\Delta M = \pm 4$, as observed experimentally in magnetization measurements. Interested readers may also wish to consult a recent review of this growing area of research [221].

1.6.4 Spin Fluctuations

Section 1.5.1.2 presented a simple formula for the scattering from a paramagnetic system of localized spins, representing elastic scattering with a $\mathbf{Q}$ dependence given by $|f(\mathbf{Q})|^2$. The situation is more complicated if the spins interact, say, with ligand electric charges in the crystal (the case discussed in the last section) or with lattice vibrations and conduction electrons via Coulomb and exchange interactions. In this case, both elastic and inelastic crystal-field transitions will experience spectral broadening as a function of temperature.

A number of metallic compounds exhibit spectra with pronounced line broadenings. These appear to be related to hybridization with conduction electrons. These interactions lead to a decay in the spin correlations with a characteristic energy $\Gamma(T)$:

$$\langle \mu_\alpha(0) \mu_\beta(t) \rangle = \frac{1}{3} \delta_{\alpha\beta} \langle \mu^2(T) \rangle e^{-\Gamma(T)t/2\hbar}, \quad (1.318)$$

Using Equation (1.296), this magnetic-moment correlation function gives rise to a susceptibility of the form

$$\chi''(\mathbf{Q}, E) = \pi |f(\mathbf{Q})|^2 \chi_s(T) \frac{\Gamma(T)/2}{\pi E^2 + [\Gamma(T)/2]^2} E, \quad (1.319)$$

where $\chi_s(T)$ is the static susceptibility. Substitution of this expression into the double differential scattering cross section (Equation 1.295) shows that the scattering spectrum is a Lorentzian function centered at $E=0$ with full-width-at-half-maximum $\Gamma(T)$.

Mixed-valence compounds such as $\text{Ce}_{0.74}\text{Th}_{0.26}$ exhibit large values of $\Gamma(T)$ and Figure 1.33 shows neutron-scattering spectra for this compound [222]. The widths are so large that it is difficult to measure the wings of the distributions with thermal neutron spectrometers, yet these have been confirmed in high-energy-transfer neutron-scattering experiments [223,224]. An extensive series of measurements in mixed-valence compounds is described by Holland-Moritz *et al.* [225]. The relevance of this class of materials in ferroelectromagnetic and colossal magnetoresistance phenomena have also been reviewed in Refs. [226] and [227], respectively.

In closing this discussion, we note that the temperature dependence of the widths in most mixed-valence compounds is often small or nonexistent, compared with the linear dependence expected for the usual Korring-type relaxation. Even more remarkable, large widths at low temperatures are observed in many actinide compounds such as UAs or UAl_2 , even down to low enough temperatures for some compounds to become magnetically ordered [228]. These have been ascribed to interactions with the conduction electrons.

1.6.5 Interband Transitions

For ferromagnetic metals where magnetic properties derive from the presence of a band of itinerant electrons, the generalized susceptibility can be written down in terms of electron energies and occupation numbers as follows:

$$\chi_0(\mathbf{Q}, E) = -\frac{(g\mu_B)^2}{N_m} \sum_{\mathbf{k}} \frac{n_{(\mathbf{k}+\mathbf{Q})\uparrow} - n_{\mathbf{k}\downarrow}}{E_{(\mathbf{k}+\mathbf{Q})\uparrow} - E_{\mathbf{k}\downarrow} + \Delta + E + i\delta}, \quad (1.320)$$

where δ is a small real number enabling one to calculate $\chi''(\mathbf{Q}, E)$ as $\delta \rightarrow 0$. Equation (1.320) neglects interactions between electrons and cannot give collective excitations such as spin waves. If the electrons are allowed to interact

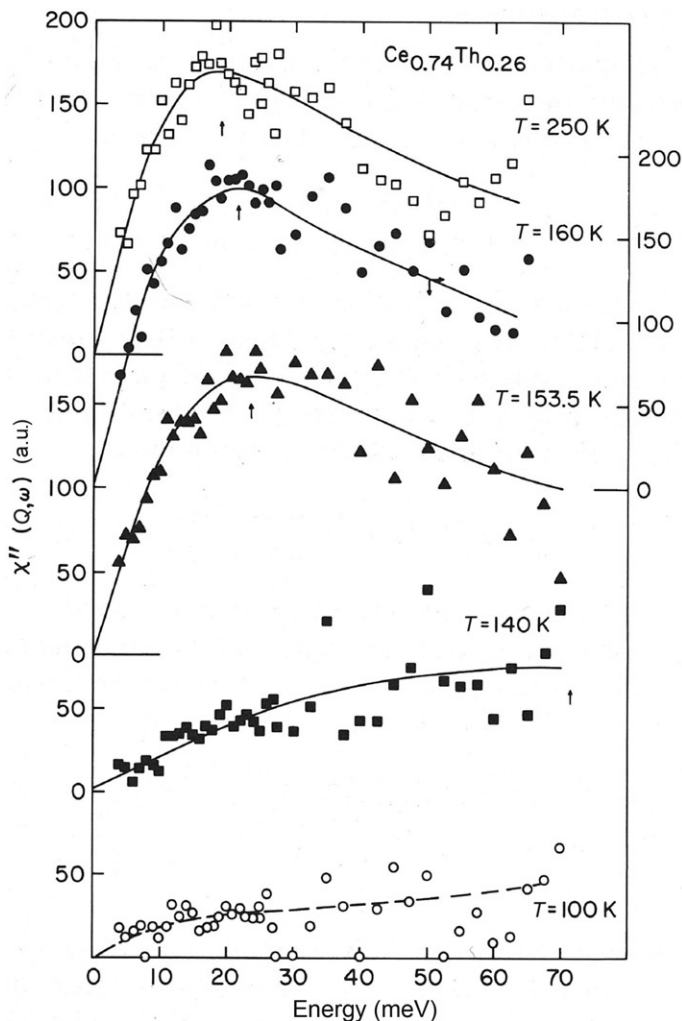


FIGURE 1.33 Susceptibilities obtained from inelastic neutron-scattering data for the mixed-valence compound $\text{Ce}_{0.74}\text{Th}_{0.26}$ at several temperatures. Reprinted with permission from Ref. [222].

electrostatically, with I representing an average value for the interaction, the resulting susceptibility can be written as

$$\chi(\mathbf{Q}, E) = \frac{\chi_0(\mathbf{Q}, E)}{1 - I\chi_0(\mathbf{Q}, E)}. \quad (1.321)$$

This susceptibility produces spin waves for $(\mathbf{Q}, E)$ values such that the denominator goes to zero. As shown in Section 9.3d of Ref. [108], this

condition leads to an expression for the double differential cross section that can be related to Equations (1.311) and (1.312), originally derived for localized spins.

We also note that at zero momentum transfer $\mathbf{Q}$, Equation (1.320) simplifies to

$$\chi_0''(\mathbf{Q}, E) = \pi(g\mu_B)^2 \frac{\Delta}{I} \delta(E - \Delta), \quad (1.322)$$

where

$$\Delta = I(n_\downarrow - n_\uparrow) \quad (1.323)$$

is the up-down band splitting, and results in a peak in the scattering at $E = \Delta$. For $|\mathbf{Q}| > 0$, these modes broaden into a band of single-particle excitations known as Stoner modes, with characteristic energies of order several electron volts. Until now, these have not been conclusively identified in neutron experiments, although indirect evidence has been presented—see, for example, Refs. [229,230].

In the paramagnetic region, the up and down spin states are degenerate and Equation (1.320) becomes

$$\chi_0''(\mathbf{Q}, E) = -\pi \frac{(g\mu_B)^2}{N_m} \sum_{\mathbf{k}} n_{(\mathbf{k}+\mathbf{Q})} (1 - n_{\mathbf{k}}) \delta(E_{(\mathbf{k}+\mathbf{Q})} - E_{\mathbf{k}} - E). \quad (1.324)$$

For a polycrystalline specimen, Equation (1.324) can be written in terms of the electronic density of states $g(\varepsilon)$ as

$$\chi_0''(\mathbf{Q}, E) = -\pi \frac{(g\mu_B)^2}{N_m} \iint d\varepsilon d\varepsilon' g(\varepsilon) g(\varepsilon') n(\varepsilon) [1 - n(\varepsilon')] \delta(\varepsilon' - \varepsilon - E). \quad (1.325)$$

It becomes therefore possible to measure (in principle) the joint density of states within some distance of the Fermi surface, perhaps ~ 1 eV. Whilst such measurements may be possible at spallation neutron sources, numerical estimates indicate that the intensities will be extremely low except in exceptionally favorable cases [231].

1.6.6 Critical Scattering

So-called magnetic critical scattering is a particular case of the anomalous scattering arising from large fluctuations near a second-order phase transition. This phenomenon has been the subject of an extremely active field, and we limit ourselves to providing a brief account of some simple cases.

For a ferromagnetic Bravais lattice characterized by a Heisenberg Hamiltonian (cf. Equation 1.301), the susceptibility defined by Equation (1.306) can be written as

$$\chi''_{\alpha\alpha}(\mathbf{Q}, E) = \pi g^2 \mu_B^2 \left(1 - e^{-E/k_B T}\right) \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \langle S_{\alpha}(\mathbf{Q}, 0) S_{\alpha}(\mathbf{Q}, t) \rangle e^{-iEt/\hbar} dt, \quad (1.326)$$

where the correlation functions are of the form

$$S_{\alpha}(\mathbf{Q}, t) = \sum_{\mathbf{l}} \mathbf{e}^{-i\mathbf{Q}\cdot\mathbf{l}} S_{\mathbf{l}}(t). \quad (1.327)$$

Defining

$$\Delta S_{\alpha}(\mathbf{Q}, t) = \sum_{\mathbf{l}} \mathbf{e}^{-i\mathbf{Q}\cdot\mathbf{l}} [S_{\mathbf{l}}(t) - \langle S_{\mathbf{l}}(t) \rangle], \quad (1.328)$$

we can then rewrite the susceptibility as

$$\chi''_{\alpha\alpha}(Q, E) = \pi g^2 \mu_B^2 \left(1 - e^{-E/k_B T}\right) \left[N^2 \delta_{Q0} \langle S_{\alpha} \rangle^2 \delta(E) + \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \langle \Delta S_{\alpha}(\mathbf{Q}, 0) \Delta S_{\alpha}(\mathbf{Q}, t) \rangle e^{-iEt/\hbar} dt \right]. \quad (1.329)$$

Since $\langle S_{\alpha} \rangle \rightarrow 0$ as $T \rightarrow T_c$, the first term within square brackets (corresponding to Bragg scattering) is small near T_c . On the contrary, the second term (related to fluctuations in the magnetization) diverges as $T \rightarrow T_c$. This behavior is most apparent in the limits $\mathbf{Q} \rightarrow 0$ (or $\mathbf{Q} \rightarrow \tau$) and $t \rightarrow 0$ (static approximation), where the second term tends to

$$\chi''_{\alpha\alpha, \text{cr}} = \pi g^2 \mu_B^2 \left(1 - e^{-E/k_B T}\right) \langle \Delta S_{\alpha}(0, 0)^2 \rangle = \pi \left(1 - e^{-E/k_B T}\right) \langle \Delta M_{\alpha}^2 \rangle, \quad (1.330)$$

where ΔM is the fluctuation in the magnetization and, therefore,

$$\chi''_{\alpha\alpha, \text{cr}} = \pi \left(1 - e^{-E/k_B T}\right) N k_B T \chi_{s, \alpha\alpha}, \quad (1.331)$$

where χ_s is the static susceptibility. Near the critical temperature T_c , magnetization fluctuations become large and χ_s diverges as $T \rightarrow T_c$. This divergent behavior leads to anomalously large scattering at $\mathbf{Q} = \tau$ known as *magnetic critical scattering*.

Away from Bragg reflections, the energy and wave vector dependence of critical scattering can be examined via the use of the isothermal susceptibility introduced in Equation (1.298):

$$\chi''_{\alpha\alpha, \text{cr}} = \pi N \chi_{\alpha}(\mathbf{Q}) E F_{\alpha}(\mathbf{Q}, E), \quad (1.332)$$

where $F_{\alpha}(\mathbf{Q}, E)$ is a spectral-weight function having unit area when integrated over energy E (see Equation 1.299). By analogy with the classical theory of critical opalescence, Van Hove [232] postulated the existence of instantaneous (exponentially decaying) spatial correlations between spins. The isothermal susceptibility then has the form

$$\chi_\alpha(\mathbf{Q}) = \frac{\chi_0}{r_1^2(q^2 + \kappa_1^2)}, \quad (1.333)$$

where $q = |\mathbf{Q} - \boldsymbol{\tau}|$, χ_0 is the susceptibility of a noninteracting spin system, r_1 represents the range of the exchange interactions, and $\kappa_1^2 r_1^2 = (T/T_c - 1)$. To include an implicit time dependence, Van Hove assumed that the spin fluctuations $S_\alpha(\mathbf{Q}, t)$ decay according to stochastic translational diffusion, that is,

$$F_s(\mathbf{Q}, E) = \left(\frac{1}{\pi}\right) \frac{\Gamma(q)}{\Gamma^2(q) + E^2}. \quad (1.334)$$

with

$$\Gamma(q) = \hbar A q^2, \quad (1.335)$$

where A is a diffusion constant expected to vary with temperature as $\kappa_1^{1/2}$ or $\chi_\alpha(0)^{1/4}$. Dynamical scaling theory [233,234] also gives a Lorentzian line shape, but with the width functions given more generally by

$$\Gamma(q, \kappa_1) = c q^{5/2} f\left(\frac{\kappa_1}{q}\right), \quad (1.336)$$

which goes over to Equation (1.335) only in the hydrodynamic limit $\kappa_1/q \rightarrow 0$.

Figure 1.34 from Collins *et al.* [235] shows data for the critical scattering from iron as a function of temperature near T_c . Well below T_c , the scans at constant $Q = 0.05 \text{ \AA}^{-1}$ show well-defined energy-loss and energy-gain magnon peaks. As a temperature is raised, the magnon line widths increase, and as T_c is approached only a single peak is observed, reaching maximum intensity at $T = T_c$. This critical scattering is observed to follow the behavior predicted by Equations (1.326)–(1.334). The temperature dependence of $\Gamma(q)$ was found to be much weaker than that predicted by Equation (1.335). It was pointed out by Als-Nielsen [236] that this is consistent with Equation (1.336), however, since the κ_1/q values covered in this experiment lie in a region where calculated scaling functions $f(\kappa_1/q)$ exhibit a minimum and thus vary quite slowly with temperature.

Critical scattering is also observed near the Néel temperature T_N in antiferromagnetic materials. In uniaxial crystals, two components of the susceptibility are involved, one parallel and one transverse to the anisotropy axis, with different values of κ_1 and r_1 . In MnF_2 , for example, $\chi_{\parallel}(\mathbf{Q})$ diverges at T_N , whereas $\chi_{\perp}(\mathbf{Q})$ remains finite at T_N but obeys critical behavior corresponding to a temperature $T_{\perp} < T_N$ [237]. In the planar antiferromagnetic K_2NiF_4 , critical scattering is concentrated along ridges in the direction perpendicular to the NiF_2 planes, that is, no three-dimensional critical scattering is observed [238]. Thus, even though the actual transition is three-dimensional, the long-range order is achieved simultaneously within and between the planes, and it is the ordering within the planes that drives the phase transition—for further details, see the comprehensive monograph by Collins [239].

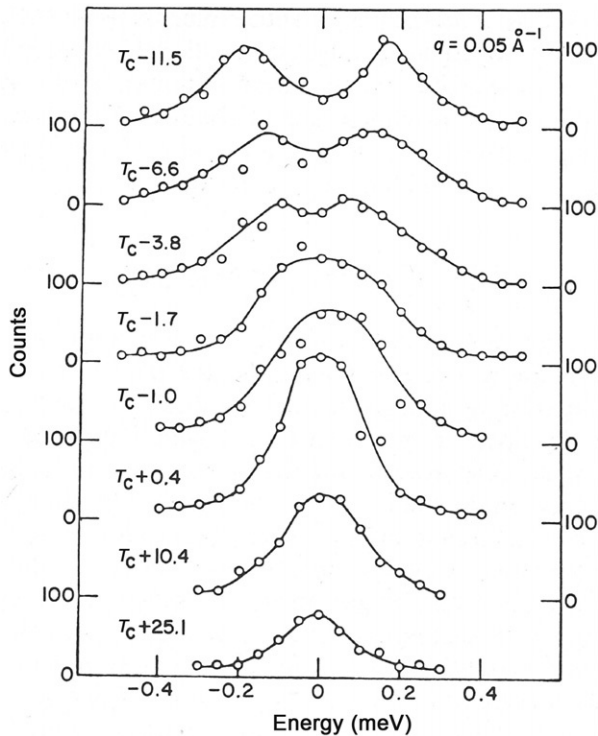


FIGURE 1.34 Magnetic critical scattering from iron. Reprinted with permission from Ref. [235].

1.7 NUCLEAR SPIN: ORDER AND DISORDER

As discussed in Section 1.5, neutron scattering from magnetic centers has an intrinsic dependence on spin orientation arising from the electromagnetic interaction between the magnetic moment of the neutron and that of the unpaired electron density, the latter giving rise to permanent magnetic moments on the order of a Bohr magneton μ_B . In the case of neutron scattering from nuclei, the magnetic moments are on the order of a nuclear magneton, $\mu_N = (m_e/m_p)\mu_B$, and the corresponding electromagnetic interaction is (while not necessarily zero) much weaker. Notwithstanding these limitations, we recall that nuclear scattering is mediated by the strong nuclear force, and for the case of non-zero-spin (i.e., NMR active nuclides), the scattering length is generally dependent on the relative orientation of neutron and nuclear spins during the collision. In cases where the nuclear spins are completely disordered, the spin dependence of the scattering process leads to the incoherent component of the scattering cross section. However, in cases where the nuclear spins are correlated, the spin dependence of the scattering amplitude can be exploited for the determination of nuclear-spin correlations. This section examines the basic features of the scattering of thermal neutrons from

ensembles of nuclear spins. The formalism follows closely the one presented in the excellent monograph of Abragam and Goldman [240].

1.7.1 A Closer Look at Nuclear Spins

Away from nuclear resonances, the scattering of thermal neutrons by nuclei does not involve the participation of partial waves with orbital angular momentum $l > 0$, that is, in the language of scattering theory, nuclear scattering is entirely s-wave in character. Physically speaking, this condition arises from the orders-of-magnitude difference between neutron wavelengths ($\text{\AA}$) and the size of nuclei ($\text{fm} = 10^{-5} \text{\AA}$). In this case, the neutron–nucleus scattering state has a total angular momentum $\mathbf{J}$ given by the quantum-mechanical vector addition of their intrinsic spins, namely,

$$\mathbf{J} = \mathbf{I} + \frac{1}{2}\boldsymbol{\sigma}, \quad (1.337)$$

where $\mathbf{I}$ and $1/2\boldsymbol{\sigma}$ correspond to nuclear and neutron spins, respectively. Since the neutron has spin $1/2$, there are only two possible values for J :

$$J = I \pm \frac{1}{2}. \quad (1.338)$$

The strength of the scattering for each ($\pm$) angular-momentum state is characterized by their corresponding scattering lengths b^+ and b^- , respectively. In operator form, the general expression for the spin-dependent neutron-scattering lengths was already introduced in Equations (1.248) and (1.249) in the context of polarized neutrons. To facilitate the present discussion, we rewrite them here

$$\hat{b} = \bar{b} + \frac{1}{2}b_N\boldsymbol{\sigma}\cdot\mathbf{I}, \quad (1.339)$$

where

$$\begin{aligned} \bar{b} &= \frac{1}{(2I+1)}[(I+1)b^+ + Ib^-], \\ b_N &= \frac{2}{(2I+1)}[b^+ - b^-], \end{aligned} \quad (1.340)$$

and we recall that $(2I+1)$ corresponds to the M_I sublevel degeneracy of the nuclear angular momentum.

1.7.2 Scattering Cross Sections

The interactions between nuclear spins are very weak and, therefore, the energies involved in their dynamics are very small. Even in the case of solid ^3He (where the exchange interaction is several orders of magnitude larger than dipolar interactions), the energies involved are on the order of a fraction of

a μeV . Information on the static correlations between nuclear spins can be obtained from elastic scattering. Following the early work by Hayter *et al.* on single crystals of lanthanum magnesium nitrate [241], several such experiments have been reported [242–257]. From a practical viewpoint, the approach offers the possibility of enhancing coherent Bragg intensities relative to incoherent scattering. Inelastic neutron-scattering studies are significantly more challenging, yet weak nuclear-spin excitations associated with hyperfine transitions in compounds such as the antiferromagnetic metal oxide CoO [258], the spin ice $\text{Ho}_2\text{Ti}_2\text{O}_7$ [259], or metallic HoAl_2 [260] have been observed at energy transfers in the μeV region using state-of-the-art neutron instrumentation. A notable exception to the considerations given above occurs in certain systems of high symmetry such as H_2 , CH_4 , and methyl (CH_3) functional groups (as well as their deuterated counterparts), where the Pauli Exclusion Principle ties the molecular and nuclear-spin wavefunctions in such a way that energy changes associated with atomic motions (primarily rotations) are tied to changes in total nuclear spin. H_2 represents the most extreme case, where the transition between $J=0$ and $J=1$ rotational levels at 15 meV is accompanied by a change in total nuclear spin from $I=0$ (*para* state) to $I=1$ (*ortho* state). While interesting in their own right, these particular cases will not be treated any further in the discussion that follows below (see discussion in Section 1.4.5.2 and Chapter 6).

To examine how neutron scattering can be used to probe nuclear-spin correlations, we recall the basic formula for the differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \left(\sum_j \langle |\hat{b}_j|^2 \rangle + \sum_{j \neq j'} \langle \hat{b}_j \hat{b}_{j'}^\dagger \rangle e^{-i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_{j'})} \right), \quad (1.341)$$

where the averages are over the spin variables. For nuclei with non-zero-spin, the scattering lengths given by Equations (1.339) and (1.340) give

$$|\hat{b}_j|^2 = \bar{b}^2 + \bar{b}b_N \sigma \cdot \mathbf{I}_j + \frac{1}{4}b_N^2 (\sigma \cdot \mathbf{I}_j)^2. \quad (1.342)$$

Using the general relation for the square of a scalar product between quantum-mechanical angular-momentum operators

$$(\sigma \cdot \mathbf{I})^2 = \mathbf{I} \cdot \mathbf{I} + i\sigma \cdot \mathbf{I} \times \mathbf{I} = I(I+1) - \sigma \cdot \mathbf{I}, \quad (1.343)$$

the two scattering-length products in Equation (1.341) can be expressed as

$$\begin{aligned} |\hat{b}_j|^2 &= \bar{b}^2 + \bar{b}b_N \sigma \cdot \mathbf{I}_j + \frac{1}{4}b_N^2 [I(I+1) - \sigma \cdot \mathbf{I}_j], \\ \hat{b}_j \hat{b}_{j'}^\dagger &= \bar{b}^2 + \frac{1}{2}\bar{b}b_N [\sigma \cdot \mathbf{I}_j + \sigma \cdot \mathbf{I}_{j'}] + \frac{1}{4}b_N^2 [\mathbf{I}_j \cdot \mathbf{I}_{j'} + i\sigma \cdot (\mathbf{I}_j \times \mathbf{I}_{j'})]. \end{aligned} \quad (1.344)$$

Substitution of these expressions into Equation (1.341) provides a general expression for the spin-dependent scattering differential cross section.

1.7.3 Uncorrelated and Correlated Spin Ensembles

The simplest case to consider involves random nuclear-spin orientations, as there is no correlation between the spin-dependent scattering amplitudes of two particles and their relative position. In this case, Equation (1.341) simplifies to

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \left(\sum_j \langle |\hat{b}_j|^2 \rangle + \sum_{j \neq j'} \langle \hat{b}_j \hat{b}_{j'}^+ \rangle e^{-i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_{j'})} \right). \quad (1.345)$$

If on the right-hand side of Equation (1.345) we add and subtract $\langle \hat{b}_j \hat{b}_{j'}^+ \rangle$ to the first and second term, respectively, we obtain

$$\frac{d\sigma}{d\Omega} = \frac{1}{N} \sum_j \left[\langle |\hat{b}_j|^2 \rangle - \langle \hat{b}_j \hat{b}_j^+ \rangle \right] + \frac{1}{N} \sum_{j, j'} \langle \hat{b}_j \hat{b}_{j'}^+ \rangle e^{-i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_{j'})}. \quad (1.346)$$

Even in the absence of spin correlations in the above sense, the spin system can have a net macroscopic polarization and the beam can also be polarized. Denoting the polarization of the target spin system by P and the polarization of the neutron beam by p , both referred to the same quantization axis, we have

$$\langle \mathbf{I}_j \cdot \mathbf{I}_{j'} \rangle = \langle \mathbf{I}_j \rangle \cdot \langle \mathbf{I}_{j'} \rangle = P^2 I^2, \quad \langle \mathbf{I}_j \times \mathbf{I}_{j'} \rangle = 0, \quad \langle \sigma \cdot \mathbf{I}_j \rangle = pPI. \quad (1.347)$$

From Equation (1.344):

$$\begin{aligned} \langle |\hat{b}_j|^2 \rangle &= \bar{b}^2 + \bar{b} b_N pPI + \frac{1}{4} b_N^2 [I(I+1) - pPI], \\ \langle \hat{b}_j \hat{b}_{j'}^+ \rangle_{j \neq j'} &= \bar{b}^2 + \bar{b} b_N pPI + \frac{1}{4} b_N^2 P^2 I^2, \end{aligned} \quad (1.348)$$

which gives a differential scattering cross section [cf. Equation (1.346)] of the form

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{1}{4} b_N^2 [I(I+1) - pPI - P^2 I^2] \\ &+ \left(\bar{b}^2 + \bar{b} b_N pPI + \frac{1}{4} b_N^2 P^2 I^2 \right) \frac{1}{N} \sum_{j, j'} e^{-i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_{j'})}. \end{aligned} \quad (1.349)$$

For a crystal with one atom per unit cell, the double sum in Equation (1.349) simplifies to

$$\frac{1}{N} \sum_{j, j'} e^{-i\mathbf{Q} \cdot (\mathbf{R}_j - \mathbf{R}_{j'})} = \sum_j e^{-i\mathbf{Q} \cdot \mathbf{R}_j}, \quad (1.350)$$

where the sum is over all N atoms (or cells). Equation (1.349) then reads

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{1}{4} b_N^2 [I(I+1) - pPI - P^2 I^2] \\ &+ \left(\bar{b}^2 + \bar{b} b_N pPI + \frac{1}{4} b_N^2 P^2 I^2 \right) \sum_j e^{-i\mathbf{Q} \cdot \mathbf{R}_j}, \end{aligned} \quad (1.351)$$

or in terms of cross sections

$$\frac{d\sigma}{d\Omega} = \frac{1}{4\pi} \left(\sigma_i + \sigma_c \frac{(2\pi)^3}{v_0} \sum_{\tau} \delta(\mathbf{Q} - \boldsymbol{\tau}) \right), \quad (1.352)$$

where the incoherent and coherent cross sections are given by

$$\sigma_i = \pi b_N^2 [I(I+1) - pPI - P^2 I^2] \quad (1.353)$$

and

$$\sigma_c = 4\pi \left(\bar{b}^2 + \bar{b} b_N pPI + \frac{1}{4} b_N^2 P^2 I^2 \right), \quad (1.354)$$

respectively.

For an unpolarized target ($P=0$) and with $\bar{b}$ and b_N expressed in terms of the up and down scattering amplitudes [cf. Equation (1.340)], the cross sections can be written as

$$\sigma_i = \pi I(I+1) b_N^2 = 4\pi \frac{I(I+1)}{(2I+1)^2} (b^+ - b^-)^2 \quad (1.355)$$

and

$$\sigma_c = 4\pi \bar{b}^2 = 4\pi \frac{1}{(2I+1)^2} [(I+1)b^+ + Ib^-]^2. \quad (1.356)$$

These two expressions are the same as those derived in Section 1.2 [cf. Equation (1.46)]. From Equation (1.353), a fully polarized target ($P=1$) will have no incoherent scattering when $pP=1$, that is, when both target spins and neutrons are polarized along the same direction.

To treat correlations between nuclear-spin orientations and the relative positions of particle pairs, it becomes necessary to generalize the treatment given above. For example, if Equation (1.347) is replaced by

$$\langle \mathbf{I}_d \cdot \mathbf{I}_{d'} \rangle = \langle \mathbf{I}_d \rangle \cdot \langle \mathbf{I}_{d'} \rangle = P_d P_{d'} I^2, \quad (1.357)$$

then we are implicitly assuming that there is only one kind of nucleus with spin I , and P_d is the polarization for a given nucleus d . The differential cross section in this case is given by

$$\frac{d\sigma}{d\Omega} = \frac{1}{4\pi} \left(\sigma_i + \sigma_c \frac{(2\pi)^3}{v_0} \sum_{\tau} |F(\boldsymbol{\tau})|^2 \delta(\mathbf{Q} - \boldsymbol{\tau}) \right) \quad (1.358)$$

with an incoherent cross section of the form

$$\sigma_i = \pi b_N^2 \sum_d [I(I+1) - pP_d I - P_d^2 I^2] \quad (1.359)$$

summed over all nuclei in the unit cell. The new structure factor $F(\boldsymbol{\tau})$ is given by

$$|F(\boldsymbol{\tau})|^2 = \sum_{d,d'} \left[\bar{b}^2 + \frac{1}{2} \bar{b} b_N p (P_d + P_{d'}) I + \frac{1}{4} b_N^2 P_d P_{d'} I^2 \right] e^{-i\mathbf{Q} \cdot (\mathbf{d} - \mathbf{d}')} \quad (1.360)$$

In the case of ferromagnetic ordering, Bragg peaks will experience an intensity modulation arising from the spin-dependent terms in Equation (1.360), but no new peaks will appear. Also, the onset of polarization will affect the incoherent scattering, as explicitly shown by Equation (1.359).

For antiferromagnetic nuclear-spin ordering, new Bragg features will appear in addition to those observed in the same structure with no nuclear-spin order. For the antiferromagnetic peaks, the differential cross section reads

$$\left(\frac{d\sigma}{d\Omega} \right)_m = \frac{(2\pi)^3}{v_m} \frac{1}{4} \sum_{d,d'} b_N^2 P_d P_{d'} I^2 e^{-i\mathbf{Q} \cdot (\mathbf{d} - \mathbf{d}')} \sum_{\boldsymbol{\tau}_m} \delta(\mathbf{Q} - \boldsymbol{\tau}_m), \quad (1.361)$$

where the subscript “m” denotes summation over the antiferromagnetic unit cell. Figure 1.35 shows neutron-diffraction data from a ^{65}Cu single crystal at ultralow (milliKelvin) temperatures. These data evince the emergence of long-range nuclear-spin antiferromagnetic order upon the application of a small magnetic field. The time evolution of the nuclear-spin antiferromagnetic (100) peak depends strongly on the magnitude of the applied magnetic field between zero and its critical value at 0.25 mT.

Although these experiments still remain quite challenging, there is clearly scope for further work in this area. In particular, the implementation of *dynamic nuclear polarization* schemes [261] including their implementation at high magnetic fields [262] offers the exciting prospects of attaining a sufficiently large nuclear polarization so as to enable a number of important applications of neutron scattering where element-specific contrast variation would be highly desirable. Signal-enhancement factors of ca. 10^{+4} have already been demonstrated in liquid-state NMR studies [263] and promising neutron experiments have been reported recently using Laue diffraction [257] and small-angle techniques [254–256]. In this context, the scheme proposed by Buckingham [264] to modulate neutron-scattering intensities via the use of well-established pulsed neutron-magnetic-resonance techniques is most enticing, and still a challenge for its successful realization in the laboratory. Further improvements in neutron sources and instrumentation would certainly accelerate the above trend.

1.8 OUTLOOK

This introductory chapter has brought together the basic theory and formalism of thermal neutron scattering for condensed matter research, and given some examples to illustrate its present use in a variety of applications. Our choices

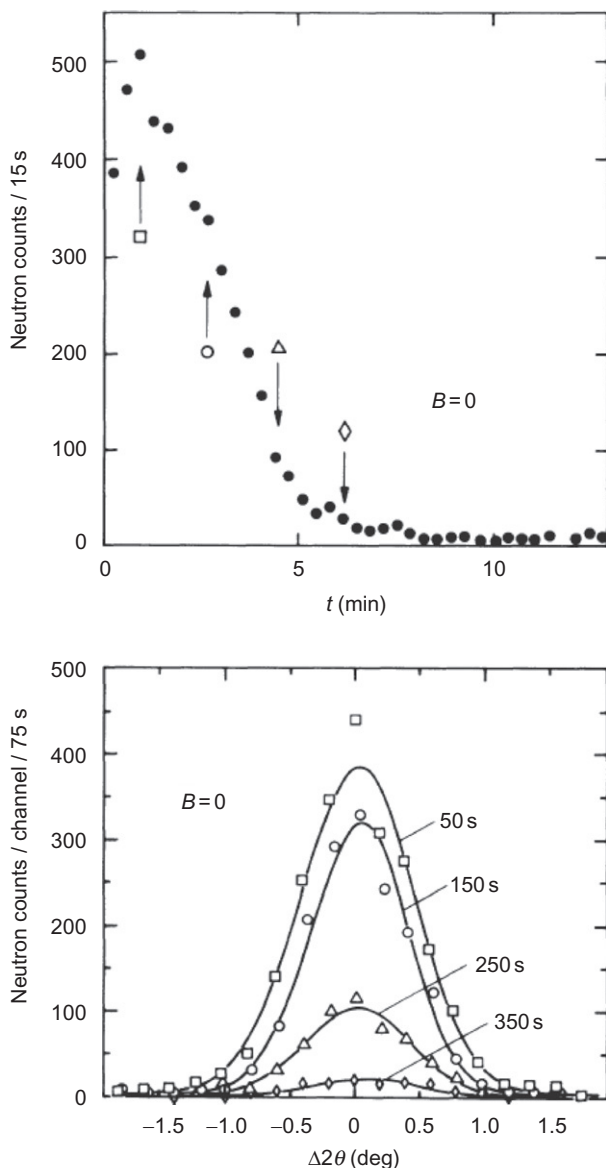


FIGURE 1.35 Top: time dependence of the integrated intensity of the (100) antiferromagnetic Bragg reflection from a ^{65}Cu single crystal following demagnetization. Bottom: (100) reflection at the points in time marked in the figure. Reprinted with permission from Ref. [247].

in the selection of this material are unavoidably colored by our personal preferences, appreciation, and knowledge of the vast and rapidly growing number of applications of neutron scattering. It is our hope that the in-depth treatments of specific research areas given in the subsequent chapters of this and

future volumes will constitute a base reference for the assessment, planning, and successful execution of neutron-scattering experiments in all of these diverse applications.

The advances made in the quarter-century following the publication of the previous version of this volume [265] have been spectacular, and it is tempting, perhaps foolhardy, to speculate about what the next quarter-century will bring in this field. Some exciting and imaginative proposals have been suggested by Furrer [266]. We could add some more obvious possibilities as follows.

First, qualitative scientific advances generally follow developments in sources and instrumentation. New innovations in these areas are respectively described in Chapters 2 and 3. In particular, the third-generation spallation sources at Oak Ridge and Tokai are getting well underway, and the novel type of source represented by the European Spallation Source provides an exciting challenge for new kinds of instruments.

Instrumental developments such as innovative uses of polarized neutrons, new detectors, and the exploitation of the explosion in computing and data-handling capabilities will likewise pave the way for major scientific advances.

No less important are unabated developments in sample-environment capabilities. Of the many that we could mention, the access to ever-higher pressures, and larger samples at moderately high pressures, is making it possible to obtain new science on solids and liquids and relate it to important issues in geology. Recent advances in the various levitation techniques are making it possible to study materials at previously inaccessible temperatures and in a greater range of metastable states, while *in situ* measurements in real time and under technologically and industrially relevant conditions can connect neutron measurements with the important problems in battery and fuel-cell materials or catalysis.

Finally, the excitement of a neutron-scattering measurement derives from the materials and phenomena being investigated. As one example, recent advances in nanoscale science and technology provides a host of opportunities for neutron experiments, ranging from the study of atomic and magnetic structure as one goes to smaller particle dimensions and confinement, to the measurement of size distributions with small-angle scattering and the investigation of nm-thin films and coatings with reflectivity techniques.

We hope that the background to neutron-scattering techniques outlined in this volume will help our increasingly interdisciplinary scientific community to exploit these exciting opportunities beyond the imaginable at the present time.

REFERENCES

Section 1

- [1] Wietfeldt FE, Greene GL. *Rev Mod Phys* 2011;83:1173.
- [2] For recommended values of all constants defined in this chapter, see the *List of Commonly Used Symbols* and references therein.
- [3] Chadwick J. *Nature* 1932;129:312.

- [4] URL: www.nobelprize.org/nobel_prizes/physics/laureates/1935/ [last accessed on July 1, 2013].
- [5] Segrè E. *Rev Mod Phys* 1955;27:257.
- [6] Dubbers D, Schmidt MG. *Rev Mod Phys* 2011;83:1111.
- [7] Alexandrova YuA. *Fundamental properties of the neutron*. Oxford: Oxford University Press; 1992.
- [8] Magán M, Sordo F, Zanini L, Terrón S, Ghiglini A, Martínez F, et al. *Nucl Instr Methods A* 2013;729:417.
- [9] See Chapter 4 in Windsor CG. *Pulsed neutron scattering*. London: Taylor and Francis; 1981.
- [10] Price DL, Fu L, Bermejo FJ, Fernandez-Alonso F, Saboungi M-L. *Chem Phys* 2013; <http://dx.doi.org/10.1016/j.chemphys.2013.02.020>, in press.
- [11] Enderby JE. *Chem Soc Rev* 1995;24:159.
- [12] Sears VF. *Neutron optics: an introduction to the theory of neutron optical phenomena and their applications*. Oxford: Oxford University Press; 1989.
- [13] Shull CG, Yamada Y. *J Phys Soc Jpn* 1962;17(Suppl. B-III):1.
- [14] Word RE, Trammell GT. *Phys Rev B* 1981;24:2430.
- [15] Newton RG. *Scattering theory of waves and particles*. New York: McGraw-Hill; 1966.
- [16] Schimmel HG, Huot J, Chapon LC, Tichelaar FD, Mulder FM. *J Am Chem Soc* 2005;127:14348.
- [17] Als-Nielsen J, McMorrow D. *Elements of modern X-ray physics*. 2nd ed. Chichester: John Wiley & Sons; 2011.
- [18] Margaritondo G. *Elements of synchrotron light for biology, chemistry, and medical physics*. Oxford: Oxford University Press; 2002.
- [19] Rehr JJ, Albers RC. *Rev Mod Phys* 2000;72:621.
- [20] Parsons JG, Aldrich MV, Gardea-Torresdey JL. *Appl Spectrosc Rev* 2002;37:187.
- [21] Rehr JJ, Amkudinov AL. *Coord Chem Rev* 2005;249:131.
- [22] Filipponi A, Di Cicco A, Natoli CR. *Phys Rev B* 1995;52:15122.
- [23] Filipponi A, Di Cicco A. *Phys Rev B* 1995;52:15135.
- [24] Cole JM, Wright AC, Newport RJ, Sinclair RN, Fischer HE, Cuello GJ, et al. *J Phys Condens Matter* 2007;19:056002.
- [25] Coldea R, Tennant DA, Wheeler EM, Wawrzynska E, Prabhakaran D, Telling M, et al. *Science* 2010;327:177.
- [26] See Chapter 2 of the ESS Technical Design Report. URL: eval.esss.lu.se/cgi-bin/public/DocDB/ShowDocument?docid=274 [Last accessed on July 4, 2013].
- [27] Burkel E. *Rep Prog Phys* 2000;63:171.
- [28] Scopigno T, Ruocco G, Sette F. *Rev Mod Phys* 2005;77:881.
- [29] Toellner TS, Alatas A, Said AH. *J Synchrotron Radiat* 2011;18:605.
- [30] Sinn H. *J Phys Condens Matter* 2001;13:7525.
- [31] Ament LJP, van Veenendaal M, Devereaux TP, Hill JP, van den Brink J. *Rev Mod Phys* 2011;83:705.
- [32] Kotani A, Shin S. *Rev Mod Phys* 2001;73:203.
- [33] Grubel G, Zontone F. *J Alloys Compd* 2011;363:3.
- [34] Leheny RL. *Curr Opin Colloid Interface Sci* 2012;17:3.
- [35] portal.slac.stanford.edu/sites/lcls_public/Pages/Default.aspx [last accessed on August 21, 2013].
- [36] Berne BJ, Pecora R. *Dynamic light scattering, with applications to chemistry, biology, and physics*. Mineola: Dover Publications; 2000.
- [37] Hayes W, Loudon R. *Scattering of light by crystals*. Mineola: Dover Publications; 2004.

- [38] Harrys DC, Bertolucci MD. Symmetry and spectroscopy—an introduction to vibrational and electronic spectroscopy. Mineola: Dover Publications; 1989.
- [39] Baxter JB, Guglietta GW. *Anal Chem* 2011;83:4342.
- [40] Ebbinghaus S, Kim SK, Heyden M, Yu X, Heugen U, Gruebele M, et al. *Proc Natl Acad Sci USA* 2007;104:20749.
- [41] Kutteruf MR, Brown CM, Iwaki LK, Campbell MB, Korter TM, Heilweil EJ. *Chem Phys Lett* 2003;375:337.
- [42] Damascelli A. *Phys Scr* 2004;T109:61.
- [43] Damascelli A, Hussain Z, Shen ZX. *Rev Mod Phys* 2003;75:473.
- [44] Sanders JKM, Hunter BK. *Modern NMR spectroscopy—a guide for chemists*. Oxford: Oxford University Press; 1994.
- [45] Price WS. *Concept Magn Res* 1998;10:197.
- [46] Parella T. *Magn Reson Chem* 1998;36:467.
- [47] Jardine AP, Hedgeland H, Alexandrowicz G, Allison W, Ellis J. *Prog Surf Sci* 2009;84:323.
- [48] Jardine AP, Dworski S, Fouquet P, Alexandrowicz G, Riley DJ, Lee GYH, et al. *Science* 2004;304:1790.
- [49] Fouquet P, Hedgeland H, Jardine AP. *Z Phys Chem* 2010;224:61.
- [50] Yaouanc A, Dalmas de Reotier P. *Muon spin rotation, relaxation, and resonance: applications to condensed matter*. Oxford: Oxford University Press; 2011.
- [51] Nagamine K. *Introductory muon science*. Cambridge: Cambridge University Press; 2003.
- [52] Lee SL, Cywinski R, Kilcoyne SH, editors. *Muon science: muons in physics, chemistry, and materials*. New York: Taylor & Francis; 1999.
- [53] Schenck AGE. *Muon spin rotation spectroscopy: principles and applications in solid state physics*. New York: Taylor & Francis; 1985.
- [54] www.isis.stfc.ac.uk/groups/muons/ [last accessed on August 20, 2013].
- [55] j-parc.jp/researcher/MatLife/en/instrumentation/ms.html [Last accessed on August 20, 2013].
- [56] lmu.web.psi.ch/ [last accessed on August 20, 2013].
- [57] <http://musr.ca/> [last accessed on August 20, 2013].
- [58] Kremer F, Schönhals A, editors. *Broadband dielectric spectroscopy*. Berlin: Springer; 2002.
- [59] Frenkel D, Smit B. *Understanding molecular simulation: from algorithms to applications*. 2nd ed. San Diego: Academic Press; 2001.
- [60] Martin RM. *Electronic structure: basic theory and practical methods*. Cambridge: Cambridge University Press; 2008.
- [61] Attinger S, Koumoutsakos P, editors. *Multiscale modelling and simulation*. Berlin: Springer; 2004.

Section 2

- [62] Squires GL. *Introduction to the theory of thermal neutron scattering*. 3rd ed. Cambridge: Cambridge University Press; 2012.
- [63] Lovesey SW. *Theory of neutron scattering from condensed matter*, vols. I and II. Oxford: Oxford University Press; 1986.
- [64] Mezei F. *Z Phys* 1972;255:146.

Section 3

- [65] Glyde HR. *Excitations in liquid and solid helium*. Oxford: Oxford University Press; 1995.
- [66] Born M, Huang K. *Dynamical theory of crystal lattices*. Oxford: Oxford University Press; 1988.
- [67] Wallace DC. *Thermodynamics of crystals*. New York: John Wiley & Sons; 1972.

- [68] Dove MT. Introduction to lattice dynamics. Cambridge: Cambridge University Press; 1993.
- [69] Mezei F. In: Hansen JP, Levesque D, Zinn-Justin J, editors. Liquids, freezing, and the glass transition. Amsterdam: North Holland; 1991. p. 629–87.
- [70] Willis BTM, Pryor AW. Thermal vibrations in crystallography. Cambridge: Cambridge University Press; 1975.
- [71] Sears VF, Shelley SA. *Acta Cryst A* 1991;47:441.
- [72] Zaccai G. *Science* 2000;288:1604.
- [73] Dove MT, Tucker MG, Keen DA. *Eur J Mineral* 2002;14:331.
- [74] Tucker MG, Keen DA, Dove MT, Goodwin AL, Hui Q. *J Phys Condens Matter* 2007;19:335218.
- [75] Serrano J, Manjón FJ, Romero AH, Ivanov A, Cardona M, Lauck R, et al. *Phys Rev B* 2010;81:174304.
- [76] Taraskin SN, Elliott SR. *Phys Rev B* 1997;55:117.
- [77] Mitchell PCH, Parker SF, Ramirez-Cuesta AJ, Tomkinson J. Vibrational spectroscopy with neutrons, with applications in chemistry, biology, materials science and catalysis. Singapore: World Scientific; 2005.
- [78] <http://www.isis.stfc.ac.uk/instruments/tosca/ins-database/> [last accessed on August 8, 2013].
- [79] Sjölander A. *Ark Fys* 1958;14:315.
- [80] Parker SF, Bennington SM, Ramirez-Cuesta AJ, Auffermann G, Bronger W, Herman H, et al. *J Am Chem Soc* 2003;125:11656.
- [81] Andreani C, Colognesi D, Pace E. *Phys Rev B* 1999;60:10008.
- [82] Fultz B. *Prog Mater Sci* 2010;55:247.
- [83] Cowley RA. *Rep Prog Phys* 1968;31:123.
- [84] Cowley RA. *Adv Phys* 1963;12:421.
- [85] Safarik DJ, Llobet A, Lashley JC. *Phys Rev B* 2012;85:174105.
- [86] Maradudin AA, Ambegaokar V. *Phys Rev* 1964;13:A1070. Cowley RA, Buyers WJL. *J Phys C* 1969;2:2262.
- [87] Glyde HR. *Can J Phys* 1974;52:2281. Varma CM, Werthamer NR. In: Bennemann KH, Ketterson JB, editors. The physics of liquid and solid helium, part I. New York: Wiley; 1974. p. 503.
- [88] Aynajian P, Keller T, Boeri L, Shapiro SM, Habicht K, Keimer B. *Science* 2008;319:1509.
- [89] Bayrakci SP, Keller T, Habicht K, Keimer B. *Science* 2006;312:1926.
- [90] Fernandez-Alonso F, Cabrillo C, Fernandez-Perea R, Bermejo FJ, Gonzalez MA, Mondelli C, et al. *Phys Rev B* 2012;86:144524.
- [91] Mesot J. *Science* 2006;312:1888.
- [92] Paulatto L, Mauri F, Lazzeri M. *Phys Rev B* 2013;87:214303.
- [93] Baroni S, de Gironcoli S, Dal Corso A, Gianozzi P. *Rev Mod Phys* 2001;73:515.

Section 4

- [94] Van Hove L. *Phys Rev* 1954;95:249.
- [95] Egami T, Billinge SJL. Underneath the Bragg peaks: structural analysis of complex materials. Amsterdam: Pergamon; 2003.
- [96] Yarnell JL, Katz MJ, Wenzel RG, Koenig SH. *Phys Rev A* 1973;7:2130.
- [97] Sköld K, Rowe JM, Ostrowski GE, Randolph PD. *Phys Rev A* 1972;6:1107.
- [98] de Gennes PG. *Physica* 1959;25:825.
- [99] de Gennes PG. *J Phys Chem Solids* 1958;4:223.
- [100] <http://www.isis.stfc.ac.uk/groups/disordered-materials/database/> [last accessed on August 10, 2013].

- [101] Sjölander A. In: Egelstaff PA, editor. Inelastic neutron scattering. New York: Academic Press; 1965. p. 301–3, and references therein.
- [102] Sjölander A. In: Egelstaff PA, editor. Thermal neutron scattering. New York: Academic Press; 1965. p. 291.
- [103] Cusack NE. The physics of structurally disordered matter, an introduction. Bristol: Adam Hilger; 1988.
- [104] Rahman A, Singwi KS, Sjölander A. Phys Rev 1962;126:986.
- [105] Placzek G. Phys Rev 1952;86:377.
- [106] Placzek G. Phys Rev 1954;93:895.
- [107] Placzek G. Phys Rev 1957;105:1240.
- [108] Marshall W, Lovesey SW. Theory of thermal neutron scattering. Oxford: Oxford University Press; 1971.
- [109] Mayers J. Phys Rev B 1990;41:41.
- [110] Sears VF. Phys Rev B 1984;30:44.
- [111] Andreani C, Colognesi D, Mayers J, Reiter GF, Senesi R. Adv Phys 2005;54:377.
- [112] Hilleke RO, Chaddah P, Simmons RO, Price DL, Sinha SK. Phys Rev Lett 1984;52:847. Sokol PE, Simmons RO, Price DL, Hilleke RO, Eckern U, Schmid A, Weber W, Wahl H, editors. Proceedings of the 17th international conference on low temperature physics. Amsterdam: North-Holland Publishing Company; 1984. p. 1213.
- [113] Sokol PE, Sköld K, Price DL, Kleb R. Phys Rev Lett 1985;54:909. Mook H. Phys Rev Lett 1985;55:2452.
- [114] Whitlock PA, Panoff RM. Proceedings of the 1984 workshop on high-energy excitations condensed matter. Rep. LA-10227, vol. II; 1984. p. 430, and references therein.
- [115] Glyde HR, Azuah RT, Stirling WG. Phys Rev B 2000;62:14337.
- [116] Watson GI. J Phys Condens Matter 1996;8:5955.
- [117] Seel AG, Ceriotti M, Edwards PP, Mayers J. J Phys Condens Matter 2012;24:365401.
- [118] Krzystyniak M, Richards SE, Seel AG, Fernandez-Alonso F. (submitted).
- [119] Adams MA, Mayers J, Kirichek O, Down RBE. Phys Rev Lett 2007;98:085301.
- [120] Diallo SO, Pearce JV, Azuah RT, Kirichek O, Taylor JW, Glyde HR. Phys Rev Lett 2007;98:205301.
- [121] Reiter GF, Mayers J, Platzman P. Phys Rev Lett 2002;89:135505.
- [122] Krzystyniak M, Fernandez-Alonso F. Phys Rev B 2011;83:134305.
- [123] Krzystyniak M, Adams MA, Lovell A, Skipper NT, Bennington SM, Mayers J, et al. Faraday Discuss 2011;151:171.
- [124] Berne BJ, Thirumalai D. Annu Rev Phys Chem 1986;37:401.
- [125] Lin L, Morrone JA, Car R, Parrinello M. Phys Rev Lett 2010;105:110602.
- [126] Ceriotti M, Markland TE. J Chem Phys 2013;138:014112.
- [127] Pantalei C, Pietropaolo A, Senesi R, Imberti S, Andreani C, Mayers J, et al. Phys Rev Lett 2008;100:177801.
- [128] Pietropaolo A, Senesi R, Andreani C, Botti A, Ricci MA, Bruni F. Phys Rev Lett 2008;100:127802.
- [129] Feng YJ, Ancona-Torres C, Rosenbaum TF, Reiter GF, Price DL, Courtens E. Phys Rev Lett 2006;97:145501.
- [130] Homouz D, Reiter G, Eckert J, Blinc R. Phys Rev Lett 2007;98:115502.
- [131] Ceriotti M, Miceli G, Pietropaolo A, Colognesi D, Nale A, Catti M, et al. Phys Rev B 2010;82:174306.
- [132] Rahman A. In: Woods Halley J, editor. Correlation functions and quasiparticle interactions in condensed matter. New York: Plenum; 1977. p. 417.

- [133] Allen MP, Tildesley DJ. Computer simulation of liquids. Oxford: Clarendon Press; 1996.
- [134] Schofield P. Phys Rev Lett 1960;4:239.
- [135] Price DL, Bermejo FJ, Saboungi M-L. Rep Prog Phys 2003;66:407.
- [136] www.visiongain.com/Report/721/The-Oil-Gas-Corrosion-Prevention-Market-2011-2021
[last accessed on August 23, 2013].
- [137] Krivoglaz MA. Theory of X-ray and thermal-neutron scattering by real crystals. New York: Plenum; 1969.
- [138] Ziman JM. Models of disorder: the theoretical physics of homogeneously disordered systems. Cambridge: Cambridge University Press; 1979.
- [139] Nield VM, Keen DA. Diffuse neutron scattering from crystalline materials. Oxford: Oxford University Press; 2001.
- [140] Welberry TR. Diffuse X-ray scattering and models of disorder. Oxford: Oxford University Press; 2004.
- [141] Welberry TR, Goossens DJ, David WIF, Gutmann MJ, Bull MJ, Heerdegen AP. J Appl Cryst 2003;36:1440.
- [142] Goodwin AL, Tucker MG, Dove MT, Keen DA. Phys Rev Lett 2004;93:075502.
Goodwin AL, Tucker MG, Dove MT, Keen DA. Phys Rev Lett 2005;95:119901.
- [143] Goodwin AL, Tucker MG, Cope ER, Dove MT, Keen DA. Physica B 2006;385–386:285.
- [144] Parsonage NG, Staveley LAK. Disorder in crystals. Oxford: Clarendon Press; 1978.
- [145] Angell CA. J Non-Cryst Solids 1991;131:13.
- [146] Brand R, Lunkenheimer P, Loidl A. J Chem Phys 2002;116:10386.
- [147] Bée M. Quasielastic neutron scattering, principles and applications in solid state chemistry, biology and materials science. Bristol: Adam Hilger; 1988.
- [148] Hempelmann R. Quasielastic neutron scattering and solid state diffusion. Oxford: Oxford University Press; 2000.
- [149] Vineyard G. Phys Rev 1959;110:999.
- [150] Nijboer BRA, Rahman A. Phys Rev 1962;126:989.
- [151] Hansen JP, McDonald IR. Theory of simple liquids. Boston: Academic Press; 1986.
- [152] Kubo R. In: Meixner J, editor. Statistical mechanics of equilibrium and non-equilibrium. Amsterdam: North Holland; 1965.
- [153] Egelstaff PA. An introduction to the liquid state. Oxford: Oxford University Press; 1992.
- [154] Balucani U, Zoppi M. Dynamics of the liquid state. Oxford: Oxford University Press; 1994.
- [155] Fernandez-Alonso F, Bermejo FJ, McLain SE, Turner JFC, Molaison JJ, Herwig KW. Phys Rev Lett 2007;98:077801.
- [156] Sears VF. Can J Phys 1966;44:1279.
- [157] Sears VF. Can J Phys 1966;44:1299.
- [158] Sears VF. Can J Phys 1967;45:237.
- [159] Volino F, Dianoux AJ. Mol Phys 1981;41:271.
- [160] Hall PL, Ross DK. Mol Phys 1981;42:673.
- [161] Bicout DJ. Phys Rev E 2000;62:261.
- [162] Bicout DJ. Phys Rev E 2001;64:011910.
- [163] Morkel Chr, Gronemeyer Chr, Gläser W, Bosse J. Phys Rev Lett 1987;58:1873, Phys Rev Lett 1987;58:2609.
- [164] Sköld K. Phys Rev Lett 1967;19:1023.
- [165] Chudley CT, Elliot RJ. Proc Phys Soc Lond 1961;77:353.

- [166] Karlsson M, Engberg D, Björketun ME, Matic A, Wahnström G, Sundell PG, et al. *Chem Mater* 2010;22:740.
- [167] Karlsson M. *Dalton Trans* 2013;42:317.
- [168] Hull S. *Rep Prog Phys* 2004;67:1233.
- [169] Höch A, Funke K, Lechner RE, Ohachi T. *Solid State Ionics* 1983;9/10:1353, and references therein.
- [170] Goff JP, Hayes W, Hull S, Hutchings MT, Klausen KN. *Phys Rev B* 1999;59:14202.
- [171] Price DL. *High-temperature levitated materials*. Cambridge: Cambridge University Press; 2010.
- [172] Press W. *Single-particle rotations in molecular crystals*. Berlin: Springer-Verlag; 1981.
- [173] Prager M, Heidemann A. *Chem Rev* 1997;97:2933.
- [174] Colmenero J, Mukhopadhyay R, Alegría A, Frick B. *Phys Rev Lett* 1998;80:2350.
- [175] Larese JZ, Martin y Marero D, Sivia DS, Carlile CJ. *Phys Rev Lett* 2002;87:206102.
- [176] Lovell A, Fernandez-Alonso F, Skipper NT, Refson K, Bennington SM, Parker SF. *Phys Rev Lett* 2008;101:126101.
- [177] Fernandez-Alonso F, Bermejo FJ, Cabrillo C, Loutfy RO, Leon V, Saboungi M-L. *Phys Rev Lett* 2007;98:215503.
- [178] Yang S, Sun J, Ramirez-Cuesta AJ, Callear SK, David WIF, Anderson DP, et al. *Nat Chem* 2012;4:887.
- [179] Feigin LA, Svergun DI. *Structure analysis by small-angle X-Ray and neutron scattering*. New York and London: Plenum Press; 1987.
- [180] Hollanby MJ. *Phys Chem Chem Phys* 2013;15:10566.
- [181] Guinier A. *Ann Phys (Paris)* 1939;12:161.
- [182] Porod G. *Kolloid Z* 1951;124:83.
- [183] Eastoe J, Gold S, Rogers SE, Paul A, Welton T, Heenan RK, et al. *J Am Chem Soc* 2005;127:7302.
- [184] Engelman DM, Moore PB. *Sci Am* 1976;235:44.
- [185] Petoukhov MV, Svergun DI. *Biophys J* 2005;89:1237.
- [186] Svergun DI, Richard S, Koch MHJ, Sayers Z, Kuprin S, Zaccai G. *Proc Natl Acad Sci USA* 1998;95:2267.
- [187] Müller-Buschbaum P. *Polym J* 2013;45:34.
- [188] Penfold J, Thomas RK. *J Phys Condens Matter* 1990;2:1368.
- [189] Sinha SK, Sirota EB, Garoff S, Stanley HB. *Phys Rev B* 1988;38:2297.
- [190] Pynn R. *Phys Rev B* 1992;45:602.
- [191] Dietrich S, Haase A. *Phys Rep* 1995;260:1.
- [192] Dalgliesh R. *Curr Opin Colloid Interface Sci* 2002;7:244.
- [193] Bowron DT, Soper AK, Jones K, Ansell S, Birch S, Norris J, et al. *Rev Sci Instrum* 2010;81:033905.
- [194] Neuefeind J, Feygenson M, Carruth J, Hoffmann R, Chipley KN. *Nucl Instr Methods B* 2012;287:68.
- [195] j-parc.jp/MatLife/en/instrumentation/ns_spec.html#bl21 [last accessed on August 21, 2013].

Section 5

- [196] Merzbacher E. *Quantum mechanics*. 3rd ed. New York: John Wiley & Sons; 1998.
- [197] Brown JM, Bunker RJ, Carrington A, di Lauro C, Dixon RN, Field RW, et al. *Mol Phys* 2000;98:1597.

- [198] Gaspar AM, Busch S, Appavou M-S, Haeussler W, Georgii R, Su Y, et al. *Biochim Biophys Acta* 2010;1804:76.
- [199] Penfold J, Webster J, Bucknall D. The use of polarisation analysis in polarised neutron reflection studies. Rutherford Appleton Laboratory Technical Report RAL-93-018, Didcot: Rutherford Appleton Laboratory; 1993.
- [200] Böni P. *Physica B* 1997;234–236:1038.
- [201] Dalglish RM, van Well AA, Boag S, Charlton TR, Frost CD, de Haan VO, et al. *Physica B* 2007;397:176.
- [202] Andersen KH, Jullien D, Petoukhov AK, Mouveau P, Bordenave F, Thomas F, et al. *Physica B* 2009;404:2652.
- [203] Surkau R, Becker J, Ebert M, Grossmann T, Heil W, Hofmann D, et al. *Nucl Instr Methods A* 1997;384:444.
- [204] Freeman FF, Williams WG. *J Phys E Sci Instrum* 1978;11:459.
- [205] Felcher GP, Kleb R, Hinks DG, Potter WH. *Phys Rev B* 1984;29:4843.
- [206] Koehler WC, Cable JW, Wilkinson MK, Wollan EO. *Phys Rev* 1961;151:414.
- [207] Low GGE, Collins MF. *J Appl Phys* 1963;34:1195. Collins MF, Low GG. *Proc Phys Soc Lond* 1965;86:535.
- [208] Hicks JH. In: Kostorz G, editor. *Neutron scattering*. New York: Academic Press; 1979. p. 337.
- [209] Binder K, Young AP. *Rev Mod Phys* 1985;58:801.
- [210] Dormann JL, Fiorani D, Tronc E. *Adv Chem Phys* 1997;98:283.
- [211] Holm C, Weiss JJ. *Curr Opin Colloid Interface Sci* 2005;10:133.
- [212] Fennell T, Deen PP, Wildes AR, Schmalzl K, Prabhakaran D, Boothroyd AT, et al. *Science* 2009;326:415.
- [213] Mühlbauer S, Binz B, Jonietz F, Pfleiderer C, Rosch A, Neubauer A, et al. *Science* 2009;323:915.
- [214] Hughes J, Burgoyne MT, Helier RB, Wallace JW. *Phys Rev* 1949;75:565.
- [215] Felcher GP, Kampwirth RT, Gray KE, Felici R. *Phys Rev Lett* 1984;52:1539.
- [216] Chakhalian J, Feland JW, Srajer G, Stremper J, Khaliullin G, Cezar JC, et al. *Nat Phys* 2006;2:244.

Section 6

- [217] Lipscombe OJ, Chen GF, Fang C, Perring TG, Abernathy DL, Christianson AD, et al. *Phys Rev Lett* 2011;106:057004.
- [218] Birgeneau RJ, Bucher E, Passel L, Price DL, Turberfield KC. *J Appl Phys* 1970;41:900.
- [219] Pfleiderer C. *Rev Mod Phys* 2009;81:1551.
- [220] Mirebeau I, Hennion M, Casalta H, Andres H, Güdel HU, Irodova AV, et al. *Phys Rev Lett* 1999;83:628.
- [221] Furrer A, Waldmann O. *Rev Mod Phys* 2013;85:367.
- [222] Shapiro SM, Axe JD, Birgeneau RJ, Lawrence JM, Parks RD. *Phys Rev B* 1977;16:2275.
- [223] Loong C-K. *J Appl Phys* 1985;57:3772.
- [224] Loong C-K, Grier BH, Shapiro SM, Lawrence JM, Parks RD, Sinha SK. *Phys Rev B* 1987;35:3092.
- [225] Holland-Moritz E, Wohlleben D, Loewenhaupt M. *Phys Rev B* 1982;25:7482.
- [226] Chupis IE. *Low Temp Phys* 2010;36:477.
- [227] Loktev VM, Pogorelov YuG. *Low Temp Phys* 2000;26:171.
- [228] Loewenhaupt M, Lander GH, Murani AP, Murasik A. *J Phys C* 1982;15:5199.
- [229] Matan K, Morinaga R, Iida K, Sato TJ. *Phys Rev B* 2009;79:054526.

- [230] Hordequin C, Pierre J, Currat R. *J Magn Magn Mater* 1996;162:75.
- [231] Sinha SK, Silver RN, editor. *Proceedings of the 1984 workshop on high-energy excitations condensed matter*, Rep. LA-10227, vol. 11. Los Alamos, New Mexico: Los Alamos Natl. Lab; 1984. p. 346, Cooke JB. *ibid.*, p. 401.
- [232] Van Hove L. *Phys Rev* 1954;95:1374.
- [233] Halperin BI, Hohenberg PC. *Phys Rev* 1969;177:952.
- [234] Foster D. *Hydrodynamic fluctuations, broken symmetry, and correlation functions*. Amsterdam: W.A. Benjamin Inc.; 1975.
- [235] Collins MF, Minkiewicz VJ, Nathans R, Passel L, Shirane G. *Phys Rev* 1969;179:417, see also Minkiewicz VJ. *Int J Magn* 1971;1:149.
- [236] Als-Nielsen J. *Brookhaven Natl Lab Rep BNL* 14793; 1971.
- [237] Schulhof MP, Heller P, Nathans R, Linz A. *Phys Rev B* 1970;1:2304.
- [238] Birgeneau RD, Skalyo J, Shirane G. *Phys Rev B* 1971;3:1736.
- [239] Collins MF. *Magnetic critical scattering*. Oxford: Oxford University Press; 1989.

Section 7

- [240] Abragam A, Goldman M. *Nuclear magnetism: order and disorder*. Oxford: Clarendon Press; 1982.
- [241] Hayter JB, Jenkin GT, White JW. *Phys Rev Lett* 1974;33:696.
- [242] Roinel Y, Bouffard V, Bacchella GL, Pinot M, Meriel P, Roubeau P, et al. *Phys Rev Lett* 1978;41:1572.
- [243] Leslie M, Jenkin GT, Hayter JB, White JW, Cox S, Wagner G. *Philos Trans R Soc Lond B* 1980;290:497.
- [244] Steiner M, Bevaart L, Ajiro Y, Millhause AJ, Ohlhoff K, Rahn G, et al. *J Phys C* 1981;14: L597.
- [245] Suzuki H, Ohtsuka T, Kawarazaki S, Kunitomi N, Moon RM, Nicklow RM. *Solid State Commun* 1984;49:1157.
- [246] Benoit A, Bossy J, Flouquet J, Schweizer J. *J Phys Lett* 1985;46:923.
- [247] Jyrkkio TA, Huiku MT, Lounasmaa OV, Siemensmeyer K, Kakurai K, Steiner M, et al. *Phys Rev Lett* 1988;60:2418.
- [248] Steiner M. *Phys Scr* 1990;42:367.
- [249] Tuoriniemi JT, Nummila KK, Vuorinen RT, Lounasmaa OV, Metz A, Siemensmeyer K, et al. *Phys Rev Lett* 1995;75:3744.
- [250] Oja AS, Lounasmaa OV. *Rev Mod Phys* 1997;69:1.
- [251] Siemensmeyer K, Clausen KN, Lefmann K, Lounasmaa OV, Metz A, Nummila KK, et al. *Physica B* 1998;241–243:506.
- [252] van den Brandt B, Glättli H, Grillo I, Hautle P, Jouve H, Kohlbrecher J, et al. *Europhys Lett* 2002;59:62.
- [253] Steiner M. *J Low Temp Phys* 2004;135:545.
- [254] van den Brandt B, Glättli H, Hautle P, Kohlbrecher J, Konter JA, Michels A, et al. *J Appl Cryst* 2007;40:S106.
- [255] Kumada T, Noda Y, Koizumi S, Hashimoto T. *J Chem Phys* 2010;133:054504.
- [256] Noda Y, Kumada T, Hashimoto T, Koizumi S. *J Appl Cryst* 2011;44:503.
- [257] Piegsa FM, Karlsson M, van den Brandt B, Carlile CJ, Forgan EM, Hautle P, et al. *J Appl Cryst* 2013;46:30.
- [258] Chatterji T, Schneider GJ. *Phys Rev B* 2009;79:212409.
- [259] Ehlers G, Mamontov E, Zamponi M, Kam KC, Gardner JS. *Phys Rev Lett* 2009;102:016405.

- [260] Chatterji T, Jalarvo N, Szytula A. Solid State Commun 2013;161:42.
- [261] Abragam A, Goldman M. Rep Prog Phys 1978;41:395.
- [262] Maly T, Debelouchina GT, Baja VS, Hu KN, Joo CG, Mak-Jurkauskas ML, et al. J Chem Phys 2008;128:052211.
- [263] Ardenkjaer-Larsen JH, Fridlund B, Gram A, Hansson G, Hansson L, Lerche MH, et al. Proc Natl Acad Sci U S A 2003;100:10158.
- [264] Buckingham AD. Chem Phys Lett 2003;371:517.

Section 8

- [265] Sköld K, Price DL, editors. Neutron scattering, experimental methods in the physical sciences, vol. 23A. New York: Academic Press; 1986.
- [266] Furrer A. Neutron News 2007;18:2.